DERIVATIVES PRICING · GREEKS · RISK · QUANTITATIVE ENGINEERING

Quant Finance From Scratch

The complete handbook: intuition, mathematics, verified numbers, code and validation for every concept

This edition merges your two sets of study notes, the scanned *Quant Notes* PDF, and the follow-up research handbook into one structured book. Nothing from the originals has been dropped. Every section has been expanded wherever it was thin, and every number has been recomputed in Python before being printed here.

Running example throughout: $S_0 = 100,\ K = 100,\ T = 1,\ r = 5\%,\ q = 0,\ \sigma = 20\%$
Call $= 10.4506$ Put $= 5.5735$ $\Delta = 0.6368$ $\Gamma = 0.01876$ Vega $= 0.3752$/vol pt $\Theta = -0.0176$/day

Prepared October 2026. Market facts (rates, lot sizes, expiry days) are dated; re-check them before relying on them.

Before you start

Why this book exists

Your original notes had exactly the right skeleton. They walked from "what is a market?" all the way to "how would I build a pricing dashboard?", and they used a good teaching pattern: intuition, then mathematics, then a numerical example, then an implementation view, then a checklist. The problem was depth. Most concepts stopped at the definition, so you could recognise a formula but not yet defend it, compute it, or test it.

Turning a definition into working knowledge takes four more layers, and this book adds them to every concept, in the same order your notes introduced them:

1. **Why it is true.** A derivation, or an arbitrage argument that would make you money if the statement were false.
2. **How big it is.** A worked example with numbers that were recomputed in code, so you can use them as unit tests.
3. **How practitioners actually quote it.** Conventions such as vega per vol point, theta per calendar day, ACT/365 time, and "ATM forward". A large share of real bugs on trading desks are convention bugs, not maths bugs.
4. **How you know your code is right.** A validation check: parity, a bound, a limit, a convergence rate, or a comparison with an independent method.

How each section is laid out

Most sections follow the same rhythm, and coloured boxes tell you which mode you are in:

> **Intuition** The plain-English picture. If you only read these boxes, you would still understand what every concept is *for*.

> **Worked example**
>
> A worked numerical example. Every figure in these boxes was recomputed in Python, so your own code should reproduce them to about four decimal places.

> **Pitfalls & conventions**
>
> The convention traps and the classic mistakes: the things that make two correct-looking programs disagree.

> **Correction to the original notes**
>
> Places where the original notes were wrong, ambiguous or incomplete, with the fix.

Check yourself

Questions you should be able to answer before moving on. If you can't, re-read the section rather than pushing ahead. Everything later is built on it.

The running example

Unless a section says otherwise, we price options on one underlying with these inputs:

$$S_0 = 100, \quad K = 100, \quad T = 1 \text{ year}, \quad r = 5\% \text{ (continuous)}, \quad q = 0, \quad \sigma = 20\%.$$

Using the same example everywhere means each new method can be checked against numbers you already trust. A tree, a Monte Carlo engine and a PDE solver should all agree with the Black–Scholes call value of 10.4506. When they don't, you have found a bug, and finding bugs early is most of the job.

Table 1. Golden values for the running example (keep these beside your test suite).

Quantity	Value	Where it is derived
d_1, d_2	0.35, 0.15	§49
$N(d_1)$, $N(d_2)$	0.636831, 0.559618	§49
European call / put	10.4506 / 5.5735	§49
Put–call parity $C - P = S - Ke^{-rT}$	4.8771	§35
Delta (call / put)	0.6368 / −0.3632	§50
Gamma	0.018762	§50
Vega	37.524 per unit = 0.3752 per vol pt	§50
Theta, call	−6.414/year = −0.01757/day	§50
Theta, put	−1.658/year = −0.00454/day	§50
Rho, call / put	0.5323 / −0.4189 per 1%	§50
Vanna, Volga	−0.2814, 9.850 (per unit σ)	§51
American put (LR tree, 5001 steps)	6.0903	§56
Black-76 call, $F = K = 100$	7.5771	§59
Implied vol of a call priced at 11	21.4616%	§54
MC standard deviation of discounted payoff	14.719	§60
Up-and-out call, $H = 120$ (continuous)	1.1761	§69
Floating-strike lookback call	17.2168	§67

Notation

Symbol	Meaning
S_t, S_0, S_T	Price of the underlying at time t, today, and at expiry
K	Strike price
T; $\tau = T - t$	Expiry (in years); time remaining to expiry
r, q	Continuously compounded risk-free rate; continuous dividend yield (or foreign rate for FX)
σ	Volatility: annualised standard deviation of log returns
F	Forward or futures price, $F = S_0 e^{(r-q)T}$
C, P, V	Call value, put value, generic derivative value

$N(x)$, $n(x)$	Standard normal CDF and density, $n(x) = e^{-x^2/2}/\sqrt{2\pi}$
W_t	Standard Brownian motion (Wiener process)
$\mathbb{P}$, $\mathbb{Q}$	Real-world probability measure; risk-neutral (pricing) measure
$\Delta, \Gamma, \nu, \Theta, \rho$	Delta, gamma, vega, theta, rho
H	Barrier level
x^+	$\max(x, 0)$

Summary of corrections to the original notes

Original notes say / omit	Correction (section)
"$\Theta = \partial V/\partial T$"	Theta is $\partial V/\partial t$ in *calendar* time, which equals $-\partial V/\partial \tau$ where τ is time to expiry. Long options usually have $\Theta < 0$. Always state per calendar day ($\div 365$) or per trading day ($\div 252$). (§50)
Vega/rho examples "0.20 for 1%"	Fine as illustrations, but analytic formulas give *per unit* values (37.52), and you divide by 100 for "per vol point". Libraries differ, so check before comparing. (§50)
Dividends section thin	Continuous yield vs discrete cash dividends; escrowed-dividend model; jump condition in trees/PDEs; early exercise of American calls before ex-dates. (§30)
Itô's lemma listed, never explained	Full statement, the Taylor-series reason for the $\frac{1}{2}\sigma^2$ term, and the worked derivation of $d \ln S$. (§40)
Black-76: no formula	Formula, Greeks, worked example, and uses (futures options, caps, swaptions). (§59)
Leisen–Reimer: no formula	Peizer–Pratt inversion written out, tree parameters, odd-N rule, a convergence table and code. (§58)
Bisection/Newton only	Adds Brent's method, initial guesses, Jäckel's *Let's Be Rational*, and the failure cases. (§54)
Variance reduction named only	Antithetic, control variates (Kemna–Vorst), importance sampling, CRN and moment matching, each with mechanics and numbers. (§64)
Monte Carlo Greeks missing	Bump-and-revalue with CRN, pathwise, likelihood-ratio, with a numerical comparison. (§65)
Smile shown, causes missing	Crash-o-phobia after 1987, leverage effect, fat tails/jumps, supply and demand. (§28)
Heston/SABR/Merton only named	SDEs, pricing formulas, what each fixes, and model-implied smiles computed here. (§41–§43)
Finite differences thin	Explicit/implicit/Crank–Nicolson, stability with a blow-up demonstration, Rannacher smoothing, boundary conditions, barrier grids. (§70)
Lookback/barrier: no formulas	Closed forms (Goldman–Sosin–Gatto, Conze–Viswanathan, Reiner–Rubinstein), checked against Monte Carlo; discrete-monitoring correction $\beta \approx 0.5826$. (§67, §69)
P&L attribution listed, not covered	Taylor-based explain with a residual, worked numerically, plus the regulatory context. (§73)
Calibration: one paragraph	Objective functions, weighting, optimisers, regularisation, stability, an SVI example. (§77)

Money-market hedge	Correct as far as it goes, but the INR borrowing rate was missing. Completed, and shown equal to covered interest parity. (§76)
Forward example "103 in 3 months"	Fine as an illustration, but under cost-of-carry it implies a rate of about 11.8%, so it is not consistent with $r = 5\%$. (§20)

The roadmap and the model map

The single most important idea in this book is that these topics are *not* separate. Each model exists because the previous one could not handle some feature of a contract:

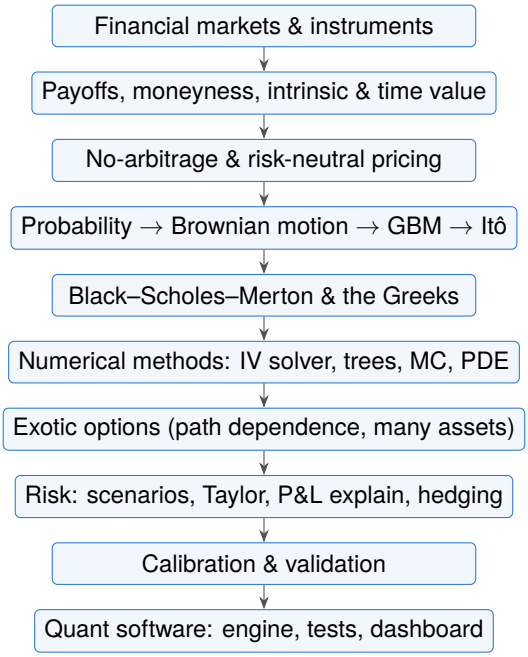


Table 4. Your project's model map: which method prices which product, and why.

Product	Model	Main challenge	Validate against
European call/put	BSM (closed form)	None, it is the benchmark	Parity, bounds, textbook values
American call/put	CRR binomial tree	Early exercise at every node	Am. $\geq$ Eur.; Am. call = Eur. call if $q = 0$
American call/put	Leisen–Reimer tree	Faster, smoother convergence	CRR at large N; BSM for Europeans
Futures option	Black-76	Underlying is F, not S	BSM with $Se^{-qT} \to Fe^{-rT}$
Asian	Monte Carlo	Payoff depends on the average	Geometric Asian closed form
Lookback	Monte Carlo	Payoff depends on max/min	Goldman–Sosin–Gatto closed form
Basket	Monte Carlo	Several correlated assets	$\rho = 1$ / single-asset limits
Barrier	Finite-difference PDE	Boundary at the barrier	Reiner–Rubinstein; in + out = vanilla

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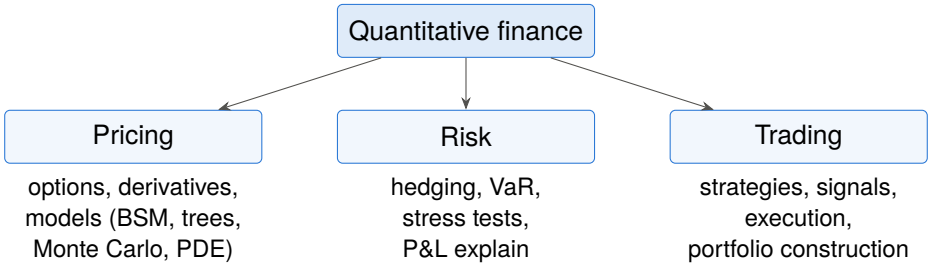
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PART I

Markets, Instruments and Contracts

1 What is quantitative finance?

Quantitative finance uses mathematics, statistics, probability and programming to understand financial markets, and to price and manage financial instruments. That is the textbook definition. In practice, the people who carry the title "quant" do three quite different jobs:



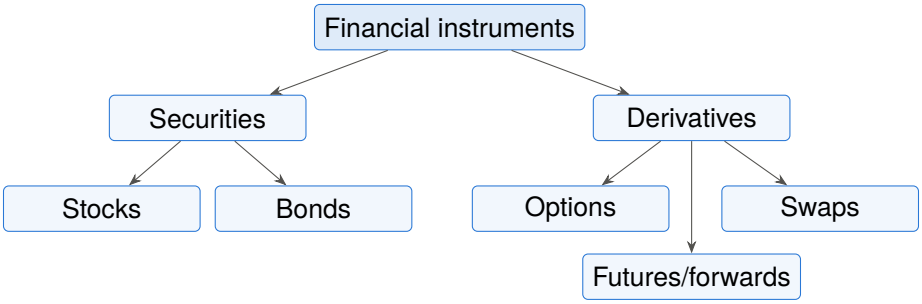
- **Pricing (“sell-side” or “desk” quants)** answer the question: *what is this contract worth today, and how does that value respond to the market?* They build the models and the code that turn market data into prices and Greeks.
- **Risk quants** answer: *how much could we lose, and why did we make or lose what we did?* They build scenario engines, VaR models and P&L-explain systems, and they validate the pricing quants' models.
- **Trading / research (“buy-side”) quants** answer: *where is there an edge, and how do we size it?* They work with forecasts, signals and portfolio optimisation.

Your project sits squarely in the first two: **derivatives pricing, Greeks, numerical methods, risk and quantitative engineering**. That determines what matters in this book. We care far more about no-arbitrage and replication than about forecasting, because, as you will see in Part V, a derivatives price does not depend on your view of where the market is going.

2 Financial markets and instruments

> **Intuition** A market is a place where *risk changes hands at a price*. Some participants want to own an economic exposure (a pension fund buying equities). Some want to get rid of one (an importer who fears a weaker rupee). Some want to transform it (a company turning floating-rate debt into fixed). Instruments are the containers that package those exposures so they can be traded.

The examples you will meet constantly are stocks, bonds, currencies, commodities, futures, options and swaps. They fall into two broad families:



2.1 Securities versus derivatives

A **security** is a direct claim on cash flows. A share is a claim on a company's residual profits after everyone else has been paid. A bond is a claim on promised coupons and the return of principal. Securities are *positive net supply*: a company issued them, and somebody in the economy holds them.

A **derivative** is a contract whose payoff is a *function of something else*, the underlying. Derivatives are mostly *zero net supply*: for every long position there is an equal and opposite short. If you buy a NIFTY call and make ₹500, someone who wrote that call loses ₹500. This symmetry is why derivatives markets are often described as "transferring" risk rather than creating wealth.

2.2 The main asset classes, a little deeper

- **Stocks (equity).** Residual ownership of a company. Their value reflects expected future cash flows and the discount rate investors apply. Many pay **dividends**, which lower the share price on the ex-date. That matters a great deal for option pricing (§30).
- **Bonds.** Promised cash flows. When yields rise, bond prices fall, because the same cash flows are discounted more heavily. The two big risks are *interest-rate risk* (measured by duration: roughly the percentage price change for a 1% change in yield) and *credit risk* (the issuer might not pay).
- **Currencies (FX).** The price of one money in terms of another, e.g. USD/INR $= 83$ means one dollar costs 83 rupees. An FX rate has *two* interest rates attached: holding dollars earns the dollar rate, holding rupees earns the rupee rate. That is why FX options are priced with the *Garman–Kohlhagen* model, which is simply Black–Scholes with the dividend yield q replaced by the foreign interest rate r_f.
- **Commodities.** Physical goods such as gold, crude oil or wheat. You cannot hold crude oil for free: it must be stored and insured. On the other hand, holding the physical good can be valuable if a shortage occurs (the *convenience yield*). So commodity forward prices reflect storage and convenience yield, not just interest (§20).
- **Futures and forwards.** Agreements to trade later at a price fixed today (Part III).
- **Options.** Rights, not obligations, to trade later at a fixed price (§5).
- **Swaps.** Agreements to exchange streams of cash flows. In an interest-rate swap (IRS), one party pays a fixed rate and the other pays a floating rate (in India, typically linked to overnight MIBOR; in the US, SOFR) on the same notional. A currency swap exchanges flows in two currencies. A useful insight: a swap is just a *portfolio of forwards*, one for each payment date, so you can value it by discounting the expected net cash flows.

Check yourself

- Why is a derivative described as "zero net supply"? Give an example.
- Why does an FX option need two interest rates while a stock option needs one rate and a dividend yield? (Hint: what does each side of the trade earn while you wait?)
- Explain in one sentence why a swap can be valued as a portfolio of forwards.

3 The underlying asset

The **underlying** is whatever the derivative's payoff references.

Derivative	Underlying
Apple call option	Apple stock
NIFTY option	NIFTY 50 index
Gold futures	Gold
USD/INR option	USD/INR exchange rate
Crude-oil option	Crude oil (usually crude-oil *futures*)
Interest-rate swap	An interest-rate index (MIBOR, SOFR)
Swaption	An interest-rate swap (a derivative on a derivative)

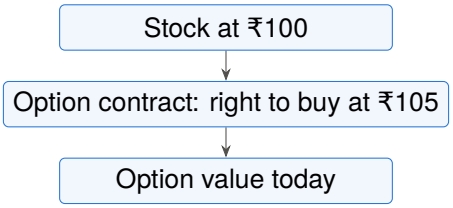
We write S_0 for today's price of the underlying. "$S_0 = 100$" means the underlying trades at 100 right now. Your project collects this as the first market input.

Pitfalls & conventions

The quantity a model really needs is often *not* the spot price. NIFTY options settle against the index close, but traders hedge with NIFTY futures, and the natural pricing variable is the *forward*, $F = S_0 e^{(r-q)T}$. Crude-oil options are usually options on futures. Interest-rate options are priced off forward rates. Get the underlying right and the choice between Black–Scholes and Black-76 (§59) becomes obvious.

4 What is a derivative?

A derivative is a contract whose value *derives* from something else. Suppose a stock trades at ₹100 and you hold a contract that says: *"I have the right to buy this stock for ₹105 three months from now."* The contract itself has value, because there is some chance the stock will be above ₹105 in three months. If it rises to ₹150 the right is worth ₹45. If it falls to ₹80 the right is worthless, and you simply do not use it.



Where does the value come from? There are two ways to answer this, and the whole of derivatives pricing is about showing that they agree:

- **Replication:** if you can build the derivative's payoff out of things that already trade (the stock and a bank account), then the derivative must cost the same as that "recipe". Otherwise there is free money (Part V).
- **Expectation:** the price is the discounted average payoff, computed under a special "risk-neutral" probability (§36).

Why do people use derivatives? For hedging (an importer locking in an exchange rate), for speculation with leverage (a small premium controls a large exposure), and for arbitrage and relative value (exploiting price inconsistencies between related contracts).

Settlement. A derivative can settle **physically** (you actually receive the asset) or in **cash** (you receive the payoff in money). Index options must be cash-settled, because you cannot deliver an index. NSE Clearing states that for index options "exercise settlement is cash settled by debiting/crediting of the clearing accounts of the relevant Clearing Members."

5 Options: calls and puts

An option gives its buyer a **right, but not an obligation**. There are two basic types:

- A **call** is the right to **buy** the underlying at the strike K.
- A **put** is the right to **sell** the underlying at the strike K.

A memory aid: you *call* the asset towards you (buy), and you *put* it away from you (sell).

> **Intuition** A call is like a refundable deposit on a house at a fixed price. If house prices boom you complete the purchase and pocket the difference. If they crash you walk away, losing only the deposit. A put is like insurance: you pay a premium so that, however far the price falls, you can still sell at K.

The asymmetry matters. The buyer's loss is capped at the premium while the upside can be large. That "heads I win a lot, tails I lose a little" shape (*convexity*) is exactly what the option premium pays for, and it is the source of every interesting property in this book.

6 Strike price

The **strike price** K (also called the exercise price) is the predetermined price at which the option can be exercised.

> **Worked example: a call with $S_0 = 100$, $K = 110$**
>
> The call gives you the right to buy at ₹110.
>
> - If the stock finishes at ₹150, you buy at 110 and can immediately sell at 150. The **gross payoff** is $150 - 110 = 40$.
> - If the stock finishes at ₹90, why would you pay 110 for something the market sells for 90? You do not exercise, and the payoff is ₹0.

In one formula: $\boxed{\text{call payoff} = \max(S_T - K, 0)}$.

Exchanges list strikes on a fixed grid around the current price. NIFTY strikes, for example, are spaced in round index points, more finely near the money for weekly expiries.

Pitfalls & conventions

When you compare strikes in a volatility smile, measure "how far away" a strike is relative to the *forward* F, not spot. With $r - q = 3\%$ and $T = 1$, the at-the-money-forward strike is $100e^{0.03} = 103.05$, not 100. Using spot as the centre makes the smile look tilted when it isn't.

7 Expiry, maturity and time measured in years

Options have a finite life. Pricing models measure time in **years**, so a three-month option has $T = 3/12 = 0.25$.

Calendar time	1 week	1 month	3 months	6 months	9 months	1 year
T (simple fraction)	$7/365 = 0.0192$	$1/12 = 0.0833$	0.25	0.50	0.75	1

7.1 Day-count conventions

Contracts specify *dates*, and models need *year fractions*. A **day-count convention** is the rule that converts one into the other:

- **ACT/365 (Fixed):** $\tau = \dfrac{\text{actual calendar days}}{365}$. This is the default for equity options and for INR markets. 30 days $\to$ 0.08219 years; a 7-day NIFTY weekly $\to$ 0.01918.
- **ACT/360:** used for USD money markets (SOFR deposits). It gives slightly *larger* year fractions for the same number of days ($30/360 = 0.0833$).
- **30/360:** pretends every month has 30 days. It is used for the accrual on many bonds.
- **Trading-day time (business/252):** volatility traders often count only trading days, because markets don't move much at weekends. Sophisticated desks go further and *weight* days: weekends near 0, ordinary days 1, event days (RBI policy, budget day, earnings) more than 1. Peter Jäckel's work on "effective volatility time" shows that this kind of weighting explains why overnight implied vol drops sharply from Thursday to Friday (by about $1/\sqrt{3}$, because Friday's option spans three calendar days but only one trading day).

Pitfalls & conventions

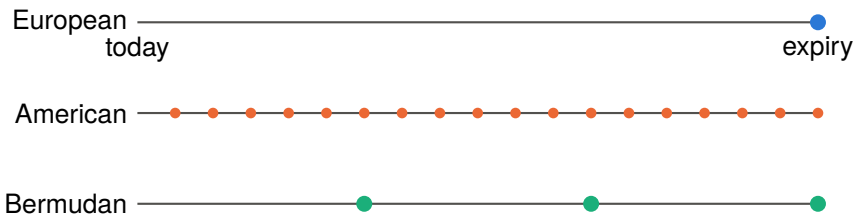
Keep the day count of your rate and of your option time consistent. A classic bug is to compute τ with ACT/365 and then plug in an ACT/360 money-market rate without converting it. The error looks tiny ($365/360 - 1 = 1.4\%$ of the rate) but it is systematic, and it shows up in every put–call parity check.

7.2 Contract specifics: NSE NIFTY options (2026)

NSE states that "Nifty 50 index options monthly contracts expire on the last Tuesday of the expiry month and weekly contracts expire on every Tuesday of the expiry week." If that Tuesday is a holiday, the contract expires on the previous trading day. Final settlement uses "the close price of the relevant underlying index in the Exchange on the last trading day", and money moves on T+1. The cash market closes at 15:30 IST. If you price near expiry, that time of day matters: a weekly option at 09:15 on expiry day still has about six hours of life, $\tau \approx 6/(24 \times 365)$, and ignoring it mis-states theta badly.

8 European, American and Bermudan exercise

- **European:** exercise only at expiry.
- **American:** exercise at any time up to and including expiry.
- **Bermudan:** exercise on a set of specific dates (common in rates products), which sits between the two.



This distinction drives your choice of model. A European option has a clean closed form (Black–Scholes–Merton). An American option needs a method that can ask, at every point in time and price, "*is it better to exercise now or to keep holding?*" Binomial trees (CRR, Leisen–Reimer) answer that question naturally at every node, which is why your project maps American options to trees.

8.1 Three facts you should be able to prove

1. **American $\geq$ European.** The American option includes every right the European one has, plus more. Extra rights can't have negative value.
2. **Never exercise an American call on a non-dividend-paying stock early** (Merton, 1973). If you exercise at time t you receive $S_t - K$. If instead you hold, the call is worth at least its lower bound $S_t - Ke^{-r\tau}$ (§34), which is strictly bigger than $S_t - K$ when $r > 0$. Intuitively, exercising early means paying K *sooner* (losing interest on it) and throwing away the insurance that the stock might fall below K. So the American call equals the European call, and our tree confirms it: 10.4506 for both (§56).
3. **American puts *can* be worth exercising early.** If a put is deep in the money, say $S = 20$ and $K = 100$, exercising gives you ₹80 now, which you can put in the bank. Waiting can gain you at most another ₹20 (the stock can't go below zero), while the interest on ₹80 is certain. At some point the certain interest beats the uncertain upside. For the running example the American put is worth 6.0903 versus 5.5735 European: the early-exercise premium is about ₹0.52.

NIFTY options are European. NSE Clearing: "For index options contracts and options contracts on individual securities, exercise style is European style. Final Exercise is Automatic on expiry." (One inconsistency in the wild: NSE's own NIFTY product page has described the options as "European style and physical settled". An index cannot be physically delivered, and NSE Clearing's settlement pages say cash. Trust the clearing house.)

9 Exotic options: an overview

A **vanilla** option's payoff depends only on the price at expiry, S_T. An **exotic** option adds conditions: on the path, on several assets, or on events.

- **Asian:** the payoff uses an *average* price instead of S_T:

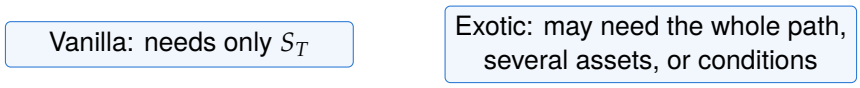
$$\bar{S} = \frac{1}{n}\sum_{i=1}^{n} S_{t_i}, \qquad \text{Asian call payoff} = \max(\bar{S} - K, 0).$$

  Averaging dampens volatility, so Asians are cheaper than vanillas. Corporates like them because their own costs are averages (an airline buys fuel every day, not once a year).

- **Barrier:** the option switches on (knock-in) or off (knock-out) if the price touches a level H. Example: "this call exists only if NIFTY never trades above 25,000." Barriers are cheaper than vanillas, but their Greeks behave wildly near the barrier.

- **Lookback:** the payoff uses the maximum or minimum price over the life, e.g. $S_T - \min_t S_t$: "buy at the lowest price, with hindsight". That perfect hindsight makes lookbacks expensive.

- **Basket:** the payoff is on a weighted sum of several assets, $B = \sum_i w_i S_i$. Correlation between the assets becomes a key input.



Your project assigns Monte Carlo to Asian, lookback and basket options and finite-difference PDEs to barriers. Part X gives each one its own section, with closed-form benchmarks to validate the numerical engines.

Check yourself

- Convert 45 calendar days into T under ACT/365 and ACT/360. Which is larger, and why?
- Prove that an American call on a non-dividend stock should not be exercised early. What changes if the stock pays a large dividend tomorrow?
- Why is an Asian option cheaper than the equivalent vanilla?

PART II

Payoffs, Positions and Moneyness

10 What is a payoff?

A payoff answers one question: *how much money does this contract produce at expiry?* It is a function of the underlying's price (and, for exotics, its path). It is **not** your profit, because it ignores what you paid to get into the trade. Keeping payoff and profit separate in your head, and in your code, avoids a whole class of mistakes.

Payoff functions are the most important objects in an option-pricing library. In a well-designed engine, an *instrument* is little more than a payoff function plus an exercise rule, and every pricing method (closed form, tree, Monte Carlo, PDE) consumes that same payoff. Write payoffs once, test them hard, and reuse them everywhere.

11 Long stock

Buy a stock at $S_0 = 100$. At the horizon:

- if $S_T = 150$, profit $= 150 - 100 = +50$;
- if $S_T = 70$, profit $= 70 - 100 = -30$.

So P&L $= S_T - S_0$. The relationship is **linear**: every ₹1 move in the stock is ₹1 of P&L, in both directions, without limit on the upside and with a loss of up to 100 on the downside.

12 Call payoff

$$\boxed{C_T = \max(S_T - K, 0) = (S_T - K)^+}$$

S_T	$S_T - K$	Call payoff ($K = 100$)
70	-30	0
90	-10	0
100	0	0
110	10	10
120	20	20
150	50	50

Below the strike the payoff is flat at zero. Above it, it rises one-for-one. The shape is the famous "hockey stick".

13 Put payoff

$$\boxed{P_T = \max(K - S_T, 0) = (K - S_T)^+}$$

S_T	$K - S_T$	Put payoff ($K = 100$)
50	50	50
70	30	30
90	10	10
100	0	0
120	−20	0
150	−50	0

Note that the put payoff is capped at K (the stock cannot go below zero), while the call payoff is unlimited.

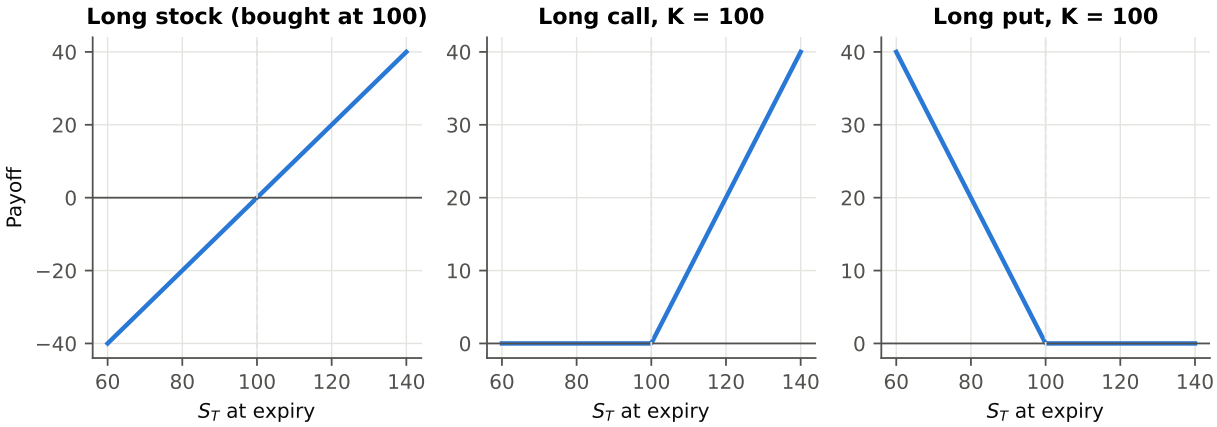


Figure 1. Payoff at expiry of a long stock, a long call and a long put. The options are "kinked" at the strike, and that kink is what you are paying for.

Intuition: why the kink matters The kink at K is *convexity*: the payoff curves upward. Before expiry, uncertainty about where S_T will land smooths that kink into a curve, the option price curve (Figure 3). The curvature of that smooth curve is **gamma**. The premium you pay for an option is, in a precise sense (§52), the price of owning that curvature.

14 Payoff versus P&L

Worked example: payoff is not profit

Strike 100, call premium ₹8, stock finishes at 130.

- Gross payoff: $130 - 100 = 30$.
- You paid ₹8 to buy the call, so the net P&L is $30 - 8 = 22$.

$$\boxed{\text{P\&L} = \text{Payoff} - \text{Premium}}$$

Two refinements turn this into something a trading desk would accept:

1. **Financing.** The premium is paid *today*, and the payoff arrives at T. Strictly, you should compare the payoff with the *future value* of the premium, premium $\times e^{rT}$. For an ₹8 premium at 5% over a year that is $8e^{0.05} = 8.41$. The 41 paise is the interest you gave up.
2. **Mark-to-market.** Before expiry, a desk's P&L is the daily change in the option's market value, not the payoff. Daily P&L is explained with the Greeks in §73.

Pitfalls & conventions

Beginners often plot *payoff* diagrams and label them "profit". Your dashboard should show both: the payoff, and the net P&L (payoff minus the future value of the premium). It should also show break-evens, maximum profit and maximum loss.

15 Break-even

The break-even is the expiry price at which the net P&L is zero.

Long call: $S_T - K - \text{premium} = 0 \Rightarrow \boxed{S_T^* = K + \text{premium}}$. With $K = 100$ and premium 8, break-even is 108.

Long put: $K - S_T - \text{premium} = 0 \Rightarrow \boxed{S_T^* = K - \text{premium}}$. With $K = 100$ and premium 8, break-even is 92.

With financing included, replace the premium by premium $\times e^{rT}$: the call break-even becomes 108.41. A long straddle (§74) has two break-evens, $K \pm$ total premium.

16 The four basic positions

Every option position is long or short a call or a put. Think of it in terms of **rights and obligations**: the buyer owns a right; the seller ("writer") has sold that right to someone else and so carries an obligation if assigned.

Position	You hold…	View	Max gain	Max loss
Long call	the right to buy	bullish	unlimited	premium
Short call	an obligation to sell if assigned	bearish/neutral	premium	unlimited
Long put	the right to sell	bearish	$K-$premium	premium
Short put	an obligation to buy if assigned	bullish/neutral	premium	$K-$premium

16.1 Margin for short options

A buyer's risk is capped at the premium already paid, so buyers usually post no further margin. A seller's potential loss can be large (unlimited, for a short call), so the clearing house demands **margin**: collateral sized to cover a bad day. Exchanges compute it with portfolio-based risk models. The classic is SPAN, which revalues the whole portfolio on a grid of spot and volatility shocks and charges the worst loss. India's clearing corporations have been moving towards VaR-based frameworks built on the same idea.

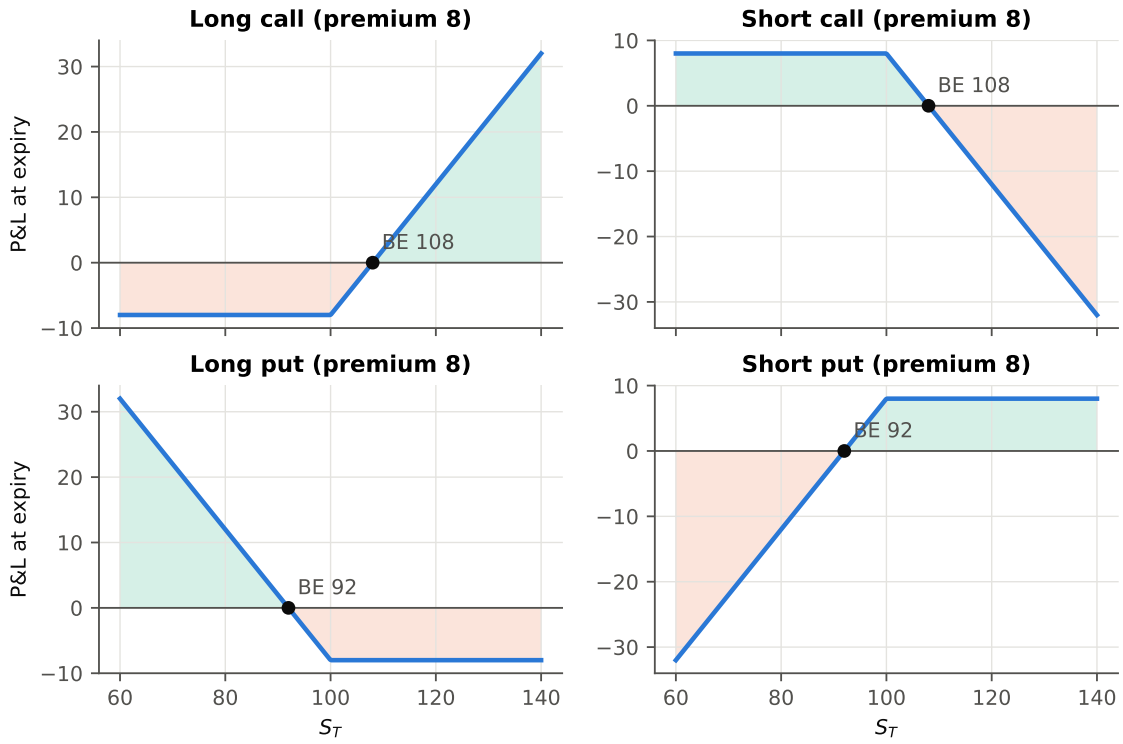


Figure 2. Expiry P&L of the four basic positions with $K = 100$ and premium 8. Green shading is profit, orange is loss. The short positions are mirror images of the long ones, because options are zero-sum.

> **Intuition** Margin is the clearing house asking: *"If tomorrow is a bad day, what could you lose?"* You will compute exactly that number yourself in the scenario-analysis section (§71).

17 Moneyness: ITM, ATM, OTM, and better measures

Moneyness describes where the strike sits relative to the underlying, which tells you whether exercising *right now* would pay.

	In the money (ITM)	At the money (ATM)	Out of the money (OTM)
Call	$S > K$	$S \approx K$	$S < K$
Put	$S < K$	$S \approx K$	$S > K$

Example: with $S = 100$ and $K = 110$, the call is OTM (you would not pay 110 for something worth 100) and the put is ITM (you could sell for 110 something worth 100).

Spot moneyness is the beginner's measure. Practitioners use more precise ones:

- **Forward moneyness:** compare K with $F = S_0 e^{(r-q)T}$. "ATMF" means $K = F$. That is the natural centre of the volatility smile, and by put–call parity it is the strike at which the call and put have *equal* prices.
- **Log-moneyness:** $k = \ln(K/F)$. Symmetric (a strike 10% above and 10% below give ± 0.095), and the coordinate used in SVI (§29). Dividing by $\sigma\sqrt{T}$ gives a "standardised" moneyness: how many standard deviations away the strike is.

- **Delta moneyness:** FX and volatility desks quote strikes by their delta, e.g. the "25-delta put". This automatically adjusts for volatility and maturity. A 25-delta put on a quiet one-week option and a 25-delta put on a volatile one-year option are equally "far away" in probability terms. FX smiles are quoted as ATM, 25Δ risk reversal and 25Δ butterfly.

Pitfalls & conventions

"ATM" is ambiguous. Always ask which one: ATM spot ($K = S$), ATM forward ($K = F$), or delta-neutral straddle (DNS, the strike where call delta + put delta = 0). For a one-year option at 5% rates they differ by several rupees of strike.

18 Intrinsic value

Intrinsic value is what you would get by exercising *immediately*:

$$\text{IV}_{\text{call}} = \max(S - K, 0), \qquad \text{IV}_{\text{put}} = \max(K - S, 0).$$

It is never negative, because you would never exercise into a loss.

19 Time value

$$\boxed{\text{Option price} = \text{Intrinsic value} + \text{Time value}}$$

Worked example: splitting a price

Stock at ₹120, call strike ₹100, market price ₹25. Intrinsic value $= 120 - 100 = 20$, so time value $= 25 - 20 = 5$. The ₹5 is what the market charges for the *possibility* that things improve further before expiry, while your downside stays limited. A second example: $K = 95$, $S = 100$, price 8 gives intrinsic 5 and time value 3.

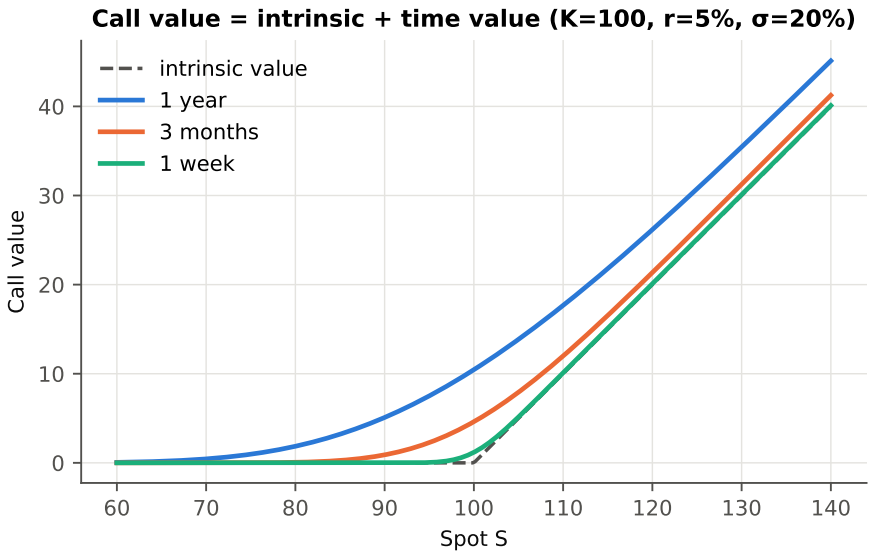


Figure 3. Call value against spot for three maturities. The gap between each curve and the dashed intrinsic line is the time value. It is largest at the money and shrinks as expiry approaches.

Three facts about time value:

1. **It is largest at the money**, where uncertainty about finishing ITM or OTM is greatest. Deep ITM options behave almost like the stock and deep OTM options are almost worthless, so in both cases there is little "optionality" left to pay for.
2. **It decays to zero at expiry.** That decay is **theta** (§50). The decay is not linear: for an ATM option, time value is roughly proportional to $\sqrt{\tau}$, so it melts faster and faster near the end (Figure 4).
3. **It can be negative for deep ITM European puts** (and for calls when dividends are large) if you measure intrinsic value against *spot*. Example: $S = 100$, $K = 150$, $T = 1$, $r = 5\%$, $\sigma = 20\%$. The European put is worth 43.04, below the spot intrinsic of 50. The reason: the ₹150 you would receive arrives only at expiry, so it is worth $150e^{-0.05} = 142.68$ today. Measure intrinsic against the forward, $\max(Ke^{-rT} - S, 0) = 42.68$, and the time value is positive again (0.36).

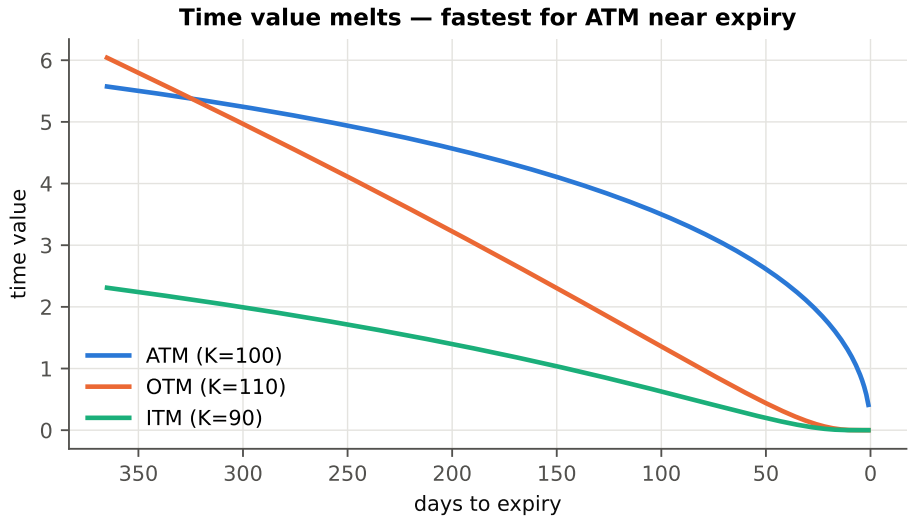


Figure 4. Time value (measured against the discounted strike) as expiry approaches, for ITM, ATM and OTM calls. ATM time value collapses fastest in the final weeks.

Check yourself

- A put has $K = 100$, $S = 93$ and trades at 9.5. Split the price into intrinsic and time value.
- Why is time value largest at the money?
- Why can a deep ITM European put trade below its spot intrinsic value, and why is that not an arbitrage?
- What strike is ATM-forward for $S = 100$, $r = 6\%$, $q = 1\%$, $T = 0.5$? (Answer: $100e^{0.025} = 102.53$.)

PART III

Forwards, Futures and Swaps

20 Forward contracts and cost-of-carry pricing

A **forward** is an agreement made today to buy or sell an asset at a future date T at a price F fixed today. Unlike an option, *both* sides are obliged: the buyer must buy and the seller must sell, whatever happens to the price. Entering a forward costs nothing upfront. The forward price F is chosen so that the contract is worth zero at inception.

Your notes' example is: "today the asset is ₹100; I agree to buy it in three months for ₹103." That is a perfectly good illustration of the *mechanics*. The next question a quant asks is: *is 103 the right number?* No-arbitrage answers it.

20.1 Deriving the forward price

> **Intuition** If you will own the asset in the future anyway, you could also buy it today and hold it. Buying today costs you interest on the money (you borrowed it, or you could have earned interest on it) but earns you any dividends. The forward price must make "buy now and carry" cost exactly the same as "agree now, buy later". Otherwise one of the two is free money.

Cash-and-carry argument (asset with continuous dividend yield q). Today:

1. Borrow $S_0 e^{-qT}$ at rate r.
2. Buy e^{-qT} units of the asset. Reinvest the dividends in more units, so that by T you hold exactly one unit.
3. Enter a short forward at price F (agree to sell one unit at T).

Net cost today: zero. At T: deliver your one unit, receive F, repay the loan $S_0 e^{-qT} e^{rT}$. You had no risk at all, so this final cash flow must be zero:

$$\boxed{F = S_0 e^{(r-q)T}}$$

If the market quoted F above this, you would run the strategy above and lock in a riskless profit. If F were below, you would run it in reverse: short the asset, lend the proceeds, go long the forward.

> **Worked example: forward prices**
>
> - $S_0 = 100$, $r = 5\%$, $q = 2\%$, $T = 1$: $F = 100e^{0.03} = 103.045$.
> - $S_0 = 100$, $r = 5\%$, $q = 0$, $T = 0.25$: $F = 100e^{0.0125} = 101.26$.

Correction to the original notes: the "₹103 in three months" example

Under cost-of-carry, $F = 103$ for three months implies $r - q = \ln(1.03)/0.25 = 11.8\%$ a year. As a description of what a forward *is*, the example is fine. Just do not combine it with the $r = 5\%$ used elsewhere in the notes: at 5% and no dividends, the fair three-month forward is ₹101.26.

20.2 Commodities: storage and convenience yield

For physical commodities, holding the asset has a storage cost u (warehousing, insurance) but may also provide a *convenience yield* y, the benefit of having the physical good on hand during a shortage. Both enter like dividends:

$$F = S_0 e^{(r+u-y)T}.$$

When convenience yield is high (a tight market), forwards can sit *below* spot. This is called *backwardation*. The opposite, forwards above spot, is *contango*.

20.3 Valuing a forward after inception

After inception the forward price moves, and an existing contract gains or loses value. A long forward struck at K_F is worth

$$f_t = (F_t - K_F)e^{-r(T-t)} = S_t e^{-q(T-t)} - K_F e^{-r(T-t)}.$$

This is the same formula as the left side of put–call parity (§35): a long call plus a short put with the same strike *is* a long forward.

21 Futures contracts

A **futures** contract is a standardised forward traded on an exchange.

Feature	Forward	Futures
Where traded	OTC (bilateral)	Exchange
Terms	Customised	Standardised (lot size, expiry dates)
Counterparty risk	On the other party	On the clearing house (CCP)
Liquidity	Generally lower	Generally higher
Settlement of gains/losses	Usually once, at maturity	Daily, by marking to market
Collateral	Negotiated (CSA)	Initial and maintenance margin

21.1 How futures actually work

- **Daily marking to market.** At the end of each day the exchange sets a settlement price, and your gain or loss for the day is credited or debited to your margin account in cash. The contract is effectively closed and reopened at the new price every day.
- **Initial and maintenance margin.** You post initial margin to open the position. If losses push your account below the maintenance level, you receive a *margin call* and must top it up, or your position is closed.

- **The clearing house as central counterparty.** The CCP *novates* every trade: it becomes the buyer to every seller and the seller to every buyer. Your counterparty risk is on the CCP, which is protected by everyone's margin and a default fund, not on an unknown trader.
- **Convergence.** As expiry approaches, the futures price converges to spot. If it didn't, you could buy the cheaper one and sell the dearer one at the final settlement.

> **Worked example: three days of marking to market**
>
> Long one NIFTY futures lot (65 units) at 25,000. Settlement prices on the next three days are 25,100, 24,800 and 24,900. Daily cash flows: $65 \times 100 = +6{,}500$, $65 \times (-300) = -19{,}500$, $65 \times 100 = +6{,}500$. The cumulative P&L is $-6{,}500$, the same as $65 \times (24{,}900 - 25{,}000)$. Unlike a forward, though, the cash moved every day. If initial margin were, say, 12% of notional ($0.12 \times 25{,}000 \times 65 \approx 1.95$ lakh), the day-2 loss would have eaten 10% of it.

NSE Clearing's theoretical futures price is $F = Se^{rt}$, with "r = rate of interest (MIBOR)". For an index, in practice you also subtract expected dividends.

21.2 Forward price versus futures price

- With **deterministic** interest rates, forward and futures prices are identical (Cox, Ingersoll and Ross, 1981).
- With **stochastic** rates that are *positively correlated* with the underlying, a long futures position gains cash exactly when rates are high (so the gains are reinvested at high rates), and loses when rates are low (so the losses are financed cheaply). That is an advantage, so the futures price exceeds the forward price.

The difference is called the **convexity adjustment**. It is material for long-dated interest-rate futures and negligible for short-dated equity index futures. For your project you can treat NIFTY futures and forwards as the same thing.

22 Swaps in one page

An **interest-rate swap** exchanges fixed payments K for floating payments L_i on a notional N at dates $T_1, \dots, T_n$. Each exchange is a forward on a rate, so

$$V_{\text{pay fixed}} = N \sum_{i=1}^{n} \alpha_i \, P(0, T_i) \, \big(F_i - K\big),$$

where α_i is the accrual fraction, $P(0, T_i)$ is the discount factor and F_i is the forward rate. The *par swap rate* is the K that makes $V = 0$:

$$K_{\text{par}} = \frac{\sum_i \alpha_i P(0,T_i) F_i}{\sum_i \alpha_i P(0,T_i)} = \frac{\text{PV of floating leg}}{\text{annuity}}.$$

The denominator, the **annuity** $A = \sum_i \alpha_i P(0, T_i)$, reappears when we price swaptions with Black-76 (§59).

Check yourself

- Prove $F = S_0 e^{(r-q)T}$ by writing out the cash-and-carry trade and its cash flows at 0 and T.
- What arbitrage would you run if NIFTY futures traded well *below* $S_0 e^{(r-q)T}$?
- Why does daily settlement create a futures/forward price difference *only* when interest rates are random and correlated with the asset?
- Show that long call + short put (same K, T) is a long forward struck at K.

PART IV

Market Inputs, Rates, Volatility and Dividends

23 The market inputs a pricing engine needs

A real pricing engine is only as good as its inputs. Your project specification lists ten; here is what each one is, where it comes from, and how it goes wrong.

Input	Where it comes from	Pitfalls
Underlying price S	Exchange last trade or mid-quote; for indices, the computed index value	Stale prints; mismatched timestamps between option and underlying quotes. Often better: use the futures-implied forward rather than spot plus a guessed dividend.
Strike K, expiry T	Contract specifications	Expiry *time of day* (NSE closes 15:30 IST); holidays move expiry; day-count (§7).
Risk-free rate r	India: overnight MIBOR, T-bill yields, OIS. US: SOFR / SOFR OIS	Compounding and day-count mismatches; using a tenor that doesn't match the option.
Dividend yield q	Forecasts, or implied from put–call parity or futures	Discrete vs continuous (§30).
Implied volatility	Solved from the market price (§54)	Bid vs ask vs mid; European vs American model; wings with no real liquidity.
Market option price	Mid of best bid and ask	Wide spreads in OTM wings; the last trade can be hours old.
Bid-ask spread	Order book	Tells you how much model precision is *meaningful*: a model error smaller than half the spread is invisible to the market.
Volume	Exchange	Activity today; spikes around events.
Open interest (OI)	Exchange	Number of outstanding contracts: a measure of positioning and liquidity, not of direction (every long has a short).

24 The risk-free rate and discounting

24.1 Why money has a time value

Worked example: discounting

At $r = 5\%$ a year, what is ₹100 received in one year worth today?

- Annual compounding: $PV = 100/1.05 = 95.24$. Check: $95.24 \times 1.05 = 100$.
- Continuous compounding: $PV = 100e^{-0.05} = 95.12$.

Growing 100 for a year at 5%: simple 105.00, annual 105.00, monthly $(1 + 0.05/12)^{12} \times 100 = 105.116$, continuous 105.127.

The general conventions for growing one unit at rate R over T years are:

$$\text{simple: } 1 + RT, \qquad m \text{ times a year: } \left(1 + \frac{R}{m}\right)^{mT}, \qquad \text{continuous: } e^{RT}.$$

- **Simple** interest is used for money-market instruments under a year: T-bills, MIBOR deposits.
- **Discrete compounding** is used for bonds (semi-annual coupons, $m = 2$).
- **Continuous** compounding is used inside models.

Converting to continuous: $r_c = m \ln(1 + R_m/m)$. For a 91-day T-bill at simple yield y:

$$r_c = \frac{\ln(1 + y \cdot 91/365)}{91/365}.$$

A 91-day yield of 5.52% converts to $r_c = 5.48\%$.

Why continuous compounding in models? Because e^{-rT} composes cleanly ($e^{-rt_1}e^{-rt_2} = e^{-r(t_1+t_2)}$), differentiates nicely, and is the natural language of continuous-time stochastic calculus. Convert market quotes to continuous at the boundary of your code, and never mix conventions inside it.

Discount factor. $P(0, T) = e^{-rT}$ is the price today of ₹1 paid at T. On real desks there is no single "risk-free rate". Collateralised derivatives are discounted on the overnight-index-swap (OIS) curve, because collateral earns the overnight rate.

24.2 Indian rate proxies

- **Overnight MIBOR** is administered by FBIL (Financial Benchmarks India Pvt Ltd). It is computed from actual interbank call-money trades on CCIL's NDS-Call platform in the first hour of trading. The methodology uses "all trades executed on NDS-Call system excluding Reciprocal and Reported Deals within the first hour of trading (currently at 9.00AM to 10.00AM)... Only T+0 Settlement deals are to be picked up". FBIL publishes it by about 10:45. NSE specifies MIBOR as the rate in its contract base-price calculations.
- **Policy anchor.** The RBI MPC's August 3–5, 2026 resolution kept the policy repo rate unchanged at 5.25%, with the SDF at 5.00% and the MSF at 5.50%. The next meeting was scheduled for October 5–7, 2026, so check for changes.
- **T-bills.** A CEIC series sourced from RBI shows a 91-day cut-off yield of 5.52% on 30 Sep 2026 (5.26% on 2 Sep 2026). Verify against RBI's own auction release before relying on it.

- **Practical choice.** For a NIFTY option under three months, use a T-bill or OIS yield of matching tenor, converted to continuous compounding. Better still, *imply* $r - q$ directly from NIFTY futures or from put–call parity on liquid strikes. That way you never need to guess dividends.

24.3 US convention: SOFR

SOFR, administered by the Federal Reserve Bank of New York, is "a broad measure of the cost of borrowing cash overnight collateralized by Treasury securities". It is a volume-weighted median of tri-party, GCF and bilateral repo trades, published around 8:00 a.m. ET, and has been published since April 2018. Because it is an overnight rate, for option maturities you build a SOFR OIS curve and read off the rate for your tenor.

24.4 NIFTY contract size

NSE circular NSE/FAOP/70616 (3 October 2025) revised the NIFTY lot size from 75 to 65. The first weekly contract with the new size expired on 6 January 2026 and the first monthly on 27 January 2026. A contract's notional is index level $\times$ 65. SEBI's framework sets a minimum contract value of ₹15 lakh at introduction, and lot sizes are periodically revised to keep contracts near that value. Re-check before relying on them.

25 Volatility

Intuition Volatility measures *how much* a price moves, not *which way*. Two stocks can both end the month where they started. If one wandered gently and the other lurched up and down, the second is more volatile, and an option on it is worth more, because large moves make the option's upside bigger while its downside stays capped.

Worked example: two stocks

Stock A: $100 \to 101 \to 100 \to 102 \to 101$. Stock B: $100 \to 120 \to 90 \to 130 \to 80$.

- Daily log returns of A: $+1.00\%, -1.00\%, +1.98\%, -0.99\%$, standard deviation 1.49%, annualised $1.49\% \times \sqrt{252} \approx 23.6\%$.
- Daily log returns of B: $+18.2\%, -28.8\%, +36.8\%, -48.6\%$, standard deviation 39.8%, annualised about 631%.

B is roughly 27 times more volatile. No real stock behaves like B: it is an extreme illustration.

Definition. Volatility σ is the **annualised standard deviation of log returns**. A 20% vol means one-standard-deviation annual moves of about $\pm 20\%$ in log terms, or about $20\%/\sqrt{252} \approx 1.26\%$ a day.

Why the square root of time? If daily returns are independent with variance σ_d^2, the variance of a sum of n of them is $n\sigma_d^2$ (variances of independent variables add). So the standard deviation grows like $\sqrt{n}$, and $\sigma_{\text{annual}} = \sigma_{\text{daily}}\sqrt{252}$. This rule assumes independent, identically distributed returns. Autocorrelation (trending or mean-reverting markets) and volatility clustering break it, which is why risk managers are cautious about scaling a one-day VaR to ten days with $\sqrt{10}$.

The kinds of volatility you will meet:

- **Realised / historical:** measured from past prices. Backward-looking.
- **Implied:** backed out of option prices. Forward-looking, and it carries a risk premium.
- **Local:** a function $\sigma(S,t)$ chosen to fit today's whole smile exactly (Dupire).
- **Stochastic:** volatility as its own random process (Heston, SABR; §41).
- **Volatility risk premium:** implied minus subsequently realised. It is usually positive for indices, because option sellers demand compensation for crash risk.

26 Historical volatility estimators

The basic recipe is: historical prices → returns → standard deviation → annualise.

Simple vs log returns. The simple return is $R_t = (S_t - S_{t-1})/S_{t-1}$. The log return is $r_t = \ln(S_t/S_{t-1})$. We use log returns because they *add* over time ($\ln(S_2/S_0) = r_1 + r_2$), they are symmetric (a fall from 100 to 50 and a rise from 50 to 100 have equal size, ∓ 0.693), and they are the quantity that GBM makes normally distributed (§39). For small moves the two are nearly equal.

26.1 Close-to-close, step by step

$$\hat{\sigma}^2_{\text{daily}} = \frac{1}{n-1}\sum_{i=1}^{n}(r_i - \bar{r})^2, \qquad \hat{\sigma}_{\text{annual}} = \hat{\sigma}_{\text{daily}}\sqrt{252}.$$

Worked example: six closing prices

Closes: 100, 101.5, 100.8, 102.3, 101.1, 103.0.

1. Log returns: 0.01489, −0.00692, 0.01477, −0.01180, 0.01862.
2. Mean $\bar{r} = 0.005912$; sum of squared deviations $= 0.000799$.
3. Sample variance $= 0.000799/4 = 0.0001997$; daily sd $= 1.413\%$.
4. Annualise: $1.413\% \times \sqrt{252} = 1.413\% \times 15.875 = \mathbf{22.4}\%$.

With only five returns this estimate is very noisy. Its own standard error is roughly $\hat{\sigma}/\sqrt{2n} \approx 7$ vol points. Real estimates use 20 to 250 days.

For short windows, many practitioners set $\bar{r} = 0$: the mean is pure noise over a few weeks, and subtracting it removes a degree of freedom for nothing.

26.2 Range-based estimators

Closing prices throw away information. A day that ranged from 98 to 104 and closed unchanged was volatile, but close-to-close sees zero. Estimators that use open, high, low and close (OHLC) are far more statistically efficient:

- **Parkinson (1980):** $\hat{\sigma}^2 = \dfrac{1}{4\ln 2}\,\overline{(\ln H/L)^2}$. It uses the daily range, assumes zero drift and no overnight gaps, and Parkinson reported it as roughly 2.5 to 5 times as efficient as close-to-close. It *underestimates* vol when markets gap at the open.

- **Garman–Klass (1980):** $\hat{\sigma}^2 = \overline{\frac{1}{2}(\ln H/L)^2 - (2\ln 2 - 1)(\ln C/O)^2}$. Reported efficiency 7.4. Still assumes zero drift and no gaps.
- **Rogers–Satchell (1991):** $\hat{\sigma}^2 = \overline{\ln \frac{H}{C} \ln \frac{H}{O} + \ln \frac{L}{C} \ln \frac{L}{O}}$. Allows for drift.
- **Yang–Zhang (2000):** $\hat{\sigma}^2_{YZ} = \hat{\sigma}^2_{\text{overnight}} + k\,\hat{\sigma}^2_{\text{open-to-close}} + (1-k)\hat{\sigma}^2_{RS}$, with k chosen to minimise variance. It handles both drift and opening gaps, and reviews describe it as up to 14 times more efficient than close-to-close. NIFTY often gaps overnight on global news, so use Yang–Zhang if you have OHLC data.

Worked example: Parkinson vs Garman–Klass

Five days with highs 101.8, 102.0, 101.6, 102.8, 103.4, lows 99.6, 100.4, 100.1, 100.9, 101.0, opens 100, 101.2, 101.0, 101.9, 101.5 and the closes above. Parkinson gives 18.3% and Garman–Klass 18.3%, compared with 22.4% close-to-close. With so few days, disagreement of a few points is normal. The range estimators are the steadier ones.

26.3 EWMA and GARCH: volatility that changes over time

EWMA (RiskMetrics): $\sigma_t^2 = \lambda\sigma_{t-1}^2 + (1-\lambda)r_{t-1}^2$, with $\lambda = 0.94$ for daily data. Yesterday's squared return gets weight 6%, and older ones decay geometrically. Example: if $\sigma_{t-1} = 1\%$ a day and yesterday's return was 2%, then $\sigma_t = \sqrt{0.94(0.01)^2 + 0.06(0.02)^2} = 1.086\%$. EWMA reacts quickly but has no long-run level to return to.

GARCH(1,1): $\sigma_t^2 = \omega + \alpha r_{t-1}^2 + \beta\sigma_{t-1}^2$. The long-run variance is $\omega/(1-\alpha-\beta)$, and forecasts mean-revert towards it at a speed set by $\alpha+\beta$. Fit it by maximum likelihood; in Python, the `arch` package does it in a few lines. Typical equity-index estimates have $\alpha \approx 0.08$ and $\beta \approx 0.90$: volatility is highly persistent.

Pitfalls & conventions

- Window length trades noise against staleness: 20 days reacts but jumps around, 250 days is smooth but slow.
- Treat holidays and weekends consistently (trading-day returns, $\sqrt{252}$).
- Intraday realised vol needs care with microstructure noise (bid-ask bounce). Sampling every 5 minutes is a common compromise.

27 Implied volatility

Historical volatility goes *forward* from prices to a number: past prices → volatility. Implied volatility goes *backwards*: market option price → pricing model → the σ that reproduces it. You know S, K, T, r and q. Volatility is the one input you cannot observe, so you solve

$$\boxed{\text{ModelPrice}(\sigma) = \text{MarketPrice}}$$

for σ. That is a root-finding problem (§54). For the running example, a call trading at ₹11 implies $\sigma = 21.46\%$; at ₹10, 18.80%; at ₹12, 24.11%.

Intuition Implied volatility is best thought of as a *quoting convention*, not a forecast. Emanuel Derman's analogy: it is like quoting houses in price per square foot. Nobody thinks a house *is* its price per square foot, but the unit makes very different houses comparable. Implied vol does the same for options with different strikes, maturities and underlyings. Traders say "I paid 18 vol" rather than "I paid ₹8.42".

28 The volatility smile and skew

If Black–Scholes were literally true, every option on the same underlying would have the *same* implied volatility, whatever its strike. Real markets disagree. Plot implied vol against strike and you get a curve: a **smile** if both wings are raised, a **skew** (or smirk) if it slopes mostly one way.

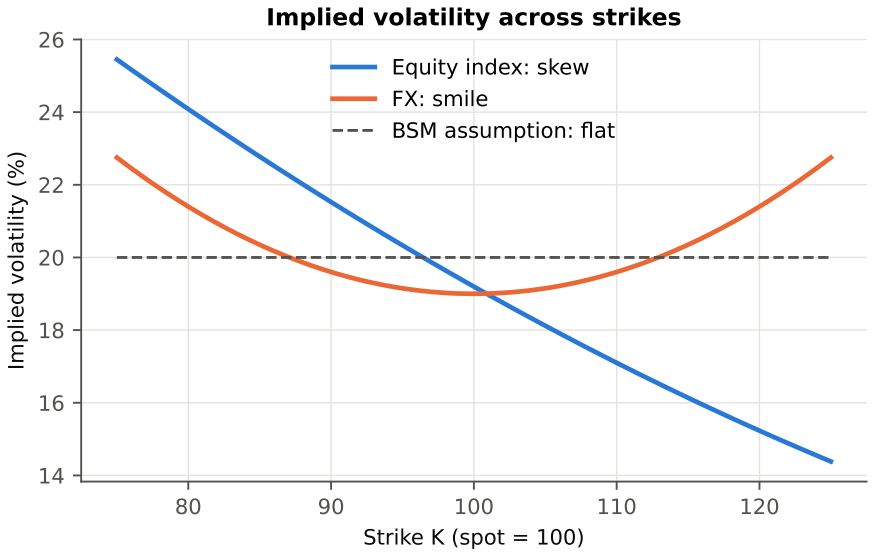


Figure 5. Equity index options show a downward skew (low-strike puts are expensive). FX options show a more symmetric smile. Black–Scholes assumes a flat line. The equity curve here is the one implied by a Heston model with $\rho = -0.7$ (§41).

28.1 A little history

Derman: "Prior to the stock market crash of October 1987, the volatility surface of index options was indeed fairly flat. Since the crash, the volatility surface of index options has become skewed." The shock was enormous. On 19 October 1987 the two-month S&P 500 futures fell 29%. Under a lognormal model with 20% vol, that is a move of more than 20 standard deviations: something that should essentially never happen in the lifetime of the universe. Rubinstein called the persistent fear that followed "crash-o-phobia". Later work found crash fears were updated upward again after the 2000 dot-com crash and the 2008 crisis.

28.2 Why the smile exists

1. **Fat tails and jumps.** Real returns have more extreme moves (excess kurtosis) than a normal distribution allows. A lognormal model underprices both tails, so the market quotes higher implied vols in the wings.

2. **The leverage effect.** When a company's stock falls, its debt-to-equity ratio rises, and the equity becomes riskier. So volatility tends to *rise as prices fall*: spot and vol are negatively correlated. That lifts low-strike implied vols and produces a downward skew.
3. **Crash-o-phobia and risk premia.** Investors pay up for crash insurance (OTM index puts), and the dealers who are short those puts demand compensation for carrying crash risk.
4. **Supply and demand.** Structural put buying by funds and structural call selling (covered-call writing supplies upside calls) steepen the equity skew.

Different shapes in different markets. Equity indices skew *down* (crashes are down). FX smiles are more *symmetric*, because either currency can crash; Derman notes "there was always a bit of a smile in currency options markets". Commodities often skew *up*, because supply shocks spike prices (think oil or natural gas).

29 The volatility surface

Add maturity and the smile becomes a surface, $\sigma_{\text{imp}}(K, T)$: implied vol as a function of both strike and maturity.

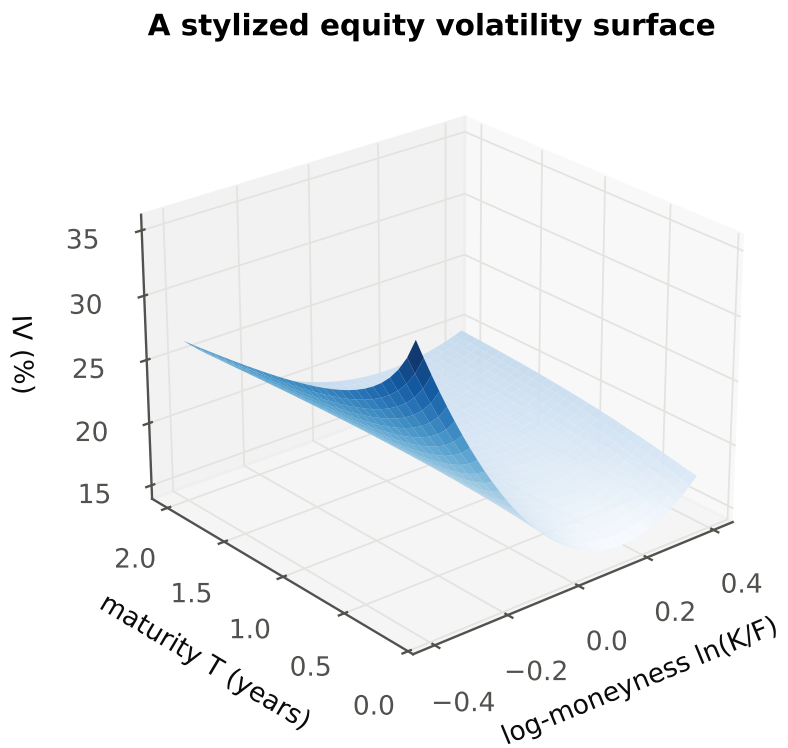


Figure 6. A stylised equity volatility surface: downward skew in strike, steepest at short maturities, and an upward-sloping ATM term structure.

- **Term structure.** In calm markets, equity ATM vol usually rises with maturity (short-dated options are cheap). In a crisis it inverts: near-term fear dominates.
- **Skew flattens with maturity.** Derman notes that skews are "steepest for small expirations as a function of strike, shallower for longer expirations". Over long horizons, the central limit theorem makes the return distribution look more normal.

29.1 Smile dynamics: sticky strike vs sticky delta

When spot moves, what happens to the smile?

- **Sticky strike:** each strike keeps its implied vol. The smile is fixed in K-space.
- **Sticky delta (sticky moneyness):** the smile moves with spot. Implied vol depends on K/S or delta.

This matters for hedging. Under sticky delta, your true delta differs from the Black–Scholes delta by vega $\times\, \partial\sigma/\partial S$. Hagan and co-authors invented SABR (§42) partly because local-vol models predicted the *wrong* direction for smile moves, and so gave bad hedges.

29.2 SVI: a parametric smile

Gatheral's SVI ("stochastic volatility inspired") models total implied variance $w(k) = \sigma_{\text{imp}}^2 T$ in log-moneyness $k = \ln(K/F)$:

$$w(k) = a + b\left\{\rho(k-m) + \sqrt{(k-m)^2 + s^2}\right\}.$$

Five parameters: a sets the overall level, b the slope of the wings, ρ the skew (left/right asymmetry), m shifts the smile sideways, and s sets the curvature at the money. The wings are *linear* in k, which matches Roger Lee's theoretical result that total variance cannot grow faster than linearly in the wings.

Worked example: an SVI slice

$T = 1$, $a = 0.02$, $b = 0.10$, $\rho = -0.5$, $m = 0$, $s = 0.15$:

$k = \ln(K/F)$	-0.3	-0.1	0	0.1	0.3
$w(k)$	0.0685	0.0430	0.0350	0.0330	0.0385
$\sigma_{\text{imp}} = \sqrt{w/T}$	26.2%	20.7%	18.7%	18.2%	19.6%

That is a typical equity skew: expensive downside, a minimum slightly above the forward, a gently rising upside wing.

29.3 No-arbitrage conditions on a surface

Gatheral and Jacquier: a surface "is free of static arbitrage if and only if (i) it is free of calendar spread arbitrage; (ii) each time slice is free of butterfly arbitrage."

- **Calendar condition:** total variance $w(k, T)$ must be non-decreasing in T for each k. Otherwise a longer option would be cheaper than a shorter one in variance terms, and you could buy the long-dated option, sell the short-dated one, and lock in a profit.
- **Butterfly condition:** call prices must be convex in K. Equivalently, the implied risk-neutral density $\partial^2 C/\partial K^2 \cdot e^{rT}$ must be non-negative. A negative density would let you buy a butterfly for a negative price.

Raw SVI can violate the butterfly condition; Axel Vogt's counterexample is the famous one. The SSVI ("surface SVI") family has explicit sufficient conditions: for instance, there is no butterfly arbitrage if $\theta\varphi(\theta) \le \min\left(4/(1+|\rho|),\, 2\sqrt{\theta/(1+|\rho|)}\right)$, where θ is the ATM total variance.

29.4 Volatility indices: VIX and India VIX

CBOE's VIX and NSE's India VIX are *model-free* 30-day implied volatility indices. They are built from a strip of OTM option prices across all strikes, not from Black–Scholes implied vols. NSE: "From the best bid-ask prices of NIFTY Options contracts, a volatility figure (%) is calculated which indicates the expected market volatility over the next 30 calendar days. India VIX uses the computation methodology of CBOE, with suitable amendments to adapt to the NIFTY options order book using cubic splines." It interpolates between near- and next-month expiries to reach exactly 30 days.

Rule of thumb: India VIX $= 15$ implies a one-standard-deviation one-month move of about $15\%/\sqrt{12} \approx 4.3\%$, or about $15\%/\sqrt{252} \approx 0.95\%$ a day.

30 Dividends

Correction to the original notes: the dividends section was too thin

The notes said only that dividends "change the economics of holding the stock". The precise statement: a dividend transfers value *out of* the share price to the shareholder. An option holder does not receive the dividend, but the stock price drops by roughly the dividend on the ex-date. So dividends *lower call values and raise put values*, and they can make early exercise of American calls optimal.

30.1 Continuous dividend yield

For a broad index, dividends trickle in from many companies, and a continuous yield q is a good model. Holding the stock earns q, so the forward is $F = S_0 e^{(r-q)T}$, and in Black–Scholes every S_0 is replaced by $S_0 e^{-qT}$ (Merton, 1973).

Worked example: effect of a 2% yield

Running example with $q = 2\%$: the forward falls from 105.13 to 103.05. The call falls from 10.4506 to 9.2270, and the put rises from 5.5735 to 6.3301.

30.2 Discrete cash dividends

A single stock pays lumpy cash dividends D on known ex-dates t_D. On the ex-date, the price drops by about D. Three ways to model this:

1. **Escrowed dividend model:** price with Black–Scholes on $S^* = S - \mathrm{PV}(D)$. Simple and widely used, but the volatility now applies to S^* rather than S, which biases prices for long maturities or large dividends.
2. **Trees and PDEs with a jump condition:** impose $V(S, t_D^-) = V(S - D, t_D^+)$ at the ex-date. This is the most accurate approach. In trees it needs interpolation or non-recombining branches.
3. **Implied dividends:** back them out of put–call parity or the futures price. This is what trading desks actually do.

30.3 Dividends and early exercise of American calls

Without dividends, never exercise an American call early. With a discrete dividend, exercising *just before* the ex-date captures D, but gives up the remaining time value and the interest on K for the remaining life. A standard sufficient condition for *not* exercising at the dividend date t_i is

$$D_i \le K\big(1 - e^{-r(t_{i+1}-t_i)}\big),$$

where t_{i+1} is the next dividend date or expiry. In words: if the dividend is smaller than the interest you save by paying K later, don't exercise. Example: $K = 90$, $r = 5\%$, with 3 months from the dividend to expiry, gives a threshold of $90(1 - e^{-0.0125}) = 1.12$. A dividend above that *may* make early exercise optimal; a tree will tell you.

Pitfalls & conventions

Treating a single large dividend as a continuous yield misprices short-dated options that straddle the ex-date. The yield spreads the drop over the whole year, while in reality it happens in one day.

Check yourself

- Convert a 91-day T-bill simple yield of 6% into a continuously compounded rate.
- Compute close-to-close vol (annualised) from five prices of your choice, by hand, then in code.
- Give three economic reasons for the equity skew.
- State the two static no-arbitrage conditions on a vol surface and the trade that exploits each violation.
- Why do dividends lower call values? When might they make early exercise of an American call optimal?

PART V

No-Arbitrage, Put–Call Parity and Risk-Neutral Pricing

31 What is arbitrage?

An **arbitrage** is a trading strategy that costs nothing (or less than nothing) today, can never lose money, and has a positive chance of making money. It is a free lottery ticket.

> **Worked example: the simplest arbitrage**
>
> Market A quotes a share at ₹100 and market B quotes the same share at ₹105. Buy in A, sell in B, simultaneously: you lock in ₹5 per share without bearing any price risk. In practice, transaction costs, settlement timing, short-selling constraints and execution risk eat into it.

In liquid markets such opportunities vanish quickly, because arbitrageurs' buying and selling pushes prices back into line. Derivatives pricing *assumes* they don't exist, and then asks what that assumption forces prices to be. The principle in one sentence:

Two things with identical future cash flows in every possible state of the world must have the same price today.
(The law of one price.)

32 Why quants care: no-arbitrage as unit tests

No-arbitrage gives you relationships that *any* correct model must satisfy, whatever its assumptions about volatility or jumps. That makes them perfect automated tests for a pricing engine. Your project lists them explicitly:

- put–call parity;
- no-arbitrage price bounds;
- American price $\geq$ European price;
- CRR and Leisen–Reimer trees converge to Black–Scholes;
- Monte Carlo converges, with error falling like $1/\sqrt{N}$;
- Greeks are stable under small perturbations;
- barrier options behave correctly at the limits (in + out = vanilla; a far-away barrier gives the vanilla).

These are not just theory. In Part XII each one becomes a line in your test suite.

33 The Fundamental Theorem of Asset Pricing

Two theorems make "no arbitrage" operational:

- **First FTAP** (Harrison–Kreps 1979; Harrison–Pliska 1981; the general version by Delbaen–Schachermayer 1994). There is no arbitrage if and only if there exists a probability measure $\mathbb{Q}$, equivalent to the real-world measure $\mathbb{P}$ (it agrees on what is possible and impossible), under which *discounted* asset prices are martingales: their expected future discounted value equals today's value.
- **Second FTAP.** The market is **complete** (every payoff can be replicated by trading) if and only if $\mathbb{Q}$ is *unique*.

Black–Scholes is a complete market: one stock, one bank account, one source of randomness. Every option can be replicated, so there is exactly one arbitrage-free price. Heston (two sources of randomness, but only one traded stock) and jump models are *incomplete*: many $\mathbb{Q}$'s are consistent with no-arbitrage. You choose one by **calibrating** to the market prices of liquid options (§77).

34 No-arbitrage bounds

For European options on a stock with dividend yield q:

$$\max\bigl(S_0e^{-qT} - Ke^{-rT}, 0\bigr) \;\le\; C \;\le\; S_0e^{-qT}, \qquad \max\bigl(Ke^{-rT} - S_0e^{-qT}, 0\bigr) \;\le\; P \;\le\; Ke^{-rT}.$$

Proof of the call lower bound. Compare (A) one call plus Ke^{-rT} in cash with (B) e^{-qT} shares, with dividends reinvested so that you hold one share at T. At T, A is worth $\max(S_T, K)$ and B is worth S_T. A is always worth at least B, so it must cost at least as much today: $C + Ke^{-rT} \ge S_0e^{-qT}$. And $C \ge 0$, because an option can never be a liability to its holder.

Proof of the call upper bound. A call can never pay more than one share is worth, so $C \le S_0e^{-qT}$.

Further shape constraints, which together say that call prices are *decreasing and convex in strike*:

$$-e^{-rT} \le \frac{\partial C}{\partial K} \le 0, \qquad \frac{\partial^2 C}{\partial K^2} \ge 0.$$

Convexity is the butterfly condition from §29: a butterfly has a non-negative payoff, so it cannot have a negative price.

> **Worked example: checking the running example**
>
> Lower bound $100 - 95.1229 = 4.877$, upper bound 100. The call at 10.4506 is comfortably inside. A deep ITM call with $K = 50$ must be worth at least $100 - 50e^{-0.05} = 52.44$; Black–Scholes gives 52.44 too, since it is almost exactly at its lower bound.

35 Put–call parity

For European options with the same strike and expiry, on a stock with no dividends:

$$\boxed{C - P = S_0 - Ke^{-rT}} \quad\Longleftrightarrow\quad \boxed{C + Ke^{-rT} = P + S_0}.$$

35.1 The proof by payoff table

Portfolio A: buy one call and invest Ke^{-rT} in a risk-free bond (it grows to K at T). **Portfolio B:** buy one put and one share.

Value at expiry	If $S_T \le K$	If $S_T > K$
A: call	0	$S_T - K$
A: bond	K	K
A total	**K**	$\mathbf{S_T}$
B: put	$K - S_T$	0
B: share	S_T	S_T
B total	**K**	$\mathbf{S_T}$

Both portfolios are worth $\max(S_T, K)$ in *every* state of the world. By the law of one price they must cost the same today: $C + Ke^{-rT} = P + S_0$. Notice that we never assumed anything about volatility, the distribution of returns, or investor preferences. Parity is *model-free*, which is exactly why it is such a powerful test.

Worked example: parity in numbers

From the notes: $S_0 = 100$, $K = 100$, $r = 5\%$, $T = 1$, and suppose $C = 10$. Then $P = C - S_0 + Ke^{-rT} = 10 - 100 + 95.123 = \mathbf{5.12}$.
With Black–Scholes prices: $C - P = 10.4506 - 5.5735 = 4.8771$ and $S_0 - Ke^{-rT} = 100 - 95.1229 = 4.8771$. They agree to machine precision.
If the market put were ₹15 with the call at 10, the put would be rich by about ₹9.88. Sell the put, buy the call, short the share and lend Ke^{-rT}. Before celebrating, though, check for dividends, borrow costs on the short stock, the right rate, stale quotes, and whether the options are American.

35.2 Variants

- **Continuous dividend yield:** $C - P = S_0 e^{-qT} - Ke^{-rT}$.
- **Discrete dividends:** $C - P = S_0 - \mathrm{PV}(D) - Ke^{-rT}$.
- **Futures options (Black-76):** $C - P = e^{-rT}(F - K)$.
- **American options** give only inequalities. With no dividends: $S_0 - K \le C_A - P_A \le S_0 - Ke^{-rT}$. With dividends: $S_0 - \mathrm{PV}(D) - K \le C_A - P_A \le S_0 - Ke^{-rT}$.

35.3 What parity is used for

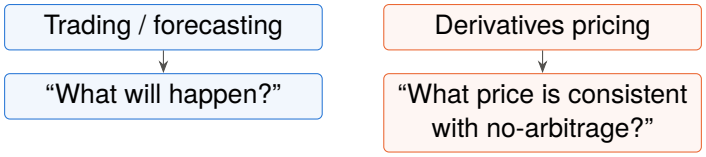
- **Implying the forward and dividends.** Regress $C(K) - P(K)$ across strikes on K. The slope is $-e^{-rT}$ (giving you the market's discount rate), and the intercept is $S_0 e^{-qT}$ (giving you the implied dividend). This is how desks get $r - q$ without guessing.
- **Detecting bad data.** A parity violation larger than the bid-ask spread usually means a stale quote, not free money.
- **Synthetics.** A long call plus a short put is a synthetic long forward. A long put plus a long share behaves like a long call. Traders use whichever version is cheapest.

36 Risk-neutral pricing

This is the most important conceptual idea in the whole subject, and the one that most surprises newcomers.

36.1 The beginner's instinct, and why it is wrong

A beginner thinks: *"To price an option I need to predict whether the stock will go up."* It feels obvious: a call should be worth more if the stock is likely to rise. Yet the Black–Scholes formula does not contain the stock's expected return at all. How can that be?



36.2 The one-step argument that makes it click

Worked example: pricing without probabilities

A stock is at 100. In one year it will be either 120 or 80. A call with $K = 100$ pays 20 or 0. The rate is 5% (continuous). You have no idea how likely "up" is.

Replicate it. Buy Δ shares and hold B in cash so that the portfolio matches the call in both states:

$$120\Delta + Be^{0.05} = 20, \qquad 80\Delta + Be^{0.05} = 0$$

$$\Rightarrow \quad \Delta = \frac{20-0}{120-80} = 0.5, \qquad B = e^{-0.05}(0 - 0.5 \times 80) = -38.049.$$

The portfolio costs $0.5 \times 100 - 38.049 = \mathbf{11.951}$. Since it pays exactly what the call pays in every state, the call *must* cost 11.951. Any other price is an arbitrage.

Now notice: we never used the probability of going up. Whether you think "up" is 90% likely or 10% likely, the price is 11.951, because you can manufacture the call from the stock and cash. Your view of the stock is already in the stock's price.

The same answer by expectation. Define $p = (e^{0.05} - 0.8)/(1.2 - 0.8) = 0.62818$. Then $e^{-0.05}[0.62818 \times 20 + 0.37182 \times 0] = \mathbf{11.951}$. Same number. This p is the *risk-neutral probability*: the probability that makes the stock earn exactly the risk-free rate on average ($0.628 \times 120 + 0.372 \times 80 = 105.13 = 100e^{0.05}$).

That is the whole idea, and it generalises. If a derivative can be hedged, its price is the cost of the hedge, and that cost can be computed as a discounted expectation under the probabilities that make every asset grow at the risk-free rate:

$$\boxed{V_0 = e^{-rT}\,\mathbb{E}^{\mathbb{Q}}[\text{Payoff}_T]}$$

36.3 $\mathbb{P}$ versus $\mathbb{Q}$

- $\mathbb{P}$ is the **real-world measure**. Under it the stock drifts at its expected return μ. You use $\mathbb{P}$ for forecasting, risk management, VaR and backtesting.

- $\mathbb{Q}$ is the **risk-neutral (pricing) measure**. Under it the stock drifts at $r - q$. You use $\mathbb{Q}$ to price.

Moving from $\mathbb{P}$ to $\mathbb{Q}$ (Girsanov's theorem) changes the *drift* but leaves *volatility* unchanged. That is why σ appears in the Black–Scholes formula but μ does not.

> **Intuition: risk-neutral does not mean investors are risk-neutral** Nobody claims investors are indifferent to risk. The claim is that, *because the option can be hedged*, risk preferences are already embedded in the stock price, and you can compute the option price *as if* everyone were risk-neutral and get the right answer. $\mathbb{Q}$ is a computational device: a set of "prices of states" rather than beliefs about them.

36.4 Numeraires, briefly

$\mathbb{Q}$ is the measure that uses the bank account as the unit of account (the "numeraire"). You can choose other numeraires. Use the stock as numeraire and you get the "stock measure"; under it, $N(d_1)$ is the probability that a call finishes in the money. Use a zero-coupon bond maturing at T and you get the T-forward measure, under which forward prices are martingales. That is the natural home of Black-76 (§59).

> **Pitfalls & conventions**
>
> - "$N(d_2)$ is the probability the option expires in the money." Only under $\mathbb{Q}$. The real-world probability uses μ instead of r and can be quite different.
> - Risk-neutral pricing is only as good as the hedging argument behind it. Where you cannot hedge (jumps, illiquid underlyings, gap risk), the price contains a risk premium that the model must take from the market via calibration.

> **Check yourself**
>
> - Prove put–call parity with a payoff table, without looking.
> - A call is quoted at 12, the put at 5, $S = 100$, $K = 100$, $r = 5\%$, $T = 1$, no dividends. Is there an arbitrage? Write down the trade and its cash flows.
> - In the one-step example, why does the real-world probability of "up" not affect the price?
> - What changes and what stays the same when moving from $\mathbb{P}$ to $\mathbb{Q}$?
> - State the first and second FTAP in one sentence each. Which applies to Heston, and what does it imply?

PART VI

Probability and Stochastic Processes

You do not need a measure-theory course to build a pricing engine, but you do need to be fluent in a handful of ideas. This part builds them in order: random variables → distributions → Brownian motion → geometric Brownian motion → Itô's lemma → richer models.

37 Probability foundations

37.1 Random variables

A **random variable** is an uncertain number: a rule that assigns a value to each possible outcome. Tomorrow's stock return R might be $+2\%$, -1%, $+5\%$, -3%, and so on. Before tomorrow we can't say which, but we can describe the *distribution*: which values are possible and how likely each is.

37.2 Expected value

The expectation is the probability-weighted average:

$$\mathbb{E}[X] = \sum_i p_i x_i \quad \text{(discrete)}, \qquad \mathbb{E}[X] = \int x\, f(x)\, \mathrm{d}x \quad \text{(continuous)}.$$

Worked example: a two-outcome return

Return $+10\%$ with probability 50%, -5% with probability 50%. Then $\mathbb{E}[R] = 0.5(10\%) + 0.5(-5\%) = \mathbf{2.5}\%$. The variance is $0.5(7.5\%)^2 + 0.5(7.5\%)^2 = 56.25\ (\%)^2$, so the standard deviation is 7.5%.

Expectation is *linear*: $\mathbb{E}[aX + bY] = a\mathbb{E}[X] + b\mathbb{E}[Y]$, always, even for dependent X and Y. That one property does a surprising amount of work in pricing (it is why a portfolio's value is the sum of its parts).

37.3 Variance and standard deviation

$$\mathrm{Var}(X) = \mathbb{E}\big[(X - \mathbb{E}X)^2\big] = \mathbb{E}[X^2] - (\mathbb{E}X)^2, \qquad \sigma_X = \sqrt{\mathrm{Var}(X)}.$$

Variance measures dispersion around the mean. Volatility is the annualised standard deviation of log returns.

37.4 Covariance and correlation

$$\mathrm{Cov}(X,Y) = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y], \qquad \rho_{XY} = \frac{\mathrm{Cov}(X,Y)}{\sigma_X \sigma_Y} \in [-1, 1].$$

$\rho = +1$ means the variables move together perfectly (linearly), $\rho \approx 0$ means little *linear* relationship, and $\rho = -1$ means they move perfectly opposite.

Worked example: two stocks

Daily returns of X: $1.0\%, -2.0\%, 1.5\%, 0.5\%, -1.0\%$. Daily returns of Y: $1.2\%, -1.5\%, 1.0\%, 0.2\%, -1.2\%$. Sample covariance $= 0.000175$; standard deviations 1.458% and 1.240%; correlation $= 0.000175/(0.01458 \times 0.01240) = \mathbf{0.968}$. These two stocks move almost in lock-step.

Pitfalls & conventions

Zero correlation does *not* mean independence. If X is standard normal and $Y = X^2$, then $\rho = 0$, yet Y is completely determined by X. Correlation only captures *linear* co-movement. That is one reason correlation-based risk models fail in crises, when assets that looked unrelated all fall together.

Variance of a sum, the formula behind every basket and portfolio:

$$\operatorname{Var}(aX + bY) = a^2\sigma_X^2 + b^2\sigma_Y^2 + 2ab\,\rho\,\sigma_X\sigma_Y.$$

Lower correlation gives lower basket variance. That is diversification, and it is why basket options get cheaper as correlation falls (§68).

37.5 The normal distribution

$X \sim \mathcal{N}(\mu, \sigma^2)$ has density $f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-(x-\mu)^2/(2\sigma^2)}$. It is symmetric and thin-tailed: 68% of outcomes fall within one standard deviation, 95% within 1.96, and 99.7% within three. We write $N(x)$ for the standard normal CDF, $\mathbb{P}(Z \le x)$, and $n(x)$ for its density. Every Black–Scholes formula is built from these two functions.

37.6 The lognormal distribution

If $\ln X \sim \mathcal{N}(m, s^2)$, then X is **lognormal**. It is always positive and skewed to the right, and

$$\mathbb{E}[X] = e^{m+s^2/2}, \qquad \text{median}(X) = e^m.$$

The mean exceeds the median because the right tail is long: a few very large outcomes drag the average up. This $+s^2/2$ is the seed of the famous $-\frac{1}{2}\sigma^2$ in GBM (§39).

37.7 The central limit theorem

Add up many independent shocks with finite variance and the sum tends to a normal distribution, whatever the shocks' own distribution. Two consequences for us:

1. It justifies modelling the log price as the sum of many small independent shocks, which is the motivation for Brownian motion.
2. It is why Monte Carlo error falls like $1/\sqrt{N}$: an average of N independent payoffs is approximately normal with standard deviation $s/\sqrt{N}$ (§60).

37.8 Calculus you will need

- **Logarithms and exponentials:** $\ln(ab) = \ln a + \ln b$, $e^{a+b} = e^a e^b$, $\frac{d}{dx}\ln x = 1/x$.
- **Partial derivatives:** the Greeks are partial derivatives of $V(S, \sigma, r, t)$, each holding the other inputs fixed.
- **Chain rule:** needed to differentiate $N(d_1(S))$ with respect to S.
- **Taylor expansion:** $f(x+h) \approx f(x) + f'(x)h + \frac{1}{2}f''(x)h^2$. This one line is the basis of delta-gamma risk (§72), finite differences (§70) and, crucially, Itô's lemma (§40).

38 Brownian motion

> **Intuition** Prices move unpredictably and (mostly) continuously. We want a mathematical object that is a continuous path built from infinitely many tiny, independent, random kicks. That object is Brownian motion, named after the botanist who watched pollen jiggle in water. Physicists call it a Wiener process.

Definition. A standard Brownian motion W_t satisfies:

1. $W_0 = 0$;
2. **independent increments:** what happens over $[s, t]$ is independent of what happened before s;
3. **normal increments:** $W_t - W_s \sim \mathcal{N}(0,\, t-s)$, so $\mathrm{Var}(W_t) = t$;
4. **continuous paths.**

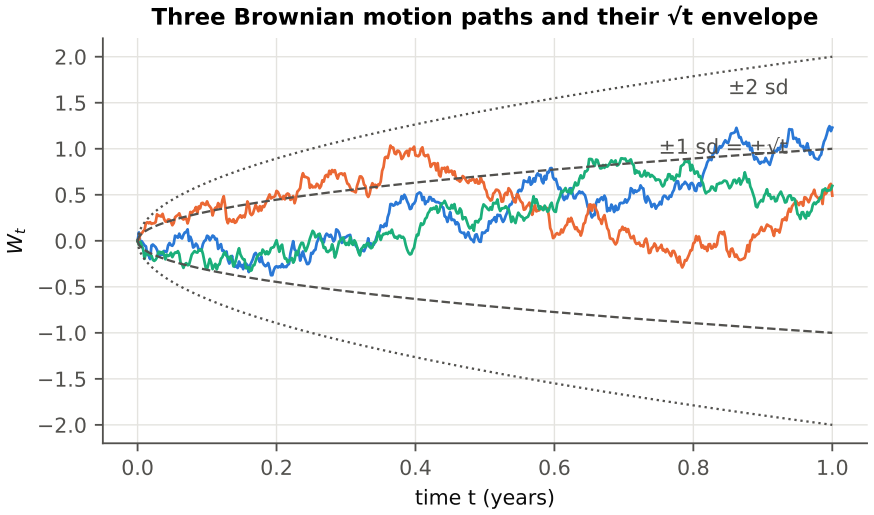


Figure 7. Three Brownian paths. The typical distance from zero grows like $\sqrt{t}$, not like t.

38.1 Four properties that matter

- **Scaling.** W_{ct} has the same distribution as $\sqrt{c}\, W_t$. Moves grow like the square root of time, which is the origin of every $\sqrt{T}$ in option pricing.

- **Martingale.** $\mathbb{E}[W_t \mid \text{information at } s] = W_s$: the best forecast of the future value is today's value. There is no drift to exploit.
- **Quadratic variation.** Chop $[0, T]$ into n pieces and add up the *squared* increments: $\sum_i (W_{t_{i+1}} - W_{t_i})^2 \to T$ as $n \to \infty$. This is not an average; it is a near-certainty for fine grids. Informally, $(dW)^2 = dt$.
- **Nowhere differentiable.** Increments over dt are of size $\sqrt{dt}$, so the slope $dW/dt \sim 1/\sqrt{dt} \to \infty$. Paths are continuous but infinitely jagged at every scale, so ordinary calculus (which needs slopes) breaks down.

Worked example: watching quadratic variation appear

Simulating one path on $[0, 1]$ with n steps:

n	$\sum(\Delta W)^2$	$\sum \lvert\Delta W\rvert$
10	0.557	1.94
100	0.986	8.23
1,000	0.974	24.6
100,000	1.0001	252.4

The sum of squares settles at $T = 1$, while the sum of absolute values (the path's "length") explodes like $\sqrt{n}$. A Brownian path has infinite length but finite quadratic variation. That one fact is why Itô calculus has an extra term.

39 Geometric Brownian motion

39.1 Why not model prices as Brownian motion directly?

If $S_t = S_0 + \sigma W_t$ (arithmetic Brownian motion), the price can go negative: a stock at ₹100 could reach -10. It also implies a ₹1 move is equally likely at a price of 10 or 1,000, whereas real prices move in *percentage* terms. So we model proportional changes instead.

39.2 The model

$$\boxed{dS_t = \mu S_t\,\mathrm{d}t + \sigma S_t\,\mathrm{d}W_t}$$

$$dS = \underbrace{\mu S\,\mathrm{d}t}_{\text{deterministic drift}} + \underbrace{\sigma S\,\mathrm{d}W}_{\text{random shock}}.$$

Here S_t is the price, μ the expected rate of return (the drift), σ the volatility, dt a tiny time step and dW a Brownian increment with mean 0 and variance dt. In words: over a short interval, the percentage change in price is a small predictable drift plus a normally distributed shock whose size is proportional to $\sigma\sqrt{dt}$.

39.3 The solution and the $-\frac{1}{2}\sigma^2$ correction

Using Itô's lemma (next section) on $\ln S$:

$$\boxed{S_T = S_0 \exp\left[\left(\mu - \tfrac{1}{2}\sigma^2\right)T + \sigma W_T\right]}, \qquad \ln S_T \sim \mathcal{N}\Big(\ln S_0 + (\mu - \tfrac{1}{2}\sigma^2)T,\ \sigma^2 T\Big).$$

So S_T is **lognormal**, always positive, and right-skewed.

Intuition: where does $-\frac{1}{2}\sigma^2$ come from? Volatility drags on compounding. Go up 10% then down 10%: $1.1 \times 0.9 = 0.99$, a loss, even though the average percentage move was zero. The *arithmetic* expected return is μ, but the expected *log* (compound) return is $\mu - \frac{1}{2}\sigma^2$. Both are true at once: the mean outcome grows at μ, the median (typical) outcome grows at $\mu - \frac{1}{2}\sigma^2$.

Worked example: mean versus median

$\mu = 10\%$, $\sigma = 20\%$, $S_0 = 100$, $T = 1$. Mean $\mathbb{E}[S_1] = 100e^{0.10} = 110.52$. Median $= 100e^{0.10-0.02} = 108.33$. The gap is the right tail at work.

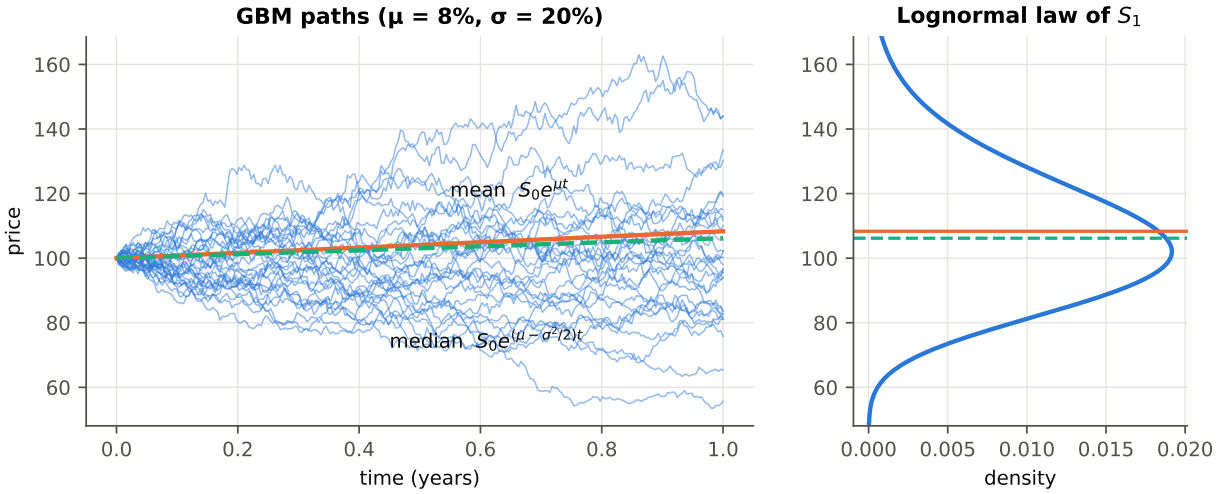


Figure 8. Thirty GBM paths with $\mu = 8\%$, $\sigma = 20\%$. The mean path grows at μ; the median path grows at $\mu - \frac{1}{2}\sigma^2$. Right: the lognormal distribution of S_1.

39.4 Simulating one step

The exact discretisation (no approximation error, because we use the solution above):

$$\boxed{S_{t+\Delta t} = S_t \exp\left[\left(\mu - \tfrac{1}{2}\sigma^2\right)\Delta t + \sigma\sqrt{\Delta t}\, Z\right]}, \qquad Z \sim \mathcal{N}(0,1).$$

Worked example: one simulated day

$S = 100$, $\mu = 8\%$, $\sigma = 20\%$, $\Delta t = 1/252$. The drift term is $(0.08 - 0.02)/252 = 0.000238$ and the shock scale is $0.2\sqrt{1/252} = 0.0126$.

- Draw $Z = 0.5$: $S = 100\exp(0.000238 + 0.0063) = \mathbf{100.66}$.
- Draw $Z = -1.2$: $S = 100\exp(0.000238 - 0.01512) = \mathbf{98.52}$.

Repeat this step 252 times and you have a one-year path. Repeat the path 100,000 times and you have a Monte Carlo engine (§60). *For pricing*, replace μ with $r - q$.

40 Stochastic calculus and Itô's lemma

Correction to the original notes: Itô's lemma was listed but never explained

It is the single tool that turns "the stock follows GBM" into "the option satisfies the Black–Scholes PDE". Here it is in full.

40.1 Statement

If X follows $dX = a\,\mathrm{d}t + b\,\mathrm{d}W$ and $f(t,x)$ is smooth, then

$$\boxed{df = \Big(\frac{\partial f}{\partial t} + a\,\frac{\partial f}{\partial x} + \tfrac{1}{2}b^2\,\frac{\partial^2 f}{\partial x^2}\Big)dt + b\,\frac{\partial f}{\partial x}\,\mathrm{d}W}$$

40.2 Why the extra $\frac{1}{2}b^2 f_{xx}$ term?

Expand f in a Taylor series to second order:

$$df = f_t\,\mathrm{d}t + f_x\,\mathrm{d}X + \tfrac{1}{2}f_{xx}(dX)^2 + \dots$$

In ordinary calculus, $(dX)^2$ is negligible: it is of order dt^2. Here

$$(dX)^2 = a^2dt^2 + 2ab\,dt\,dW + b^2(dW)^2.$$

The first two terms vanish faster than dt, but $(dW)^2 = dt$ (quadratic variation!), so $(dX)^2 = b^2dt$, which is the *same order* as dt. The second-order Taylor term survives and contributes $\frac{1}{2}b^2 f_{xx}\,dt$. That's the whole secret.

A multiplication table worth memorising: $dt\cdot dt = 0$, $dt\cdot dW = 0$, $dW\cdot dW = dt$. For two correlated Brownian motions, $dW_1\cdot dW_2 = \rho\,dt$.

40.3 Worked derivation: the log of a GBM

Take $dS = \mu S\,\mathrm{d}t + \sigma S\,\mathrm{d}W$ (so $a = \mu S$, $b = \sigma S$) and $f = \ln S$. Then $f_S = 1/S$, $f_{SS} = -1/S^2$, $f_t = 0$:

$$d\ln S = \frac{1}{S}(\mu S\,\mathrm{d}t + \sigma S\,\mathrm{d}W) + \frac{1}{2}\Big(-\frac{1}{S^2}\Big)\sigma^2S^2\,\mathrm{d}t = \boxed{(\mu - \tfrac{1}{2}\sigma^2)dt + \sigma\,\mathrm{d}W}.$$

The right side has constant coefficients, so integrate directly: $\ln S_T - \ln S_0 = (\mu - \frac{1}{2}\sigma^2)T + \sigma W_T$. That is exactly the GBM solution. The $-\frac{1}{2}\sigma^2$ is the Itô correction.

40.4 Two more uses

- **Itô on an option value** $V(S,t)$ gives $dV = (V_t + \mu S V_S + \frac{1}{2}\sigma^2S^2V_{SS})dt + \sigma S V_S\,\mathrm{d}W$. This is the starting point of the Black–Scholes derivation (§46).
- **Martingale check.** Under $\mathbb{Q}$ (drift r), Itô on $e^{-rt}S_t$ gives $d(e^{-rt}S) = e^{-rt}\sigma S\,\mathrm{d}W$: no dt term, so the discounted price is a martingale, as the FTAP requires.

41 Stochastic volatility: the Heston model

GBM assumes σ is constant. Real markets show volatility clustering (calm and turbulent regimes), a negative spot–vol correlation, and a smile. The fix is to make volatility random too:

GBM: price random, volatility fixed. Heston: price random, volatility random.

The model (under $\mathbb{Q}$):

$$dS = (r-q)S\,\mathrm{d}t + \sqrt{v}\,S\,\mathrm{d}W_1, \qquad dv = \kappa(\theta - v)\,\mathrm{d}t + \xi\sqrt{v}\,\mathrm{d}W_2, \qquad dW_1\,dW_2 = \rho\,\mathrm{d}t.$$

- v is the instantaneous variance, starting at v_0.
- κ is the speed of mean reversion towards the long-run variance θ.
- ξ is the "vol of vol"; it creates smile *curvature*.
- ρ is the spot–vol correlation; $\rho < 0$ creates the equity *skew*, which is the leverage effect.

Why Heston is famous: its 1993 paper derives a closed-form (semi-analytic) price. The call is $C = Se^{-qT}P_1 - Ke^{-rT}P_2$, where P_1 and P_2 are recovered by integrating a known characteristic function. That makes calibration fast.

Worked example: the smile Heston produces

$v_0 = \theta = 0.04$ (20% vol), $\kappa = 2$, $\xi = 0.5$, $\rho = -0.7$, $T = 1$, $r = 5\%$:

Strike	80	90	100	110	120
Heston price	25.237	17.140	10.155	4.866	1.738
Implied vol	24.05%	21.59%	19.21%	17.02%	15.27%

A 100,000-path Monte Carlo of the same model gives 10.139 ± 0.035 at $K = 100$, agreeing with the formula. The negative ρ has produced a nine-point downward skew from a model whose *average* vol is 20%.

Pitfalls & conventions

- **Feller condition:** $2\kappa\theta > \xi^2$ keeps v strictly positive. Here $2(2)(0.04) - 0.25 = -0.09 < 0$, so it is violated, which is common in calibrated parameters. Your simulation must handle v hitting zero: use "full truncation" (use $\max(v, 0)$ in the drift and diffusion) or Andersen's QE scheme.
- **The "little Heston trap"** (Albrecher et al., 2007): one branch choice for the complex logarithm in the characteristic function makes prices jump at long maturities. Use the stable formulation.
- QuantLib offers analytic, finite-difference and Monte Carlo Heston engines, so you can cross-check three ways.

42 The SABR model

$$dF = \sigma_t F^{\beta}\,\mathrm{d}W, \qquad d\sigma_t = \nu\sigma_t\,\mathrm{d}Z, \qquad dW\,dZ = \rho\,\mathrm{d}t, \qquad \sigma_0 = \alpha.$$

SABR (Hagan, Kumar, Lesniewski, Woodward 2002) models the *forward* directly and was built to get smile *dynamics* right: local-vol models fit today's smile but predict that it moves the wrong way when the forward moves, which gives bad hedges. The paper says SABR "predicts the correct dynamics of the implied volatility curves".

- $\beta \in [0,1]$ sets the "backbone", how ATM vol moves with the forward: $\sigma_{ATM} \approx \alpha / F^{1-\beta}$. $\beta = 0$ gives a steep backbone (normal-like), $\beta = 1$ a flat one (lognormal). It is usually fixed by convention: 0.5 for rates, 1 for FX.
- α sets the level, ρ the skew, ν the curvature (vol of vol).
- Hagan's expansion gives a closed-form Black implied vol $\sigma_B(K, F)$, which you plug into Black-76.

Worked example: a SABR smile

$F = 105.13$, $T = 1$, $\beta = 0.5$, $\alpha = 0.2\sqrt{F} = 2.05$, $\rho = -0.3$, $\nu = 0.4$. Hagan's formula gives implied vols of 23.9% (strike 80), 22.1% (90), 20.8% (100), 20.2% (ATM 105.13), 19.7% (110) and 18.9% (120).

Pitfall: the expansion can imply *negative* densities at low strikes and long maturities. Fixes include "arbitrage-free SABR" and shifted SABR for negative rates.

43 Jump diffusion: the Merton model

GBM paths are continuous: $100 \to 101 \to 102 \to 103$. Real markets sometimes *jump*: $100 \to 101 \to 103 \to 102 \to 140$, on an earnings surprise, a central-bank decision, a bankruptcy or a geopolitical event. You cannot delta-hedge through a jump.

Merton (1976):

$$\frac{dS}{S} = (r - q - \lambda k)\,\mathrm{d}t + \sigma\,\mathrm{d}W + (J-1)\,\mathrm{d}N.$$

Here N is a Poisson process with intensity λ (jumps per year), $\ln J \sim \mathcal{N}(\mu_J, \delta^2)$ is the jump size, and $k = \mathbb{E}[J-1] = e^{\mu_J + \delta^2/2} - 1$. The $-\lambda k$ term keeps the discounted price a martingale.

The formula is a Poisson-weighted average of Black–Scholes prices:

$$C = \sum_{n=0}^{\infty} \frac{e^{-\lambda' T}(\lambda' T)^n}{n!}\, C_{BS}\big(S, K, T, r_n, \sigma_n\big),$$

with $\lambda' = \lambda(1+k)$, $\sigma_n^2 = \sigma^2 + n\delta^2/T$ and $r_n = r - \lambda k + n\ln(1+k)/T$.

Intuition Condition on exactly n jumps happening. Then the log price is still normal, just with extra variance, so the conditional price is a Black–Scholes price. Average over the Poisson probabilities of $n = 0, 1, 2, \ldots$ jumps. Truncate the sum around $n = 50$, or when the terms become negligible.

Worked example: the smile Merton produces

Diffusion vol 15%, $\lambda = 0.5$ jumps a year, mean jump -10% ($\mu_J = -0.10$), jump vol $\delta = 15\%$:

Strike	80	90	100	110	120
Merton price	24.768	16.696	10.092	5.417	2.599
Implied vol	21.25%	19.99%	19.04%	18.42%	18.05%

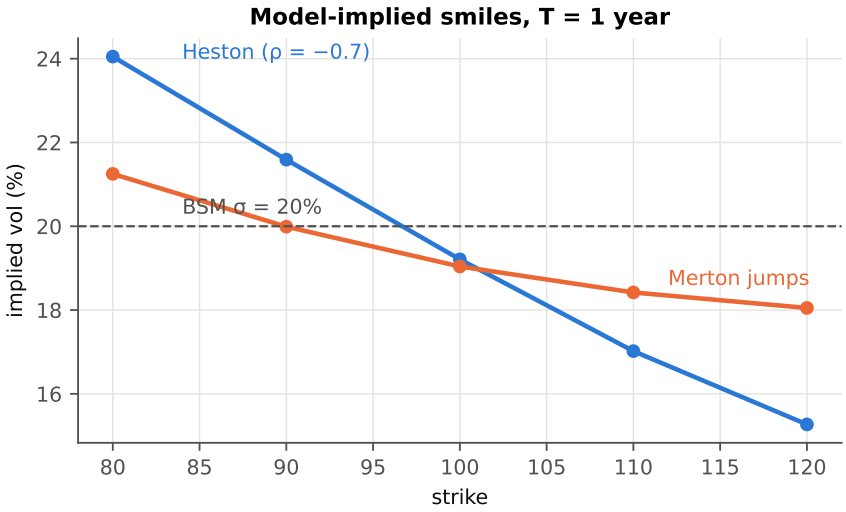


Figure 9. Smiles generated by Heston and by Merton jumps from this section's parameters, against the flat Black–Scholes assumption.

Which model fixes what? Jumps dominate *short-dated* skew (a big gap can happen tomorrow, but diffusion can't move far in a day), while stochastic volatility dominates *longer-dated* smile. Bates (1996) combines Heston with Merton jumps to get both.

Check yourself

- State the four defining properties of Brownian motion, and explain why $(dW)^2 = dt$.
- Derive $d \ln S$ from Itô's lemma without notes.
- Why is S_T lognormal under GBM, and why is its mean above its median?
- What does dW represent? Why can't we write dW/dt?
- In Heston, which parameter creates the skew and which the curvature? What is the Feller condition?
- Why do jumps matter more for short-dated options than for long-dated ones?

PART VII

Black–Scholes–Merton and the Greeks

44 What problem does BSM solve?

Given S_0, K, T, r, q and σ, find the fair price of a **European** call or put, plus all its sensitivities. Your project uses BSM for European vanillas and requires every major Greek. It is also the benchmark against which every other engine in the project (trees, Monte Carlo, PDE) is validated, so it pays to know it inside out.

45 The assumptions, and how each one fails

Assumption	Reality	Consequence
GBM with constant σ	Volatility is random and clusters	The smile; vega risk; model risk in hedges
Continuous price paths	Jumps and overnight gaps	Short-dated skew; gap risk that delta hedging can't remove
Continuous, frictionless hedging	Discrete rebalancing, transaction costs, bid-ask	Hedging error (§53); Leland's cost-adjusted vol
Constant r and q	Rates and dividends change	Rho and dividend risk
Unlimited borrowing and short selling at r	Short-sale constraints, funding spreads	Forwards that deviate from theory; funding adjustments (XVA)
No arbitrage	Mostly true in liquid markets	Parity and bounds hold to within costs

Intuition BSM is wrong as a model of the world and right as a *language*. Nobody believes volatility is constant, but implied volatility gives everyone a common unit in which to quote, compare and hedge options. A model can be false and still extremely useful, as long as you know where it breaks.

46 Derivation 1: delta hedging and the BSM PDE

Hold one option $V(S,t)$ and short Δ shares. By Itô's lemma (§40),

$$dV = \left(V_t + \mu S V_S + \tfrac{1}{2}\sigma^2 S^2 V_{SS}\right)dt + \sigma S V_S \, \mathrm{d}W.$$

The portfolio $\Pi = V - \Delta S$ changes by $d\Pi = dV - \Delta\, dS - \Delta q S \, \mathrm{d}t$ (being short the stock means you pay its dividends). Substituting $dS = \mu S \, \mathrm{d}t + \sigma S \, \mathrm{d}W$:

$$d\Pi = \left(V_t + \mu S V_S + \tfrac{1}{2}\sigma^2 S^2 V_{SS} - \Delta\mu S - \Delta q S\right)dt + \sigma S(V_S - \Delta)\, \mathrm{d}W.$$

Now choose $\boxed{\Delta = V_S}$. The dW term cancels, and so does every μ:

$$d\Pi = \Big(V_t + \tfrac{1}{2}\sigma^2 S^2 V_{SS} - qSV_S\Big)dt.$$

The portfolio is now *riskless* over the instant, so by no-arbitrage it must earn the risk-free rate: $d\Pi = r\Pi\,\mathrm{d}t = r(V - SV_S)\,\mathrm{d}t$. Equating the two gives the **Black–Scholes–Merton PDE**:

$$\boxed{\frac{\partial V}{\partial t} + \tfrac{1}{2}\sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} + (r-q)S\frac{\partial V}{\partial S} - rV = 0}, \qquad V(S,T) = \text{payoff}(S).$$

Without dividends ($q = 0$) this is the equation in your notes. Look closely: μ is nowhere in it. That is risk-neutrality emerging from the hedging argument, not assumed.

47 Derivation 2: risk-neutral expectation and the formula

Under $\mathbb{Q}$, $S_T = S_0 \exp\big((r - q - \tfrac{1}{2}\sigma^2)T + \sigma\sqrt{T}Z\big)$ with $Z \sim \mathcal{N}(0,1)$. Then

$$C = e^{-rT}\mathbb{E}^{\mathbb{Q}}\big[(S_T - K)^+\big] = e^{-rT}\Big\{\mathbb{E}^{\mathbb{Q}}\big[S_T \mathbf{1}_{S_T > K}\big] - K\,\mathbb{Q}(S_T > K)\Big\}.$$

- $\mathbb{Q}(S_T > K) = \mathbb{Q}\big(Z > -d_2\big) = N(d_2)$, where $d_2 = \dfrac{\ln(S_0/K) + (r - q - \tfrac{1}{2}\sigma^2)T}{\sigma\sqrt{T}}$.
- $\mathbb{E}^{\mathbb{Q}}[S_T \mathbf{1}_{S_T>K}] = S_0 e^{(r-q)T} N(d_1)$ with $d_1 = d_2 + \sigma\sqrt{T}$. You get this by writing the expectation as an integral and completing the square in the exponent, or, more elegantly, by switching to the stock numeraire.

Putting the pieces together:

$$\boxed{C = S_0 e^{-qT} N(d_1) - K e^{-rT} N(d_2)}, \qquad \boxed{P = K e^{-rT} N(-d_2) - S_0 e^{-qT} N(-d_1)},$$

$$d_1 = \frac{\ln(S_0/K) + (r - q + \tfrac{1}{2}\sigma^2)T}{\sigma\sqrt{T}}, \qquad d_2 = d_1 - \sigma\sqrt{T}.$$

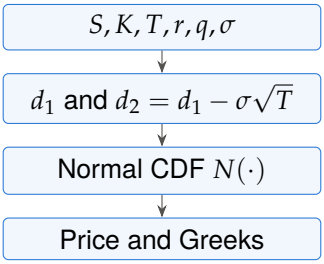
The put follows from the call by put–call parity, or by the same calculation with the indicator reversed.

48 What do d_1, d_2, $N(d_1)$ and $N(d_2)$ mean?

Your notes rightly warn against memorising "$N(d_1)$ is a probability". Here is the full picture:

- d_2 is the number of standard deviations by which the log forward exceeds the log strike, after the Itô correction. It answers: *how far in the money is the option, in risk-neutral standard deviations?*
- $N(d_2)$ is the **risk-neutral probability** that the call finishes in the money, $\mathbb{Q}(S_T > K)$.
- $e^{-qT}N(d_1)$ is the call's **delta**: the number of shares in the replicating portfolio.
- $N(d_1)$ is also the probability of finishing in the money, but under the *stock* measure.

Read the formula as a sentence: *"The call is worth the present value of the stock you receive if you exercise, minus the present value of the strike you pay if you exercise."* $S_0 e^{-qT} N(d_1)$ is the first part; $K e^{-rT} N(d_2)$ is the second.



49 The worked example

Worked example: $S = K = 100$, $T = 1$, $r = 5\%$, $q = 0$, $\sigma = 20\%$

1. $\ln(S/K) = \ln 1 = 0$ and $\frac{1}{2}\sigma^2 = 0.02$, so $d_1 = \dfrac{0 + (0.05 + 0.02) \times 1}{0.2 \times 1} = \dfrac{0.07}{0.2} = \mathbf{0.35}$.
2. $d_2 = 0.35 - 0.20 = \mathbf{0.15}$.
3. $N(0.35) = 0.636831$ and $N(0.15) = 0.559618$.
4. $Ke^{-rT} = 100e^{-0.05} = 95.1229$.
5. $C = 100(0.636831) - 95.1229(0.559618) = 63.6831 - 53.2325 = \mathbf{10.4506}$.
6. $P = 95.1229\,N(-0.15) - 100\,N(-0.35) = 95.1229(0.440382) - 100(0.363169) = \mathbf{5.5735}$.

Check: $C - P = 4.8771 = 100 - 95.1229$. Parity holds. Textbook check (Hull): $S = 42$, $K = 40$, $r = 10\%$, $\sigma = 20\%$, $T = 0.5$ gives $C = 4.7594$ and $P = 0.8086$.

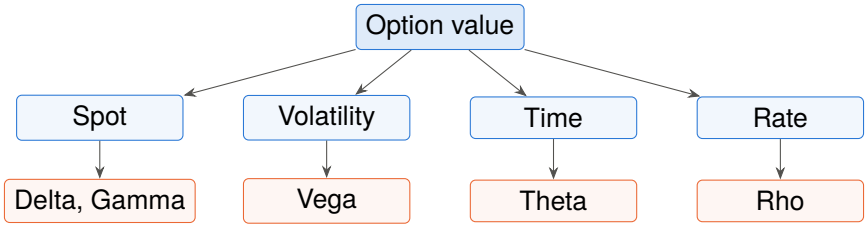
```python
import numpy as np
from scipy.stats import norm

def bsm(S, K, T, r, sigma, q=0.0, cp=1):
    """Black-Scholes-Merton price. cp=+1 call, -1 put."""
    d1 = (np.log(S/K) + (r - q + 0.5*sigma**2)*T) / (sigma*np.sqrt(T))
    d2 = d1 - sigma*np.sqrt(T)
    return cp*(S*np.exp(-q*T)*norm.cdf(cp*d1) - K*np.exp(-r*T)*norm.cdf(cp*d2))

assert abs(bsm(100, 100, 1, 0.05, 0.2) - 10.4506) < 1e-4
assert abs(bsm(100, 100, 1, 0.05, 0.2, cp=-1) - 5.5735) < 1e-4
assert abs(bsm(42, 40, 0.5, 0.10, 0.2) - 4.7594) < 1e-4      # Hull
```

50 The Greeks

The option price is a function $V = f(S, \sigma, r, t, q)$. The **Greeks** are its partial derivatives: how much the value moves when one input moves and the others stay put. Think of them as the option's risk dashboard.



50.1 Delta

$$\Delta = \frac{\partial V}{\partial S}, \qquad \Delta_{\text{call}} = e^{-qT} N(d_1) \in [0,1], \qquad \Delta_{\text{put}} = -e^{-qT} N(-d_1) \in [-1,0].$$

Meaning: roughly how much the option price changes for a ₹1 move in the underlying. With $\Delta = 0.60$, a ₹1 rise moves the option up about ₹0.60. **Running example:** call 0.6368, put -0.3632 (they differ by exactly $e^{-qT} = 1$, by parity).

Three more ways to read delta:

1. the number of shares in the replicating portfolio (the *hedge ratio*);
2. the option's equivalent position in the underlying: 100 calls with $\Delta = 0.6$ behave like 60 shares for small moves;
3. a rough (stock-measure) probability of finishing ITM. Traders loosely call a 25-delta call "a one-in-four shot".

> **Worked example: delta hedging from the notes**
>
> You own 100 calls with $\Delta = 0.60$. Portfolio delta $= 100 \times 0.60 = +60$. Short 60 shares (delta -60) and the net delta is about zero: for small moves, gains on the calls offset losses on the shares. But this neutrality does not last (see gamma), and it does not remove all risk (see §53).

50.2 Gamma

$$\Gamma = \frac{\partial^2 V}{\partial S^2} = \frac{\partial \Delta}{\partial S} = \frac{e^{-qT} n(d_1)}{S\sigma\sqrt{T}} \quad \text{(same for calls and puts)}.$$

Meaning: how fast delta changes, i.e. the curvature of the price. With $\Delta = 0.50$ and $\Gamma = 0.04$, a ₹1 rise takes delta to about 0.54 and a ₹1 fall to about 0.46. Gamma measures how quickly your hedge goes stale. **Running example:** $\Gamma = 0.018762$.

Long options are always long gamma ($\Gamma > 0$). That is the convexity you paid for: when the market moves, your delta moves in your favour (more long as it rises, more short as it falls).

50.3 Vega

$$\nu = \frac{\partial V}{\partial \sigma} = S_0 e^{-qT} n(d_1)\sqrt{T} \quad \text{(same for calls and puts)}.$$

Running example: $\nu = 37.524$ per unit of σ (i.e. per 100 vol points), so **0.3752 per vol point**. If implied vol rises from 20% to 21%, the call gains about 37.5 paise. Options are fundamentally instruments for trading volatility, and vega is how much of it you hold. Vega is largest at the money and grows with $\sqrt{T}$, so long-dated options carry far more vega than short-dated ones.

50.4 Theta

Correction to the original notes: the definition of theta

The notes wrote $\Theta = \partial V/\partial T$. That is ambiguous at best and wrong-signed at worst. If T means *maturity* or *time to expiry*, then $\partial V/\partial T$ is usually **positive** (longer options are worth more). Theta is defined with respect to **calendar time** t:

$$\boxed{\Theta = \frac{\partial V}{\partial t} = -\frac{\partial V}{\partial \tau}}, \qquad \tau = T - t.$$

Most long options have $\Theta < 0$: they lose value as each day passes, all else equal.

Closed forms (per year):

$$\Theta_{\text{call}} = -\frac{S_0 e^{-qT} n(d_1)\sigma}{2\sqrt{T}} - rKe^{-rT}N(d_2) + qS_0 e^{-qT}N(d_1),$$

$$\Theta_{\text{put}} = -\frac{S_0 e^{-qT} n(d_1)\sigma}{2\sqrt{T}} + rKe^{-rT}N(-d_2) - qS_0 e^{-qT}N(-d_1).$$

Running example: call -6.414 per year, i.e. $\mathbf{-0.01757}$ **per calendar day** ($\div 365$) or -0.02545 per trading day ($\div 252$); put -1.658 per year, -0.00454 per day. The notes' illustration, $\Theta = -0.05$ meaning "loses about ₹0.05 a day", is correct *provided* the convention is per day.

Check by revaluation: the call one calendar day later is worth 10.4330, a change of -0.01758. One week later: -0.1234. Thirty days later: -0.5346. Notice that decay accelerates (Figure 4).

50.5 Rho

$$\rho_{\text{call}} = KTe^{-rT}N(d_2), \qquad \rho_{\text{put}} = -KTe^{-rT}N(-d_2).$$

Running example: call 53.23 per unit rate, i.e. **0.5323 per 1%**; put -0.4189 per 1%. Revaluation: $+50$bp gives $+0.268$, $+100$bp gives $+0.539$. Calls gain when rates rise, because the strike you will pay later is discounted more, and because the forward rises. For short-dated equity options rho is small; for long-dated options and rates products it matters.

Pitfalls & conventions: units and conventions

- **Vega** is often quoted per 1 vol point (divide the analytic value by 100). py_vollib does this: its documentation says that "in practice vega is defined as the change in price for each 1 percent change in IV".
- **Theta** is quoted per calendar day ($\div 365$) in py_vollib ("theta is defined as the change in price for each day change in t, hence we divide by 365"), per trading day ($\div 252$) on some desks, and *per year* in QuantLib.
- **Rho** is usually per 1% (divide by 100).
- Put the conventions in one configuration object and print the units next to every Greek on your dashboard. Comparing two libraries without converting units is the most common reason "the Greeks don't match".

50.6 The Greeks at a glance

Greek	Definition	Measures sensitivity to	Call formula	Running example
Delta	$\partial V/\partial S$	spot	$e^{-qT}N(d_1)$	0.6368
Gamma	$\partial^2 V/\partial S^2$	spot (curvature)	$e^{-qT}n(d_1)/(S\sigma\sqrt{T})$	0.01876
Vega	$\partial V/\partial \sigma$	volatility	$Se^{-qT}n(d_1)\sqrt{T}$	0.3752 / vol pt
Theta	$\partial V/\partial t$	time passing	see above	-0.01757 / day
Rho	$\partial V/\partial r$	interest rate	$KTe^{-rT}N(d_2)$	0.5323 / 1%

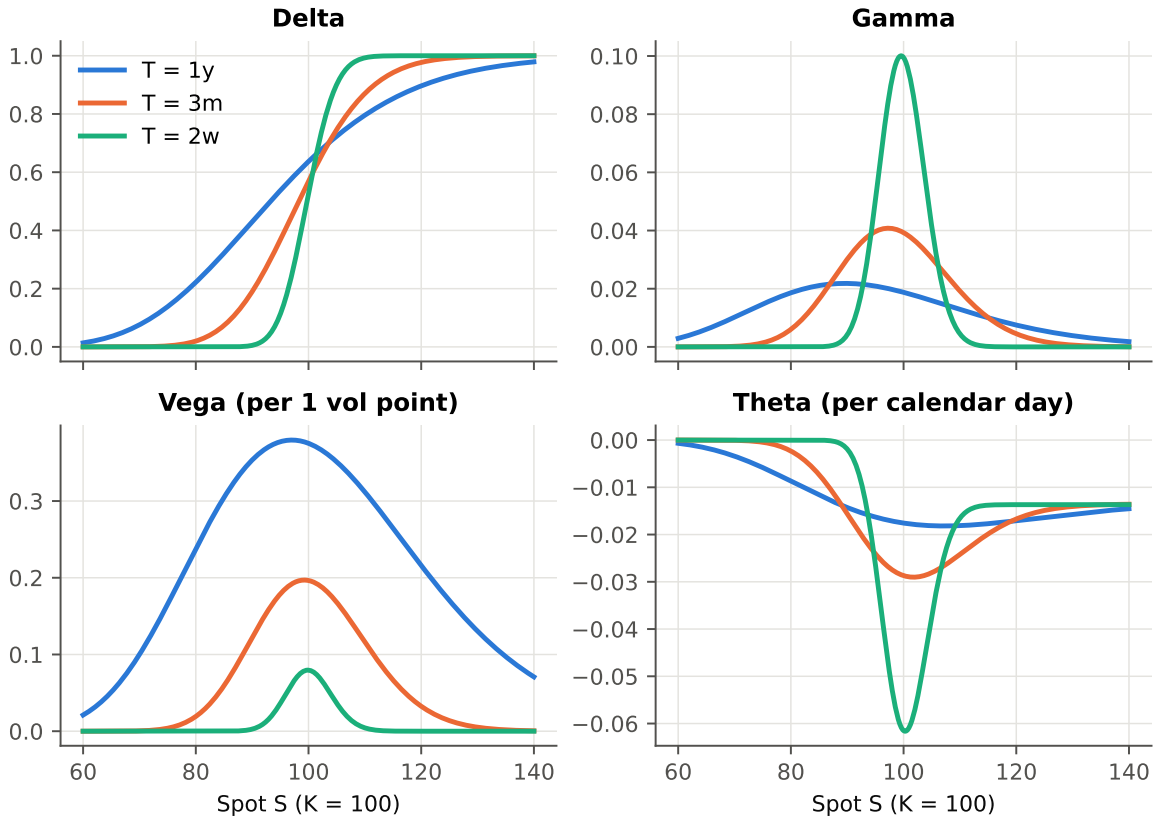


Figure 10. Greeks of a $K = 100$ call against spot for three maturities. Delta becomes a step function near expiry; gamma and theta concentrate sharply at the strike; vega shrinks with maturity.

Behaviour you should be able to explain from the picture:

- As expiry approaches, delta becomes a step: 0 below the strike, 1 above.
- ATM gamma explodes like $1/\sqrt{\tau}$ near expiry, so a short-dated ATM option is a hedging nightmare.
- Vega moves the opposite way to gamma: it shrinks like $\sqrt{\tau}$. Long-dated options are "vega instruments"; short-dated ones are "gamma instruments".
- Theta is most negative at the money near expiry: the price of all that gamma.

51 Higher-order Greeks

- **Vanna** $= \dfrac{\partial \Delta}{\partial \sigma} = \dfrac{\partial \nu}{\partial S} = -e^{-qT} n(d_1) \dfrac{d_2}{\sigma}$. Running example: -0.2814 per unit σ. It matters whenever spot and vol move together, which in equities is always (the skew).
- **Volga (vomma)** $= \dfrac{\partial \nu}{\partial \sigma} = \nu \dfrac{d_1 d_2}{\sigma}$. Running example: 9.850. It is positive for OTM options, which is why the wings carry a premium for "vol of vol".
- **Charm** $= \partial \Delta / \partial t$: how delta drifts overnight with no market move. Running example: -0.00018 per day. It matters a lot for weekly options, where a hedge can drift materially overnight.
- **Speed** $(\partial \Gamma / \partial S)$, **colour** $(\partial \Gamma / \partial t)$ and **zomma** $(\partial \Gamma / \partial \sigma)$ are used to manage large gamma books.

52 The gamma–theta relationship

Rearrange the BSM PDE with Greeks in place of derivatives:

$$\Theta + \tfrac{1}{2}\sigma^2 S^2 \Gamma + (r - q) S \Delta - rV = 0.$$

For a delta-hedged position with small r, this says

$$\boxed{\Theta \approx -\tfrac{1}{2}\sigma^2 S^2 \Gamma}.$$

This is the most important identity in option trading. **You pay theta to own gamma.** A long option, delta-hedged, bleeds time value every day. In return, every move in the stock lets you rebalance at a profit: you sell as it rises and buy as it falls. If the stock moves as much as implied vol predicted, the two exactly cancel.

> **Worked example: checking the identity**
>
> $\tfrac{1}{2}(0.04)(10{,}000)(0.018762) = 3.752$; $rS\Delta = 3.184$; $rV = 0.5225$. So $\Theta = 0.5225 - 3.752 - 3.184 = -6.414$, matching the closed form.

53 Delta hedging: how it works and where it fails

53.1 Dynamic hedging

Delta does not stay constant. Suppose in the morning your book's delta is $+500$, and after a rally it is $+620$. Your hedge (short 500) is now short by 120. Keeping a book delta-neutral means **rebalancing**, which is called dynamic hedging.

53.2 Where the P&L of a hedged option comes from

Over a small step, a long option that is delta-hedged earns approximately

$$\boxed{d\text{P\&L} \approx \tfrac{1}{2}\Gamma S^2 \left(\sigma^2_{\text{realised}} - \sigma^2_{\text{implied}}\right) dt}.$$

You make money if the stock actually moves more than the implied vol you paid for, and lose if it moves less. This is *gamma scalping*, and it is how option traders turn a view on volatility into P&L without taking a view on direction.

Worked example: paying 20 vol, realising 25

Short the running-example call at 20% implied vol, delta-hedge daily, and let the stock actually move with 25% vol. Over 20,000 simulated paths the average hedged P&L is -1.98 (standard deviation 1.00). That is close to the difference in option values, $C(20\%) - C(25\%) = -1.89$. The short-vol position lost because realised vol came in higher than implied.

53.3 The limits of delta hedging

Delta $= 0$ does not mean risk $= 0$.

After delta hedging you still carry:

- **gamma:** large moves hurt short-gamma books between rebalances;
- **vega:** implied vol can move independently of spot;
- **theta:** time decay;
- **rho** and dividend risk;
- **jump/gap risk:** you can't rebalance through a gap;
- **liquidity and transaction costs;**
- **model risk:** the wrong σ or wrong smile dynamics give the wrong delta.

53.4 Discrete hedging error

Real hedges are rebalanced N times, not continuously. Derman and Kamal (1999) showed that the leftover error has standard deviation approximately

$$\sigma_{\text{P\&L}} \approx \sqrt{\frac{\pi}{4N}}\, \nu\, \sigma .$$

Worked example: hedging frequency

For the running example ($\nu = 37.52$, $\sigma = 0.2$), simulating a short call that is delta-hedged N times over a year (20,000 paths, realised vol = implied vol):

Rebalancing	N	Simulated sd of P&L	Derman–Kamal
Monthly	12	1.910	1.920
Weekly	52	0.938	0.922
Daily	252	0.437	0.419

Daily hedging still leaves about 4% of the 10.45 premium as noise. Quadrupling the frequency halves the error, but quadruples transaction costs. QuantLib's `DiscreteHedging` example prints exactly this benchmark, so it is a ready-made test for your own hedging

simulator.

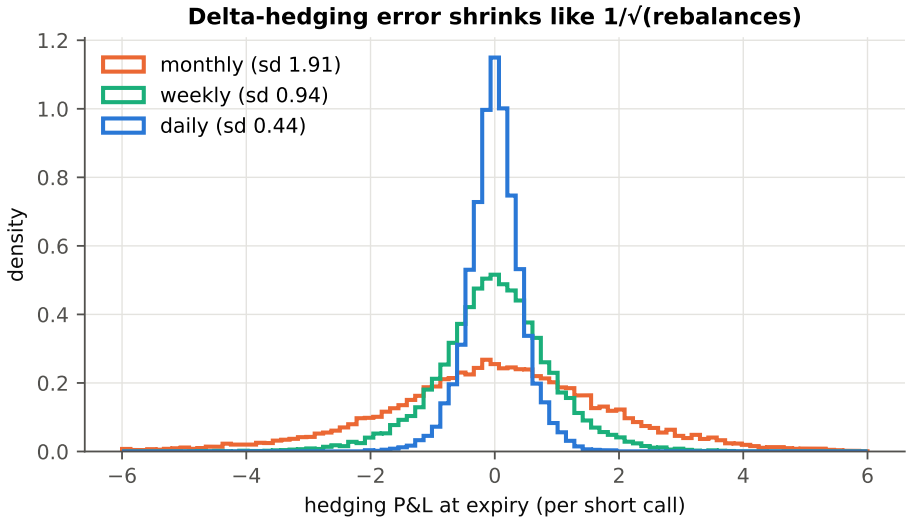


Figure 11. Distribution of the final P&L of a delta-hedged short call for three rebalancing frequencies. The mean is about zero in every case; the spread shrinks like $1/\sqrt{N}$.

Check yourself

- Derive the BSM PDE from the delta-hedging argument. Where exactly does μ disappear?
- What are d_1 and d_2? Interpret $N(d_2)$ and $e^{-qT}N(d_1)$.
- Compute the running-example call by hand to four decimals.
- Why is theta $\partial V/\partial t$ and not $\partial V/\partial T$? Convert -6.414 per year into per calendar day and per trading day.
- Explain "you pay theta to own gamma" using the PDE.
- Why does delta hedging not eliminate all risk? Name five remaining risks.

PART VIII

Implied Volatility, Trees and Black-76

54 The implied volatility solver

54.1 Why we need numerical methods

The market says a call costs ₹11. You know S, K, T, r and q, but not σ. You need to solve

$$f(\sigma) = \text{BSM}(\sigma) - 11 = 0.$$

There is no algebraic formula for σ, so this is a **root-finding problem**. The good news: BSM is strictly increasing in σ (vega > 0), so if the price lies inside the no-arbitrage bounds there is exactly one root.

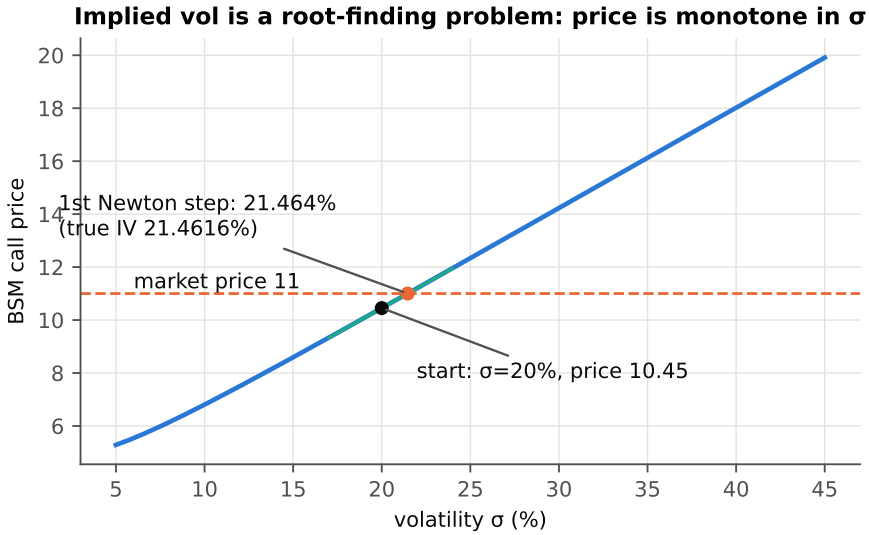


Figure 12. The call price rises monotonically with σ, so each price inside the bounds corresponds to exactly one implied vol. One Newton step from 20% already lands within 0.003 vol points of the answer.

54.2 Newton–Raphson

The general rule is $x_{n+1} = x_n - f(x_n)/f'(x_n)$: follow the tangent line to zero. For implied vol, $f'(\sigma)$ is simply vega:

$$\boxed{\sigma_{n+1} = \sigma_n - \frac{C(\sigma_n) - C_{\text{market}}}{\nu(\sigma_n)}}$$

Worked example: the notes' intuition

Current guess 20%, model price ₹8, market price ₹10: the model is too cheap, so raise vol. With vega 0.40 *per vol point*, the step is $(10 - 8)/0.40 = 5$ points, so the next guess is 25%. Repeat until the model price matches.

Worked example: the real numbers (running example, market price 11)

Iteration	σ_n	Model price	Vega (per unit)
0	20.000000%	10.450584	37.524
1	21.464172%	11.000966	37.650
2	21.461606%	11.000000002	37.650
3	21.461606%	11.000000000	37.650

Step 1: $0.20 - (10.4506 - 11)/37.524 = 0.214642$. The error shrinks from 10^{-1} to 10^{-3} to 10^{-9} to machine precision: the number of correct digits roughly *doubles* each step. That is quadratic convergence.

Where Newton fails: when vega is tiny (deep OTM options, or very near expiry), the step $(\text{error})/\nu$ explodes and can jump to negative or absurd vols.

54.3 Bisection

Bracket the root between a low and a high vol where the price errors have opposite signs, evaluate the midpoint, and keep the half that still brackets the root.

Worked example: bisection on $[10\%, 40\%]$ for a price of 11

Step	Low	High	Mid	Price at mid
1	10%	40%	25.000%	12.336 (too high, root below)
2	10%	25%	17.500%	9.516 (too low, root above)
3	17.5%	25%	21.250%	10.920 (too low)
4	21.25%	25%	23.125%	11.627 (too high)
5	21.25%	23.125%	22.188%	11.273 (too high)
6	21.25%	22.188%	21.719%	11.097 (too high)
7	21.25%	21.719%	21.484%	11.009 (too high)
8	21.25%	21.484%	21.367%	10.964 (too low)

After 8 steps we are within 0.12 vol points. Bisection halves the interval each step (linear convergence), so 30 steps give about 10^{-9} accuracy. It is slow, but it *cannot* fail once you have a valid bracket, and it never needs vega.

54.4 Brent's method: the practical default

Brent's method (`scipy.optimize.brentq`) combines bisection's guarantee with the speed of secant and inverse-quadratic interpolation. It keeps a bracket at all times, so it is robust, and usually converges in a handful of evaluations. Use it unless you have a reason not to.

54.5 Initial guesses

For an option that is ATM *forward*, $C \approx 0.4\, Fe^{-rT}\sigma\sqrt{T}$ (because $n(0) = 1/\sqrt{2\pi} \approx 0.4$). Inverting gives the Brenner–Subrahmanyam (1988) guess:

$$\sigma_0 \approx \sqrt{\frac{2\pi}{T}}\frac{C}{Fe^{-rT}}.$$

Example: the ATMF call ($K = 105.13$) in the running example costs 7.9656, so $\sigma_0 = \sqrt{2\pi} \times 7.9656 e^{0.05}/105.13 = 19.97\%$, almost exact. Corrado–Miller extends the approximation away from the money.

54.6 State of the art: Jäckel's "Let's Be Rational"

Peter Jäckel's 2015 algorithm shows "how Black's volatility can be implied from option prices with as little as two iterations to maximum attainable precision on standard (64-bit floating point) hardware for all possible inputs." It uses rational-function initial guesses in four regions of log-moneyness plus a high-order (Householder) iteration. Jäckel reports about 180 nanoseconds per implied vol for the 2024 version. py_vollib wraps it, so for production speed use that; for learning, write Brent yourself.

54.7 Failure cases you must handle

- **Price outside the arbitrage bounds:** there is no solution. Return NaN and flag it; do not clip.
- **Near expiry or deep OTM:** vega underflows, prices are a tick or two, and the implied vol is meaningless noise.
- **American options:** inverting a European formula on an American price overstates IV for puts. Use a tree-based IV, or de-Americanise first.
- **Stale or crossed quotes.**

Build smiles from **OTM options only**: puts below the forward, calls above. They are more liquid, and their price is pure time value rather than intrinsic value plus a sliver of optionality.

```
from scipy.optimize import brentq

def implied_vol(P, S, K, T, r, q=0.0, cp=1):
    lo = max(cp*(S*np.exp(-q*T) - K*np.exp(-r*T)), 0.0)   # lower bound
    hi = S*np.exp(-q*T) if cp == 1 else K*np.exp(-r*T)    # upper bound
    if not (lo < P < hi):
        return np.nan                                      # no solution: flag it
    return brentq(lambda s: bsm(S, K, T, r, s, q, cp) - P, 1e-6, 5.0, xtol=1e-12)

assert abs(implied_vol(11.0, 100, 100, 1, 0.05) - 0.214616) < 1e-6
# round trip test over a grid
for s in [0.05, 0.2, 0.6]:
    for K in [60, 100, 160]:
        p = bsm(100, K, 1, 0.05, s)
        assert abs(implied_vol(p, 100, K, 1, 0.05) - s) < 1e-7
```

55 Binomial trees

55.1 Why trees?

BSM prices European options in closed form, but it cannot handle early exercise. A tree discretises the stock's possible paths into a lattice, so at every node you can compare "exercise now" with

"keep holding", and pick the better one.

55.2 The one-step tree and replication

Worked example: the notes' one-step tree ($u = 1.10$, $d = 0.90$)

$S_0 = 100$ goes to 110 or 90 in one year. A call with $K = 100$ pays 10 or 0. With $r = 5\%$ (continuous):

- Risk-neutral probability: $p = \dfrac{e^{0.05} - 0.9}{1.1 - 0.9} = 0.7564$.
- Call value: $e^{-0.05}(0.7564 \times 10 + 0.2436 \times 0) = \mathbf{7.195}$.
- Replicating portfolio: $\Delta = (10 - 0)/(110 - 90) = 0.5$ shares and $B = e^{-0.05}(0 - 0.5 \times 90) = -42.81$ in cash. Cost: $50 - 42.81 = 7.195$. Same answer.

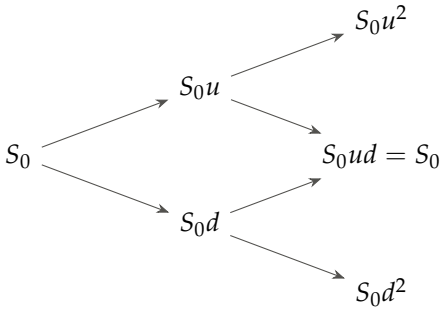
The earlier example in §36 (120/80) gives 11.951 the same way: larger moves mean higher volatility, which means a more valuable option.

55.3 The CRR model and where its parameters come from

Cox, Ross and Rubinstein (1979) chose the up and down factors so that the tree matches the risk-neutral *mean* and *variance* of the stock over each step $\Delta t = T/N$, and so that it *recombines* ($ud = 1$):

- **Mean:** $pu + (1-p)d = e^{(r-q)\Delta t} \Rightarrow \boxed{p = \dfrac{e^{(r-q)\Delta t} - d}{u - d}}$.
- **Variance:** $\approx \sigma^2 \Delta t$ with $d = 1/u$ gives $\boxed{u = e^{\sigma\sqrt{\Delta t}},\ d = e^{-\sigma\sqrt{\Delta t}}}$ (to first order in Δt).

Recombination is what makes trees efficient: an up-then-down move lands on the same price as down-then-up, so after N steps there are only $N + 1$ nodes instead of 2^N.



55.4 Backward induction

The algorithm at the heart of every tree:

1. Build the stock tree: node (i, j) has price $S_0 u^j d^{i-j}$.
2. At expiry, set each node's option value to the payoff.

3. Step backwards. At each node,

$$\text{European: } \boxed{V = e^{-r\Delta t}\left[pV_u + (1-p)V_d\right]}$$

$$\text{American: } \boxed{V = \max\left(\text{exercise value},\ \text{continuation value}\right)}$$

4. The value at the root is the price.

Worked example: a two-step American put

$S = K = 100$, $T = 1$, $r = 5\%$, $\sigma = 20\%$, $N = 2$, so $\Delta t = 0.5$: $u = e^{0.2\sqrt{0.5}} = 1.1519$, $d = 0.8681$, $p = (e^{0.025} - 0.8681)/(1.1519 - 0.8681) = 0.5539$.

- Stock at expiry: 132.69, 100, 75.36. Put payoffs: 0, 0, 24.64.
- Up node ($S = 115.19$): continuation $= 0$, exercise value negative, so $V_u = 0$.
- Down node ($S = 86.81$): continuation $= e^{-0.025}(0.5539 \times 0 + 0.4461 \times 24.64) = 10.72$, but exercise gives $100 - 86.81 = \mathbf{13.19}$. Exercise wins: $V_d = 13.19$.
- Root: European $= e^{-0.025}(0.4461 \times 10.72) = 4.66$; American $= e^{-0.025}(0.4461 \times 13.19) = \mathbf{5.74}$.

The early-exercise right at the down node added more than ₹1 of value. Two steps is far too coarse for an accurate price (the converged value is 6.09), but you can do it by hand and check your code against it.

56 American options and early exercise

The decision at every node is the one your notes describe. For a put with $K = 100$ at a node where $S = 80$, immediate exercise is worth $100 - 80 = 20$:

- if the continuation value is 17, then $V = \max(20, 17) = 20$: **exercise**;
- if the continuation value is 25, then $V = \max(20, 25) = 25$: **keep holding**.

The early-exercise boundary. For each date there is a critical stock price $S^*(t)$ below which it is optimal to exercise the American put. From a 1,000-step tree for the running example: $S^* \approx 81.7$ with 9 months left, 83.8 with 6 months, 87.0 with 3 months and 95.1 with about four days. The boundary rises towards K as expiry approaches, because there is less time value left to protect.

Results for the running example. American put $= 6.0903$ (LR, 5,001 steps) versus European put 5.5735: an early-exercise premium of 0.517. American call $= 10.4506 =$ European call, as Merton's theorem requires with no dividends.

57 Convergence and the odd–even problem

CRR prices converge to Black–Scholes with error of order $1/N$, but they **oscillate**: the strike sits at a different position relative to the terminal nodes depending on whether N is odd or even. You can see it in the saw-tooth of Figure 13. Remedies include averaging N and $N + 1$, placing the strike exactly on a node, Richardson extrapolation, or using a better tree.

58 The Leisen–Reimer tree

Correction to the original notes: no formula in the original notes

The notes said only that Leisen–Reimer "improves convergence". Here is how it works.

Intuition Instead of matching the mean and variance of each step (CRR), Leisen and Reimer (1996) choose the tree so that its probabilities of finishing in the money reproduce the Black–Scholes values $N(d_1)$ and $N(d_2)$ *exactly*. They use a very accurate normal-to-binomial inversion (Peizer–Pratt) to do it. The tree is built from the answer, so it converges smoothly and fast.

Peizer–Pratt inversion (method 2), for an odd number of steps n:

$$h^{-1}(z;n) = \frac{1}{2} + \frac{\operatorname{sign}(z)}{2}\sqrt{1 - \exp\left[-\left(\frac{z}{n+\frac{1}{3}+\frac{0.1}{n+1}}\right)^2\left(n+\frac{1}{6}\right)\right]}.$$

Tree parameters:

$$p = h^{-1}(d_2;n), \qquad p' = h^{-1}(d_1;n), \qquad u = e^{(r-q)\Delta t}\frac{p'}{p}, \qquad d = \frac{e^{(r-q)\Delta t} - pu}{1-p}.$$

The pipeline: $S_0, K, T, r, q, \sigma \to d_1, d_2 \to$ Peizer–Pratt $\to p', p \to u, d, p \to$ stock tree $\to$ terminal payoff $\to$ backward induction $\to$ price. n **must be odd.** Use the same backward-induction code as CRR, with early-exercise checks for Americans.

Worked example: LR parameters, running example, $n = 51$

$d_2 = 0.15$ gives $p = h^{-1}(0.15;51) = 0.51045$; $u = 1.02828$ and $d = 0.97250$.

58.1 The CRR vs LR experiment

Your notes suggest this as a research question: *how many steps do CRR and LR need to converge?* Here is the answer for the running example.

Table 7. Tree error for the European call (vs BSM 10.450584), and American put values.

N	CRR call error	LR call error	CRR American put	LR American put
25	+0.07038	−0.00053	6.14619	6.07567
50	−0.03989	$(N \to 51)$	6.07373	—
51	+0.03443	−0.00013	6.11795	6.08357
100	−0.01997	$(N \to 101)$	6.08235	—
101	+0.01737	−0.000034	6.10457	6.08722
201	+0.00872	−0.0000087	6.09752	6.08881
500	−0.00400	$(N \to 501)$	6.08881	—
501	+0.00350	−0.0000014	6.09328	6.08978
1001	+0.00175	−0.00000035	6.09183	6.09008

Reading the table. LR with 51 steps (error 0.00013) beats CRR with 1,001 steps (error 0.00175). LR's error falls about fourfold each time N doubles, i.e. roughly second order for Europeans. For

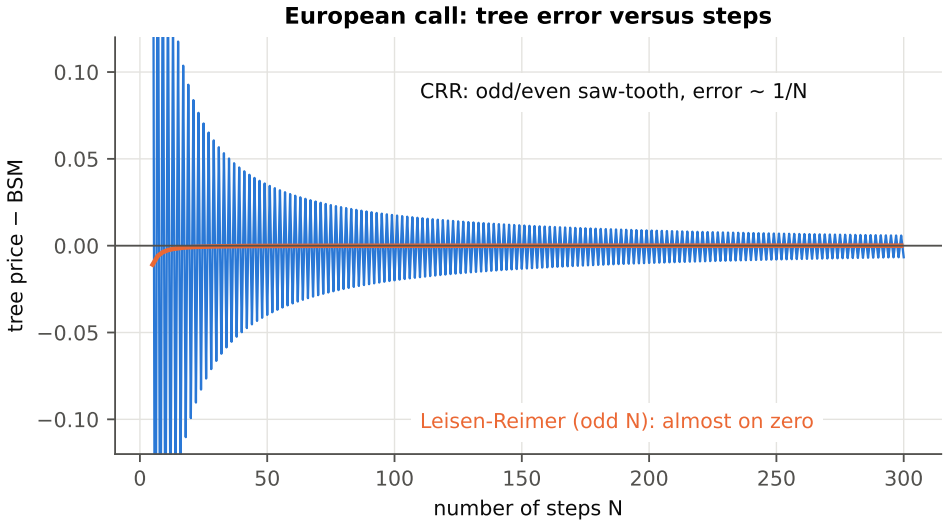


Figure 13. European call: CRR error zig-zags between positive (odd N) and negative (even N) and shrinks like $1/N$; Leisen–Reimer is almost on zero from the start.

the American put, LR also converges smoothly from below (6.0757, 6.0836, 6.0872, 6.0888, 6.0898, 6.0901), while CRR oscillates. Smooth convergence also means LR Greeks (computed from the tree nodes) are stable. A subtle point: the research question is not "is LR always better?" but "how do the two converge, for this option?" Ask that for each new product.

```
def pp_inv(z, n):
    return 0.5 + np.sign(z)*0.5*np.sqrt(1 - np.exp(-(z/(n + 1/3 + 0.1/(n+1)))**2 * (n + 1/6)
    ))

def lr_tree(S, K, T, r, sigma, q=0.0, n=201, cp=1, american=False):
    n += (n % 2 == 0)                                  # force odd n
    dt = T/n
    d1 = (np.log(S/K) + (r - q + 0.5*sigma**2)*T)/(sigma*np.sqrt(T))
    d2 = d1 - sigma*np.sqrt(T)
    p, pp = pp_inv(d2, n), pp_inv(d1, n)
    g = np.exp((r - q)*dt); u = g*pp/p; d = (g - p*u)/(1 - p)
    disc = np.exp(-r*dt)
    ST = S * u**np.arange(n, -1, -1) * d**np.arange(0, n + 1)
    V = np.maximum(cp*(ST - K), 0)
    for i in range(n - 1, -1, -1):                     # backward induction
        ST = ST[:-1]/u
        V = disc*(p*V[:-1] + (1 - p)*V[1:])
        if american:
            V = np.maximum(V, cp*(ST - K))             # early-exercise check
    return V[0]

assert abs(lr_tree(100, 100, 1, 0.05, 0.2) - 10.450584) < 1e-4
assert lr_tree(100,100,1,0.05,0.2,cp=-1,american=True) > 5.5735   # Am >= Eur
```

Pitfalls & conventions

Guard against extreme inputs. One open-source review found an LR implementation returning NaN for a near-zero spot (the Peizer–Pratt formula hits $0/0$ when p or $1-p$ underflows). Test the edges: tiny spot, tiny vol, tiny T, deep ITM and deep OTM.

59 Black-76: options on futures and forwards

Correction to the original notes: no formula in the original notes

The notes said only that Black-76 "uses F instead of S". Here is the model in full.

Intuition A futures contract costs nothing to enter, so it has no "carry": under the right measure its price has zero drift. Black's 1976 insight was that you can take Black–Scholes, replace the discounted spot $S_0 e^{-qT}$ by the discounted futures price Fe^{-rT}, and you are done.

$$\boxed{C = e^{-rT}\left[F\,N(d_1) - K\,N(d_2)\right]}, \qquad \boxed{P = e^{-rT}\left[K\,N(-d_2) - F\,N(-d_1)\right]},$$

$$d_1 = \frac{\ln(F/K) + \frac{1}{2}\sigma^2 T}{\sigma\sqrt{T}}, \qquad d_2 = d_1 - \sigma\sqrt{T}.$$

Greeks (with respect to F): $\Delta_F = e^{-rT}N(d_1)$ for a call, $-e^{-rT}N(-d_1)$ for a put; $\Gamma_F = e^{-rT}n(d_1)/(F\sigma\sqrt{T})$; vega $= Fe^{-rT}n(d_1)\sqrt{T}$. Parity: $C - P = e^{-rT}(F - K)$.

Worked example: Black-76 at the money

$F = K = 100$, $T = 1$, $r = 5\%$, $\sigma = 20\%$: $d_1 = 0.1$, $d_2 = -0.1$. $C = e^{-0.05}(100 \times 0.53983 - 100 \times 0.46017) = \mathbf{7.5771}$, and the put is the same (ATM forward, so $C = P$). Futures delta $= 0.5135$, gamma $= 0.01888$, vega $= 37.76$ (0.3776 per vol point).

Consistency check with BSM: if $F = S_0 e^{(r-q)T}$, Black-76 gives exactly the BSM price. For the running example, Black-76 with $F = 105.127$ returns 10.4506.

Where Black-76 is used:

- options on futures: commodities, index futures, bond futures;
- **caps and floors:** each caplet is a call on a forward rate. Value $= N\,\alpha_i\,P(0, T_{i+1}) \times \text{Black}(F_i, K, \sigma_i, T_i)$. Example: notional ₹1 crore, 3-month accrual, forward rate 6%, strike 6%, vol 20%, fixing in 1 year, discount factor $e^{-0.06\times 1.25}$: caplet $\approx$ 11,085;
- **swaptions:** annuity $A(0) \times \text{Black}(S_{\text{swap}}, K, \sigma, T)$, with the annuity from §20;
- when rates go negative, $\ln(F/K)$ breaks. Use shifted Black or the Bachelier (normal) model.

Your project asks for Black-76 calls, puts, Greeks, and plots against futures price and volatility. The same plotting code as BSM works: just swap the pricing function.

Check yourself

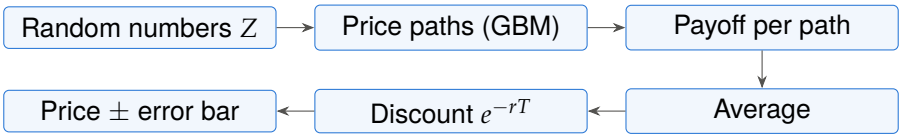
- Why does Newton's method use vega? When does it fail, and what do you do then?
- Derive p, u and d for CRR. Why must the tree recombine?
- Price a two-step American put by hand and identify the node where exercise is optimal.
- What does Leisen–Reimer match that CRR doesn't? Why must n be odd?
- Show that Black-76 with $F = S_0 e^{(r-q)T}$ reproduces BSM.

PART IX

Monte Carlo Simulation

60 What is Monte Carlo pricing?

Intuition If the price is a discounted expectation, $V_0 = e^{-rT}\mathbb{E}^{\mathbb{Q}}[\text{payoff}]$, then estimate the expectation the way a statistician would: simulate many possible futures under $\mathbb{Q}$, compute the payoff on each, average, and discount. The law of large numbers guarantees the average converges to the true expectation.



Monte Carlo's great strength is flexibility. Path-dependent payoffs (Asian, lookback, barrier), many assets (baskets), and exotic dynamics (Heston, jumps) all slot in by changing the path generator or the payoff function. Its weakness is slow convergence. Your project requires GBM path simulation, variance reduction, finite-difference Greeks and confidence intervals.

Worked example: the notes' five-path example

Simulated payoffs: $20, 0, 15, 5, 10$. Average $= 50/5 = 10$. With a discount factor $e^{-rT} = 0.95$, the estimate is $10 \times 0.95 = 9.50$. Five paths is far too few to trust; the rest of this part is about how many is enough, and how to need fewer.

61 Simulating paths

Exact (log-Euler) scheme for GBM:

$$S_{t+\Delta t} = S_t \exp\left[(r - q - \tfrac{1}{2}\sigma^2)\Delta t + \sigma\sqrt{\Delta t}\, Z\right].$$

This has *no* discretisation error, because it uses the exact solution. For a European payoff you can jump straight to T in one step. For path-dependent payoffs, simulate the monitoring dates.

Euler scheme: $S_{t+\Delta t} = S_t\big(1 + (r - q)\Delta t + \sigma\sqrt{\Delta t}Z\big)$. It has bias of order Δt and can produce negative prices. Use it only for SDEs without an exact solution (e.g. Heston's variance), and then prefer better schemes (Milstein, or Andersen's QE for Heston).

62 Standard error and confidence intervals

With N independent discounted payoffs Y_i:

$$\hat{V} = \frac{1}{N}\sum_i Y_i, \qquad \boxed{\mathrm{SE} = \frac{s}{\sqrt{N}}}, \qquad 95\%\ \mathrm{CI} = \hat{V} \pm 1.96\,\mathrm{SE},$$

where s is the sample standard deviation of the Y_i. By the central limit theorem, $\hat{V}$ is approximately normal, which justifies the 1.96.

Worked example: the notes' confidence interval

$\hat{V} = 10$, SE $= 0.10$: the 95% CI is $10 \pm 0.196 = [9.804, 10.196]$. Report it like that. A Monte Carlo number without an error bar pretends to a precision it does not have.

The square-root law. SE $\propto 1/\sqrt{N}$, so to halve the error you need **four times** as many paths, and to gain one more decimal digit you need **100 times** as many. That is why Monte Carlo can be expensive, and why variance reduction matters.

Worked example: real convergence for the running example

The standard deviation of the discounted call payoff can be computed analytically: $s = 14.719$. One seeded run gives:

Paths N	Estimate	SE	95% CI contains 10.4506?
1,000	9.912	0.448	yes
5,000	10.428	0.209	yes
10,000	10.343	0.149	yes
50,000	10.480	0.066	yes
100,000	10.446	0.047	yes
1,000,000	10.453	0.0147	yes

The SE column falls by about $\sqrt{10} \approx 3.16$ for every tenfold increase in paths, exactly as theory says. The notes' illustrative table (9.72, 9.51, 9.48, 9.45, 9.44) shows the same idea for a different option. What matters is that the estimates stabilise and the error bars shrink.

63 Random numbers

- Use a modern pseudo-random generator (`numpy.random.default_rng`, PCG64) with an **explicit seed**, so every result can be reproduced.
- For parallel runs, create independent streams with `SeedSequence.spawn`. Never reuse the same seed across workers.
- **Quasi-Monte Carlo:** Sobol sequences (`scipy.stats.qmc.Sobol`, scrambled) fill space more evenly than random points, with error closer to $O((\log N)^d/N)$ than $O(1/\sqrt{N})$. Combine them with a Brownian-bridge construction, so that the first (best-distributed) dimensions decide the overall shape of the path. Use *randomised* QMC so you can still compute error bars.

```
def mc_european(S, K, T, r, sigma, q=0.0, N=100_000, seed=42, cp=1):
```

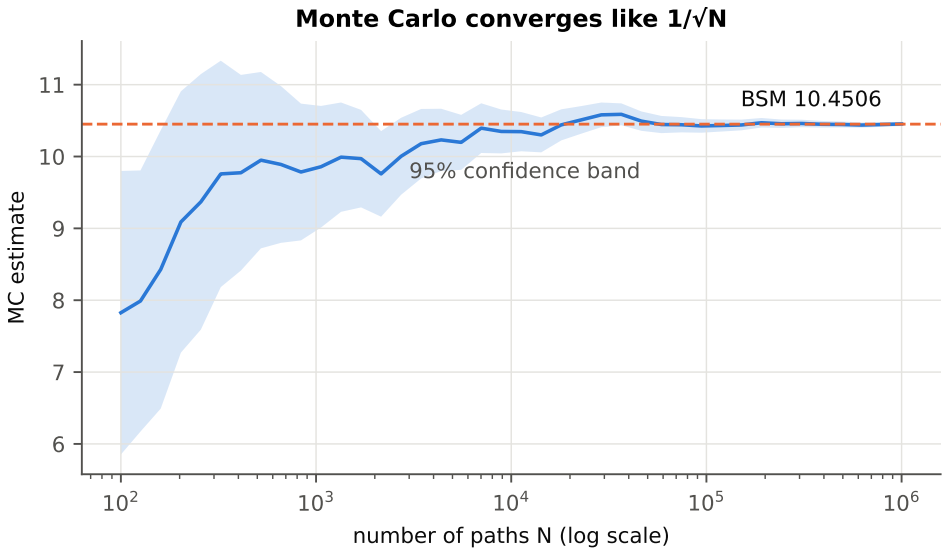


Figure 14. Monte Carlo estimate (blue) and its 95% confidence band against the number of paths. The band narrows like $1/\sqrt{N}$ and always contains the true price.

```
    rng = np.random.default_rng(seed)
    Z = rng.standard_normal(N)
    ST = S*np.exp((r - q - 0.5*sigma**2)*T + sigma*np.sqrt(T)*Z)
    Y = np.exp(-r*T)*np.maximum(cp*(ST - K), 0)
    est, se = Y.mean(), Y.std(ddof=1)/np.sqrt(N)
    return est, se, (est - 1.96*se, est + 1.96*se)

est, se, ci = mc_european(100, 100, 1, 0.05, 0.2)
assert abs(est - 10.4506) < 3*se          # statistical test, not a fixed tolerance
```

64 Variance reduction

> **Correction to the original notes: variance reduction was only named**
>
> The notes listed four techniques. Here is how each one works, with numbers.

Rather than "run ten million paths", make each path more informative. All of the results below use the *same* 10,000 normal draws, so the comparison is fair.

64.1 Antithetic variates

For every Z, also use $-Z$, and average the pair: $\frac{1}{2}[f(Z) + f(-Z)]$. If $Z = +0.8$ gives a high price, -0.8 gives a low one, and their average is less noisy than two independent draws. It works best when the payoff is monotone in Z (vanilla calls and puts). It helps little for symmetric payoffs like straddles, where $f(Z)$ and $f(-Z)$ are similar.

64.2 Control variates

Find a quantity Y that is highly correlated with your payoff X and whose expectation you know exactly. Then use

$$\hat{V}_{CV} = \bar{X} - b\,(\bar{Y} - \mathbb{E}[Y]), \qquad b^* = \frac{\mathrm{Cov}(X,Y)}{\mathrm{Var}(Y)}.$$

When the simulation happens to overshoot on Y, it probably overshot on X too, so you correct for it. The variance falls by a factor $(1 - \rho_{XY}^2)$.

- **Simple control:** the discounted stock itself, $\mathbb{E}[e^{-rT}S_T] = S_0 e^{-qT}$.
- **The classic:** use the **geometric Asian** option, which has a closed form, as the control for the **arithmetic Asian**, which doesn't (Kemna–Vorst, 1990). The two averages are so highly correlated that the variance falls by orders of magnitude (§66).
- **Exotic vs vanilla:** as your notes put it, use a known vanilla price to steady the estimate of an unknown exotic.

64.3 Importance sampling

Shift the sampling distribution so that more paths land where the payoff is non-zero (deep OTM options, tail risk), then reweight each path by the likelihood ratio so the estimate stays unbiased. For a normal shift by μ: sample $X \sim \mathcal{N}(\mu, 1)$ and weight each path by $e^{-\mu X + \mu^2/2}$. For a deep OTM call, where 99% of plain paths pay zero, this can cut variance enormously.

64.4 Common random numbers (CRN)

When you compare two scenarios (base and bumped, for Greeks; or strategy A vs strategy B), use the *same* random numbers for both. The noise is common to both runs, so it cancels in the difference. This is essential for Monte Carlo Greeks (§65).

64.5 Moment matching and stratification

Rescale the sample Z's to have exactly mean 0 and variance 1, or force the sample mean of $e^{-rT}S_T$ to equal $S_0 e^{-qT}$. This introduces a tiny bias but removes first-order noise. **Stratified sampling** and Latin hypercube sampling force the draws to cover every quantile evenly.

Worked example: the same 10,000 normals, different estimators

Estimator	Price	SE	Comment
Plain MC	10.228	0.1455	1.5 SE below the truth: an unlucky sample
Antithetic (5,000 pairs)	10.396	0.1029	variance $\approx$ halved
Control variate (S_T)	10.407	0.0561	$\rho = 0.923$, $b = 0.670$; variance cut 6.7×
Moment matching	10.442	—	corrects the unlucky sample mean

A 6.7× variance reduction is equivalent to running 6.7 times as many paths, almost for free.

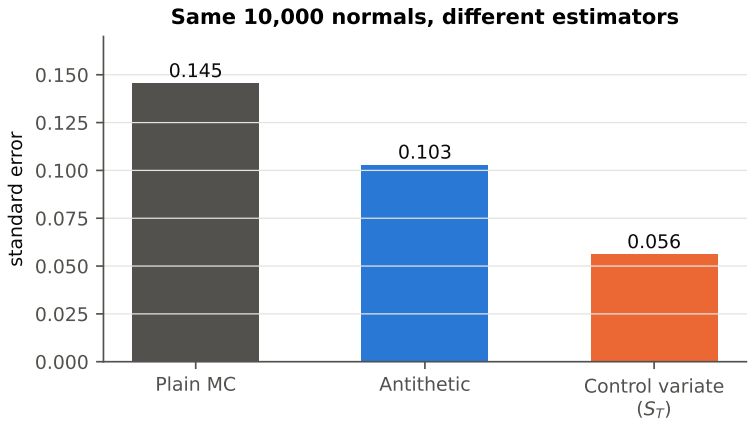


Figure 15. Standard error of three estimators on the same 10,000 normal draws.

65 Monte Carlo Greeks

> **Correction to the original notes: Monte Carlo Greeks were missing**
>
> Your project asks for "finite-difference Greeks" from Monte Carlo. There are three methods, and the naive one is dangerous.

65.1 Bump-and-revalue (finite differences)

$$\Delta \approx \frac{V(S+h) - V(S-h)}{2h}.$$

Always use common random numbers. With independent draws for the two prices, the noise in the difference is divided by $2h$ and explodes. With CRN it mostly cancels. Gamma by second differences is much noisier still, and digital or barrier payoffs need the payoff smoothed first.

65.2 Pathwise method

Differentiate inside the expectation. For a call, $\partial(S_T - K)^+/\partial S_0 = \mathbf{1}_{S_T>K}\, S_T/S_0$, so

$$\Delta = e^{-rT}\mathbb{E}\left[\mathbf{1}_{S_T>K}\, S_T/S_0\right], \qquad \nu = e^{-rT}\mathbb{E}\left[\mathbf{1}_{S_T>K}\, S_T\, \frac{\ln(S_T/S_0) - (r - q + \frac{1}{2}\sigma^2)T}{\sigma}\right].$$

It is unbiased and low-variance, but it needs a payoff that is continuous in the path. It fails for digitals (the derivative of a step is zero almost everywhere) and for gamma of a vanilla.

65.3 Likelihood-ratio (score) method

Differentiate the *density* instead of the payoff:

$$\Delta = e^{-rT}\mathbb{E}\left[\text{payoff} \times \frac{Z}{S_0\sigma\sqrt{T}}\right].$$

It works for discontinuous payoffs (digitals, barriers), and one set of paths serves any payoff. The price is higher variance. A good hybrid is to use pathwise for the smooth part of the payoff and LR for the kink.

Worked example: delta of the running-example call (true value 0.6368), 10,000 paths

Method	Estimate	SE
Bump-and-revalue, **with** CRN ($h = 1$)	0.6311	0.0057
Bump-and-revalue, **without** CRN	0.5633	0.1023
Pathwise	0.6312	0.0058
Likelihood ratio	0.6179	0.0144
Pathwise vega (true 37.52)	36.39	0.75

Without CRN the estimate is 18 times noisier and useless. Pathwise and CRN-bump are the most precise here; LR is noisier but would still work if the payoff were a digital. (All the estimates are about one SE low because they share the same slightly unlucky sample as the plain-MC price above.)

At industrial scale, banks use adjoint algorithmic differentiation (AAD; Giles and Glasserman, "Smoking adjoints", 2006), which computes *all* Greeks for a small constant multiple of the cost of one price.

Check yourself

- Why does Monte Carlo converge like $1/\sqrt{N}$? How many more paths do you need to cut the error by 10?
- Write the 95% CI for an estimate of 10.42 with SE 0.05. Does it contain the BSM price?
- Explain why antithetic variates help for a call but not much for a straddle.
- What makes the geometric Asian a good control variate for the arithmetic Asian?
- Why must bump-and-revalue Greeks use common random numbers? When does the pathwise method fail?

PART X

Exotic Options and Finite Differences

66 Asian options

66.1 What and why

An Asian option's payoff depends on an **average** price over monitoring dates $t_1, \dots, t_n$:

$$\text{Average-price call: } \max(\bar{S} - K, 0), \qquad \bar{S} = \frac{1}{n}\sum_{i=1}^{n} S_{t_i}.$$

> **Worked example: from the notes**
>
> Simulated prices on four dates: 100, 105, 110, 95. Arithmetic average $\bar{S} = 410/4 = 102.5$. With $K = 100$ the payoff is $\max(102.5 - 100, 0) = \mathbf{2.5}$. A European call on the same path would pay $\max(95 - 100, 0) = 0$. The average remembers the good days; S_T does not.

So a European option needs only S_T, but an Asian needs $S_{t_1}, S_{t_2}, \dots, S_{t_n}$: the *path*. That is why you cannot just plug the average into BSM, and why the project prices Asians by Monte Carlo.

66.2 Arithmetic versus geometric averages

The arithmetic average $A = \frac{1}{n}\sum S_{t_i}$ is what contracts use, but a sum of lognormals is not lognormal, so there is no closed form. The **geometric** average $G = \left(\prod S_{t_i}\right)^{1/n}$ is a product of lognormals, so it *is* lognormal, and it has an exact formula. By the AM–GM inequality $G \le A$ on every path, so the geometric call is a lower bound for the arithmetic call. And because G and A move almost in lock-step, it is a superb control variate.

Geometric Asian, discrete monitoring. $\ln G$ is normal with

$$\mu_G = \ln S_0 + \left(r - q - \tfrac{1}{2}\sigma^2\right)\bar{t}, \qquad \sigma_G^2 = \frac{\sigma^2}{n^2}\sum_{i,j}\min(t_i, t_j),$$

and the call is $e^{-rT}\left[e^{\mu_G + \sigma_G^2/2}N(d_1) - KN(d_2)\right]$ with $d_1 = (\mu_G - \ln K + \sigma_G^2)/\sigma_G$ and $d_2 = d_1 - \sigma_G$. For *continuous* averaging this reduces to BSM with $\sigma_G = \sigma/\sqrt{3}$ and an adjusted carry $b_G = \frac{1}{2}(r - q - \sigma^2/6)$.

Turnbull–Wakeman (1991) approximation. Match the first two moments M_1, M_2 of the arithmetic average to a lognormal, then use Black-76 with $F = M_1$ and $\sigma_A^2 = \ln(M_2/M_1^2)/T$. Levy (1992) and Curran (1992) offer refinements.

Worked example: monthly-average Asian call, running example ($n = 12$, $K = 100$)

Method	Price	SE
Geometric Asian, closed form	5.9402	exact
Arithmetic Asian, plain MC (100,000 paths)	6.1258	0.0267
Arithmetic Asian, MC + Kemna–Vorst control	**6.1558**	**0.00074**
Turnbull–Wakeman approximation	6.1742	—
Continuous geometric (for comparison)	5.5468	exact
European call (for comparison)	10.4506	exact

The control variate (correlation 0.9996) cuts the standard error by a factor of 36, equivalent to running about 1,300 times as many paths. Turnbull–Wakeman overprices slightly (by 0.018) at 20% vol. Averaging has cut the option's value by about 41% relative to the European.

67 Lookback options

Worked example: from the notes

Path: $100 \to 110 \to 90 \to 120 \to 115$. The maximum is $S_{\max} = 120$ and the minimum is $S_{\min} = 90$. A floating-strike lookback call pays $S_T - S_{\min} = 115 - 90 = 25$: you "bought at the low". A fixed-strike lookback call with $K = 100$ pays $\max(S_{\max} - K, 0) = 20$: you "sold at the high".

- **Floating strike:** the call pays $S_T - \min_t S_t$, the put pays $\max_t S_t - S_T$. Always exercised.
- **Fixed strike:** the call pays $\max(\max_t S_t - K, 0)$, the put pays $\max(K - \min_t S_t, 0)$.

67.1 Closed forms (continuous monitoring, GBM)

With carry $b = r - q$, running minimum m and running maximum M (both equal to S at inception):

Floating-strike call (Goldman–Sosin–Gatto, 1979):

$$c = Se^{(b-r)T}N(a_1) - me^{-rT}N(a_2) + Se^{-rT}\frac{\sigma^2}{2b}\Big[\Big(\frac{S}{m}\Big)^{-2b/\sigma^2}N\Big(-a_1 + \frac{2b\sqrt{T}}{\sigma}\Big) - e^{bT}N(-a_1)\Big],$$

$$a_1 = \frac{\ln(S/m) + (b + \tfrac{1}{2}\sigma^2)T}{\sigma\sqrt{T}}, \qquad a_2 = a_1 - \sigma\sqrt{T}.$$

Floating-strike put:

$$p = Me^{-rT}N(-b_2) - Se^{(b-r)T}N(-b_1) + Se^{-rT}\frac{\sigma^2}{2b}\Big[-\Big(\frac{S}{M}\Big)^{-2b/\sigma^2}N\Big(b_1 - \frac{2b\sqrt{T}}{\sigma}\Big) + e^{bT}N(b_1)\Big],$$

with $b_1 = \frac{\ln(S/M) + (b + \frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}}$ and $b_2 = b_1 - \sigma\sqrt{T}$.

Fixed-strike call (Conze–Viswanathan, 1991), for $K \geq M$:

$$c = Se^{(b-r)T}N(d_1) - Ke^{-rT}N(d_2) + Se^{-rT}\frac{\sigma^2}{2b}\Big[-\Big(\frac{S}{K}\Big)^{-2b/\sigma^2}N\Big(d_1 - \frac{2b\sqrt{T}}{\sigma}\Big) + e^{bT}N(d_1)\Big],$$

with the usual BSM d_1, d_2. The first two terms are a vanilla call; the bracket is the extra value of hindsight.

Worked example: lookbacks on the running example

Option	Closed form	MC, daily monitoring	MC daily, BGK-shifted
Floating-strike call	17.2168	16.66	17.27
Floating-strike put	14.2906	13.47	14.31
Fixed-strike call, $K = 100$	19.1676	18.41	19.25

(100,000 paths, SE about 0.07. A 2,520-step run gives 17.21 and 19.13, matching the closed forms.) Two lessons. First, lookbacks are expensive: the floating call costs 65% more than the vanilla. Second, **monitoring frequency matters**. Daily monitoring misses the true continuous extremes, so it undervalues by 0.5 to 0.8. Shifting the simulated extremes by $e^{\pm 0.5826\sigma\sqrt{\Delta t}}$ (the Broadie–Glasserman–Kou correction, §69) closes most of the gap.

68 Basket options and correlation

A basket option depends on several assets. With $S_1 = 100$, $S_2 = 200$ and weights 0.5 each, $B = 0.5(100) + 0.5(200) = 150$, and a basket call pays $\max(B_T - K, 0)$. The complication is that the assets don't move independently, and the basket's volatility depends on how they co-move:

$$\sigma_B^2 \approx w_1^2\sigma_1^2 + w_2^2\sigma_2^2 + 2w_1w_2\rho\,\sigma_1\sigma_2 \quad \text{(in value-weighted terms)}.$$

68.1 Correlation

$\rho = 0.8$: the stocks usually rise and fall together. $\rho \approx 0$: little linear relationship. $\rho < 0$: they tend to move in opposite directions. Higher correlation means less diversification, a more volatile basket, and a more valuable basket call.

Worked example: correlation sensitivity of a basket call

$S_1 = 100$ ($\sigma_1 = 20\%$), $S_2 = 200$ ($\sigma_2 = 30\%$), weights 0.5/0.5, $K = 150$, $T = 1$, $r = 5\%$; 200,000 paths, SE about 0.05.

ρ	-0.5	0	0.3	0.5	0.8	1.0
Basket call	14.19	16.23	17.29	17.95	18.89	19.48

Moving from independent to perfectly correlated assets adds 20% to the price. That is why your project asks for a correlation sensitivity: correlation is a genuine, hard-to-hedge risk factor. (For "worst-of" products the sign flips: higher correlation makes them *cheaper*.)

68.2 Generating correlated paths: Cholesky

Given a correlation matrix C, factor it as $C = LL^\top$ with L lower-triangular (`np.linalg.cholesky`). Draw independent $\varepsilon \sim \mathcal{N}(0, I)$ and set $Z = L\varepsilon$. Then Z has correlation C. For two assets this is simply

$$Z_1 = \varepsilon_1, \qquad Z_2 = \rho\varepsilon_1 + \sqrt{1-\rho^2}\,\varepsilon_2.$$

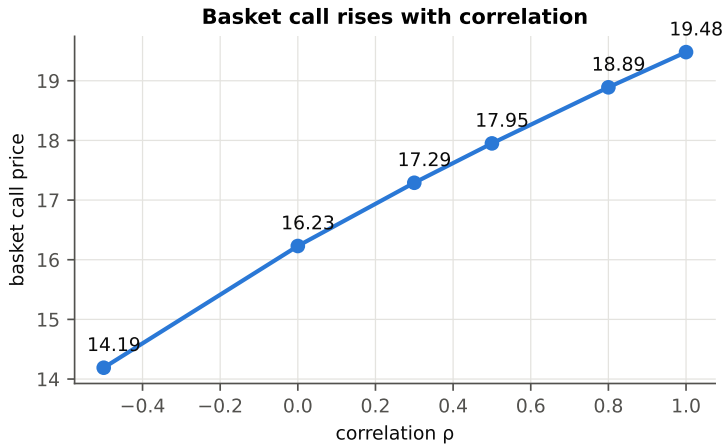


Figure 16. Basket call price against correlation.

Pitfalls & conventions

- Cholesky fails if C is not positive definite, which estimated or hand-edited matrices often aren't. Repair with a nearest-correlation-matrix projection (Higham) or by clipping negative eigenvalues.
- Estimate correlation from synchronous log returns. Non-synchronous closes (NIFTY vs S&P 500) bias it towards zero.
- Correlations jump towards 1 in sell-offs, exactly when you need diversification most.
- Correlation can be *implied* from index-versus-component option prices; trading the difference is called dispersion trading.

```
def basket_call_mc(S, w, sig, corr, K, T, r, N=200_000, seed=5):
    rng = np.random.default_rng(seed)
    L = np.linalg.cholesky(corr)                    # corr = L @ L.T
    Z = rng.standard_normal((N, len(S))) @ L.T      # correlated normals
    ST = S*np.exp((r - 0.5*sig**2)*T + sig*np.sqrt(T)*Z)
    Y = np.exp(-r*T)*np.maximum(ST @ w - K, 0)
    return Y.mean(), Y.std(ddof=1)/np.sqrt(N)

S=np.array([100.,200.]); w=np.array([.5,.5]); sig=np.array([.2,.3])
print(basket_call_mc(S, w, sig, np.array([[1,.5],[.5,1]]), 150, 1, .05))  # ~17.95
```

69 Barrier options

69.1 Up/down and in/out

A barrier is a price level H that, if touched, switches the option on or off.

- **Up** barrier: $H > S_0$. **Down** barrier: $H < S_0$.
- **Knock-out:** the option dies when the barrier is hit (sometimes paying a small rebate).
- **Knock-in:** the option comes alive only when the barrier is hit.

That gives eight vanilla barriers: up/down × in/out × call/put. The key point, again: *the path matters, not merely* S_T.

Worked example: a down-and-out call (from the notes)

$S_0 = 100$, $K = 100$, $H = 60$. The option stays alive only if $S_t > 60$ throughout. Payoff: $(S_T - K)^+ \mathbf{1}\{\min_{t \le T} S_t > H\}$.

- **Path A** $100 \to 120 \to 140 \to 160$: never touches 60, so the payoff is $160 - 100 = 60$.
- **Path B** $100 \to 110 \to 90 \to 120$: never touches 60, so the payoff is 20.
- **Path C** $100 \to 80 \to 60 \to 90 \to 150$: touches 60, knocked out, payoff **0**, even though $S_T = 150$.
- **Path D** $100 \to 90 \to 80 \to 70 \to 80$: alive, but $S_T = 80 < K$, so the payoff is 0.

Paths C and A both finish high. Only the route differs, and the route decides.

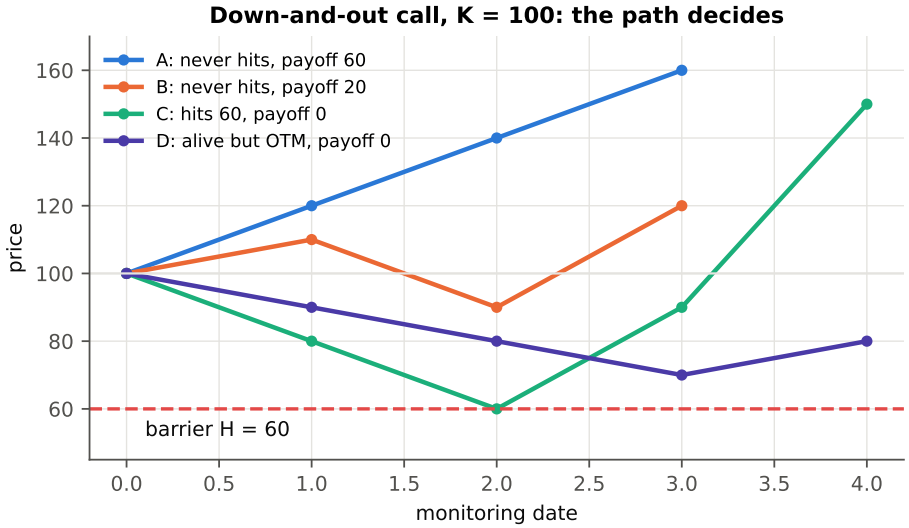


Figure 17. The four paths of the down-and-out call example.

69.2 In–out parity

For the same strike, barrier and expiry (and no rebate):

$$\boxed{\text{knock-in} + \text{knock-out} = \text{vanilla}}.$$

Every path either hits the barrier (and the knock-in pays the vanilla payoff) or doesn't (and the knock-out pays it). Price one and you get the other free, and you have an excellent test.

69.3 Closed forms (Reiner–Rubinstein, 1991)

Under continuous monitoring and GBM, all eight types are combinations of four building blocks. With $\mu = (b - \frac{1}{2}\sigma^2)/\sigma^2$, $b = r - q$ and $\phi = \pm 1$ for call/put, $\eta = \pm 1$ for down/up:

$$x_1 = \frac{\ln(S/K)}{\sigma\sqrt{T}} + (1+\mu)\sigma\sqrt{T}, \qquad x_2 = \frac{\ln(S/H)}{\sigma\sqrt{T}} + (1+\mu)\sigma\sqrt{T},$$
$$y_1 = \frac{\ln(H^2/(SK))}{\sigma\sqrt{T}} + (1+\mu)\sigma\sqrt{T}, \qquad y_2 = \frac{\ln(H/S)}{\sigma\sqrt{T}} + (1+\mu)\sigma\sqrt{T},$$

$$
\begin{aligned}
A &= \phi S e^{(b-r)T} N(\phi x_1) - \phi K e^{-rT} N(\phi x_1 - \phi\sigma\sqrt{T}),\\
B &= \phi S e^{(b-r)T} N(\phi x_2) - \phi K e^{-rT} N(\phi x_2 - \phi\sigma\sqrt{T}),\\
C &= \phi S e^{(b-r)T} (H/S)^{2(\mu+1)} N(\eta y_1) - \phi K e^{-rT} (H/S)^{2\mu} N(\eta y_1 - \eta\sigma\sqrt{T}),\\
D &= \phi S e^{(b-r)T} (H/S)^{2(\mu+1)} N(\eta y_2) - \phi K e^{-rT} (H/S)^{2\mu} N(\eta y_2 - \eta\sigma\sqrt{T}).
\end{aligned}
$$

For calls ($\phi = 1$), no rebate:

Type	$K > H$	$K < H$
Down-and-out ($\eta = 1$)	$A - C$	$B - D$
Down-and-in ($\eta = 1$)	C	$A - B + D$
Up-and-out ($\eta = -1$)	0	$A - B + C - D$
Up-and-in ($\eta = -1$)	A	$B - C + D$

(For puts and rebates, see Haug's *Complete Guide to Option Pricing Formulas*. The `(H/S)` reflection terms come from the reflection principle for Brownian motion.)

Worked example: barrier calls on the running example (continuous monitoring)

Option	Price	Check
Up-and-out call, $H = 120$	1.1761	
Up-and-in call, $H = 120$	9.2745	$1.1761 + 9.2745 = 10.4506 =$ vanilla ✓
Down-and-out call, $H = 90$	8.6655	
Down-and-in call, $H = 90$	1.7851	sum $= 10.4506$ ✓
Down-and-out call, $H = 60$	10.4506	barrier so far away it is the vanilla ✓
Up-and-out call, $H = 200$	10.3658	approaching the vanilla as $H \to \infty$ ✓

The up-and-out call costs only 11% of the vanilla. It dies in exactly the scenarios where the vanilla pays the most.

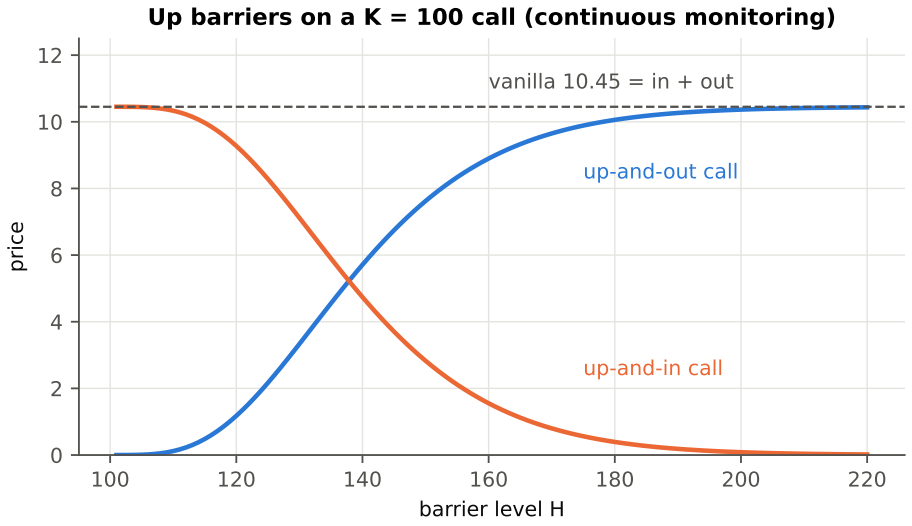


Figure 18. Up-and-out and up-and-in call prices against the barrier level. They always sum to the vanilla, and the knock-out approaches the vanilla as the barrier moves away.

69.4 Discrete monitoring: why barriers are computationally hard

Suppose you simulate daily prices 100, 110, 119, 115 with a barrier at 120. You conclude the barrier was not hit. But the continuous path could have gone $119 \to 121 \to 115$ between day 3 and day 4. Daily monitoring *misses* crossings, so analytic (continuous) formulas and discretely monitored contracts give different prices. A discretely monitored knock-out is worth *more* than its continuous counterpart, because it is harder to knock out.

The Broadie–Glasserman–Kou correction (1997). Price the discrete barrier with the continuous formula, but shift the barrier away from spot by $e^{\beta\sigma\sqrt{\Delta t}}$, with

$$\beta = -\frac{\zeta(1/2)}{\sqrt{2\pi}} \approx 0.5826$$

(ζ is the Riemann zeta function). Use $He^{+\beta\sigma\sqrt{\Delta t}}$ for up barriers and $He^{-\beta\sigma\sqrt{\Delta t}}$ for down barriers.

Worked example: daily monitoring of the up-and-out call

$\Delta t = 1/252$: shift factor $e^{0.5826\times 0.2\times 0.0630} = 1.00734$, so $H = 120$ acts like 120.88.

Continuous-monitoring formula, $H = 120$	1.1761
MC, daily monitoring (100,000 paths)	1.3290 ± 0.011
BGK: continuous formula at $H = 120.88$	1.3321
MC with Brownian-bridge crossing correction	1.1771 ± 0.010

Daily monitoring is worth 13% more than continuous, and BGK captures it almost exactly. The correction degrades when the barrier is very close to spot.

Monte Carlo for continuous barriers. Naively checking only the simulated dates is biased. Between two simulated points S_i and S_{i+1} that are both below an up barrier, the probability that the continuous path crossed is

$$\mathbb{P}(\text{cross}) = \exp\Big(-\frac{2\ln(H/S_i)\ln(H/S_{i+1})}{\sigma^2\Delta t}\Big).$$

Multiply the payoff by the product of $(1-\mathbb{P})$ across steps (the Brownian-bridge correction). The table above shows it recovering the continuous price.

Pitfalls & conventions: barrier Greeks and numerics

- Near the barrier, delta can flip sign and gamma spikes: a knock-out call just below an up barrier has *negative* delta, because a rise may kill it.
- Pathwise MC Greeks fail at the barrier discontinuity; use LR or smoothing.
- In trees and grids, place nodes *exactly on the barrier*. Misaligned barriers cause large, oscillating errors.

70 PDEs and finite-difference methods

70.1 The PDE

The Black–Scholes PDE (§46),

$$\frac{\partial V}{\partial t} + \tfrac{1}{2}\sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} + (r-q)S\frac{\partial V}{\partial S} - rV = 0,$$

describes how a derivative's value evolves across time and stock price. Different products are the same PDE with different terminal conditions (the payoff) and boundary conditions (what happens at the edges, such as a barrier). Many products have no closed form, so we solve the PDE numerically on a grid.

70.2 The grid and the finite differences

Set up a grid $S_j = j\Delta S$ $(j = 0, \dots, M)$ and time-to-expiry $\tau_n = n\Delta\tau$ $(n = 0, \dots, N)$, and write $V_j^n \approx V(S_j, \tau_n)$. Replace derivatives with differences (each follows from a Taylor expansion):

$$\frac{\partial V}{\partial S} \approx \frac{V_{j+1} - V_{j-1}}{2\Delta S}, \qquad \frac{\partial^2 V}{\partial S^2} \approx \frac{V_{j+1} - 2V_j + V_{j-1}}{(\Delta S)^2}, \qquad \frac{\partial V}{\partial \tau} \approx \frac{V_j^{n+1} - V_j^n}{\Delta\tau}.$$

In time-to-expiry $\tau = T - t$ the PDE becomes $V_\tau = \frac{1}{2}\sigma^2 S^2 V_{SS} + (r - q)SV_S - rV$, solved *forwards* in τ from the payoff at $\tau = 0$. A continuous maths problem has become a linear-algebra problem.

70.3 Three schemes

The θ-scheme evaluates the spatial terms as a blend of the old and new time levels: $\theta = 0$ is explicit, $\theta = 1$ implicit, $\theta = \frac{1}{2}$ Crank–Nicolson.

- **Explicit (forward Euler, $\theta = 0$):** each new value comes directly from three old ones. Simple, but only **conditionally stable**: roughly $\Delta\tau \le (\Delta S)^2/(\sigma^2 S_{\max}^2)$. Refine the price grid and you must shrink the time step quadratically (a CFL-type condition). The explicit scheme is equivalent to a trinomial tree.
- **Implicit (backward Euler, $\theta = 1$):** solve a tridiagonal linear system each step (Thomas algorithm, $O(M)$). **Unconditionally stable**, first-order accurate in time.
- **Crank–Nicolson ($\theta = \frac{1}{2}$):** the average of the two. Second-order in time and unconditionally stable, *but* it does not damp high-frequency errors. A non-smooth payoff (the kink at K, a digital, a barrier) creates spurious oscillations, especially in gamma.

Worked example: stability, made visible

Explicit scheme for the running-example call with $M = 100$ price steps ($S_{\max} = 400$). The stability limit is $\Delta\tau \le 1/(\sigma^2 M^2) \approx 0.0025$, i.e. $N \ge 392$ time steps:

N (time steps)	Explicit price
100	−19,578,224 (blown up)
200	−74,848,516,321 (blown up)
350	10.4138
400	10.4134

Below the limit, rounding errors are amplified at every step until they swamp everything. (At $N = 350$ the instability has not yet had time to grow.) The implicit and CN schemes never do this.

Worked example: convergence of implicit vs Crank–Nicolson ($S_{\max} = 400$)

Grid $M \times N$	Implicit error	CN error	CN error ratio
100×100	-0.0505	-0.0398	
200×200	-0.0152	-0.0099	4.0
400×400	-0.0051	-0.0025	4.0

Halving both steps cuts CN's error by 4: second-order convergence, as theory predicts. This ratio test is exactly how you validate a PDE solver.

70.4 Rannacher smoothing

The fix for Crank–Nicolson's oscillations is **Rannacher smoothing**: replace the first one or two CN steps with four half-size implicit steps, which damp the kink, then continue with CN. It works well for European options, producing smooth Greeks and restoring good convergence. A 2021 study notes that it is less effective for American options. For barriers, re-smooth after each discrete monitoring date.

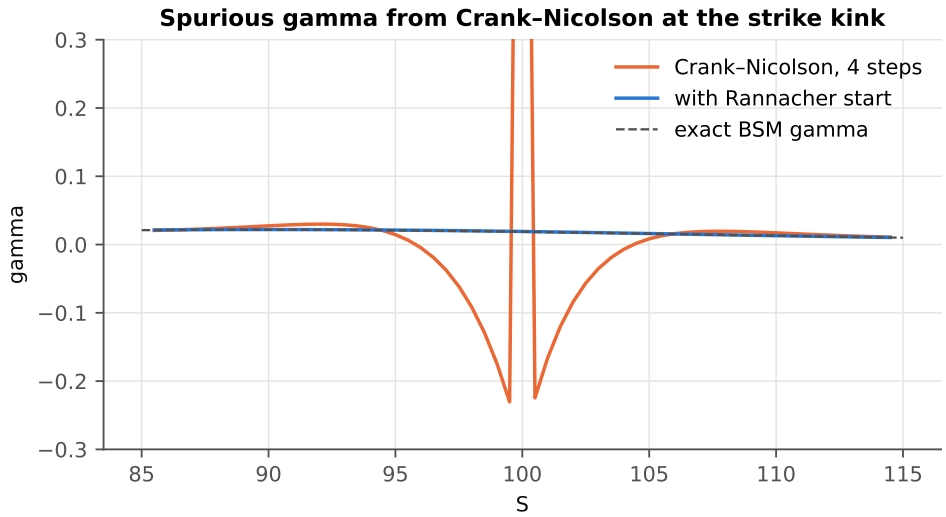


Figure 19. Gamma near the strike from a coarse Crank–Nicolson run (4 time steps over a year, $\Delta S = 0.5$). Plain CN produces a spike of 1.70 and negative gammas; two Rannacher start-up steps give a gamma that tracks the exact BSM curve.

70.5 Boundary conditions and grid design

- **Call:** $V(0,\tau) = 0$; $V(S_{\max},\tau) \approx S_{\max}e^{-q\tau} - Ke^{-r\tau}$, or impose linearity $V_{SS} = 0$.
- **Put:** $V(0,\tau) = Ke^{-r\tau}$; $V(S_{\max},\tau) = 0$.
- **Up-and-out barrier:** end the grid *at* H and impose $V(H,\tau) = 0$ (or the rebate). This is exactly why PDEs suit barriers: the barrier is just a boundary.
- $S_{\max}$: three to five standard deviations above the forward.
- Log-space ($x = \ln S$) gives constant coefficients. Concentrate nodes near K and the barrier (sinh-stretched grids), and place K consistently on a node or a midpoint.

- **American options:** apply $V = \max(V, \text{payoff})$ after each step (simple), or use projected SOR or the Brennan–Schwartz solver for a properly implicit treatment. An implicit 400×400 grid gives 6.0814 for the American put (tree: 6.0903).

Worked example: the up-and-out call by Crank–Nicolson (grid ending at $H = 120$)

Grid $M \times N$	Price (exact 1.17607)
60×100	1.16808
120×200	1.17407
240×400	1.17557
480×800	1.17594

Clean, monotone convergence to the Reiner–Rubinstein value, because the barrier sits exactly on the grid edge. That is the payoff from matching method to product.

```
from scipy.linalg import solve_banded

def fd_uo_call(S0, K, H, T, r, sigma, M=240, N=400, rannacher=2):
    """Crank-Nicolson for an up-and-out call; grid ends at the barrier H."""
    S = np.linspace(0, H, M + 1); dt = T/N; j = np.arange(1, M)
    a = 0.5*(sigma**2*j**2 - r*j); b = -(sigma**2*j**2 + r); c = 0.5*(sigma**2*j**2 + r*j)
    V = np.maximum(S - K, 0.0); V[-1] = 0.0                  # payoff; knocked out at H
    def step(V, th, h):
        rhs = V[1:-1] + (1-th)*h*(a*V[:-2] + b*V[1:-1] + c*V[2:])
        ab = np.zeros((3, M-1))
        ab[0,1:] = -th*h*c[:-1]; ab[1] = 1 - th*h*b; ab[2,:-1] = -th*h*a[1:]
        Vn = np.zeros_like(V); Vn[1:-1] = solve_banded((1,1), ab, rhs)
        return Vn                                            # V(0)=0, V(H)=0
    for n in range(N):
        if n < rannacher:                                    # implicit half steps
            V = step(step(V, 1.0, dt/2), 1.0, dt/2)
        else:
            V = step(V, 0.5, dt)                             # Crank-Nicolson
    return np.interp(S0, S, V)

print(fd_uo_call(100, 100, 120, 1, 0.05, 0.2))   # ~1.1756 (exact 1.1761)
```

Check yourself

- Why does an Asian option need the whole path? Why is the geometric Asian an excellent control variate?
- Price a floating-strike lookback by MC with daily and with 10-step-a-day monitoring. Explain the difference.
- Show with Cholesky how to generate two normals with correlation -0.3.
- State in–out parity and use it to test your barrier engine.
- Why does daily monitoring make a knock-out *more* valuable? What is β and how is it used?
- What is the stability condition of the explicit scheme? Why does Crank–Nicolson oscillate, and how does Rannacher fix it?

- Why are PDEs natural for barrier options?

PART XI

Risk: Scenarios, Taylor Approximations, P&L Explain, Strategies and Hedging

71 Scenario analysis and stress testing

A pricing engine should answer one question beyond "what is it worth?": *"What happens to my position if the market moves?"*

Worked example: the simplest scenario (from the notes)

An option is worth ₹10 with $S = 100$ and $\Delta = 0.60$. If spot moves to 105, a first-order estimate is $\Delta V \approx 0.60 \times 5 = 3$, so $V \approx 13$. That is a delta approximation. The rest of this section is about doing better.

71.1 The project's shock set

Your project specifies:

- **spot:** $\pm 1\%$, $\pm 5\%$, $\pm 10\%$;
- **volatility:** ± 5 and ± 10 vol points;
- **rates:** ± 50 and ± 100 basis points;
- **time decay:** 1, 7 and 30 days;
- **basket correlation** shocks and **barrier-level** shocks.

Worked example: a spot $\times$ vol grid for the long running-example call (full revaluation, P&L in ₹)

Vol shock \ Spot	-10%	-5%	-1%	0	$+1\%$	$+5\%$	$+10\%$
-10 pts	-8.77	-6.72	-4.34	-3.65	-2.92	$+0.28$	$+4.76$
-5 pts	-7.11	-4.83	-2.51	-1.86	-1.19	$+1.72$	$+5.78$
0	-5.36	-2.94	-0.63	0.00	$+0.65$	$+3.41$	$+7.21$
$+5$ pts	-3.58	-1.06	$+1.27$	$+1.89$	$+2.52$	$+5.20$	$+8.85$
$+10$ pts	-1.79	$+0.82$	$+3.16$	$+3.78$	$+4.41$	$+7.05$	$+10.61$

Other shocks: one day of decay -0.018, seven days -0.123, thirty days -0.535; rates -100bp -0.526, -50bp -0.264, $+50$bp $+0.268$, $+100$bp $+0.539$. The worst cell (-10% spot, -10 vol) loses 84% of the premium. That is the number a margin system would focus on.

71.2 Beyond the grid

- **Historical stresses:** replay real episodes (2008, March 2020, local events, 19 October 1987, when the S&P 500 futures fell 29%), including the accompanying vol and skew moves, not just spot.
- **Hypothetical stresses:** a weekend or holiday gap; correlation going to 1; a liquidity shock where bid-ask spreads widen.
- **Reverse stress test:** "what move would lose ₹X?"
- Report the **shape**, not just the worst number. A short-gamma book looks harmless for small moves and terrible for large ones.

72 Delta and delta–gamma approximations

The second-order Taylor expansion of option value in spot:

$$\boxed{\Delta V \approx \Delta \cdot \Delta S + \tfrac{1}{2}\Gamma(\Delta S)^2}$$

Delta alone treats the option as if it were a straight line. Options are curved, and gamma adds the curvature.

Worked example: from the notes

$\Delta = 0.60$, $\Gamma = 0.04$, $\Delta S = 5$. First order: $0.60 \times 5 = 3$. Second order: $\frac{1}{2}(0.04)(25) = 0.5$. Total $\Delta V \approx 3.5$, instead of just 3.

Worked example: against full revaluation, running example

For $S : 100 \to 105$: delta only 3.184; delta–gamma $3.184 + \frac{1}{2}(0.018762)(25) = \mathbf{3.419}$; full revaluation $C(105) - C(100) = 13.858 - 10.451 = \mathbf{3.407}$.

ΔS	Full revaluation	Delta only	Delta + gamma
-20	-8.591	-12.737	-8.984
-10	-5.359	-6.368	-5.430
-5	-2.940	-3.184	-2.950
-1	-0.627	-0.637	-0.627
$+1$	$+0.646$	$+0.637$	$+0.646$
$+5$	$+3.407$	$+3.184$	$+3.419$
$+10$	$+7.212$	$+6.368$	$+7.306$
$+20$	$+15.718$	$+12.737$	$+16.489$

For small moves delta–gamma is excellent. At $\pm 20\%$ it is off by 0.4 to 0.8 (third-order "speed" effects). That is why stress tests use full revaluation, and the Taylor version is reserved for quick explain and VaR approximations.

The fuller expansion includes the other risk factors:

$$\Delta V \approx \Delta\,\delta S + \tfrac{1}{2}\Gamma\,\delta S^2 + \nu\,\delta\sigma + \Theta\,\delta t + \rho\,\delta r + \text{vanna}\,\delta S\,\delta\sigma + \tfrac{1}{2}\text{volga}\,\delta\sigma^2.$$

Delta–gamma VaR plugs a distribution for δS into this expansion (using Cornish–Fisher or Monte Carlo) to get a fast VaR for an options book.

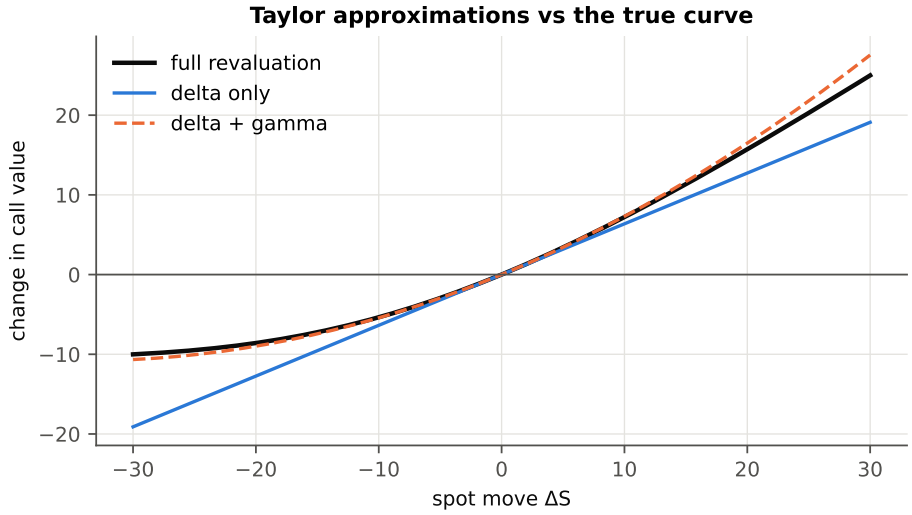


Figure 20. Change in call value against spot move: the true curve, the delta tangent line, and the delta–gamma parabola. Your project asks for exactly this plot.

73 P&L attribution (P&L explain)

Correction to the original notes: listed in the notes, never covered

P&L attribution is the daily discipline of explaining *why* a book made or lost money, factor by factor. It is how desks prove they understand their risk.

Method. Actual P&L $= V(S_1, \sigma_1, t_1, r_1) - V(S_0, \sigma_0, t_0, r_0)$. The explained P&L is the Taylor expansion above, evaluated with yesterday's Greeks and today's market moves. The unexplained residual is actual minus explained. A small, random residual means the risk model is capturing what matters. A large or persistent residual means a missing risk factor (skew moves, dividends), stale data, a booking error, or a convention bug such as using trading days in one place and calendar days in another.

Worked example: one day on the running-example call

Spot 100 $\to$ 101, implied vol 20% $\to$ 21%, one calendar day passes.

Component	₹	Calculation
Delta	$+0.6368$	0.6368×1
Gamma	$+0.0094$	$\frac{1}{2} \times 0.018762 \times 1^2$
Vega	$+0.3752$	37.524×0.01
Theta	-0.0176	$-6.414/365$
Vanna	-0.0028	$-0.2814 \times 1 \times 0.01$
Volga	$+0.0005$	$\frac{1}{2} \times 9.850 \times 0.01^2$
Explained	$+1.0016$	
Actual (full revaluation)	$+1.0006$	$C(101,\ 21\%,\ 364/365) = 11.4512$
Unexplained	$\mathbf{-0.0009}$	less than 0.1% of the move

Spot and vol together explain 99% of the day. (An earlier draft of these notes quoted an actual P&L of 1.0013 and a residual of $+0.0003$; recomputing in code gives 1.0006 and -0.0009. Both are tiny, but small slips like that are exactly what a P&L-explain process exists to catch.)

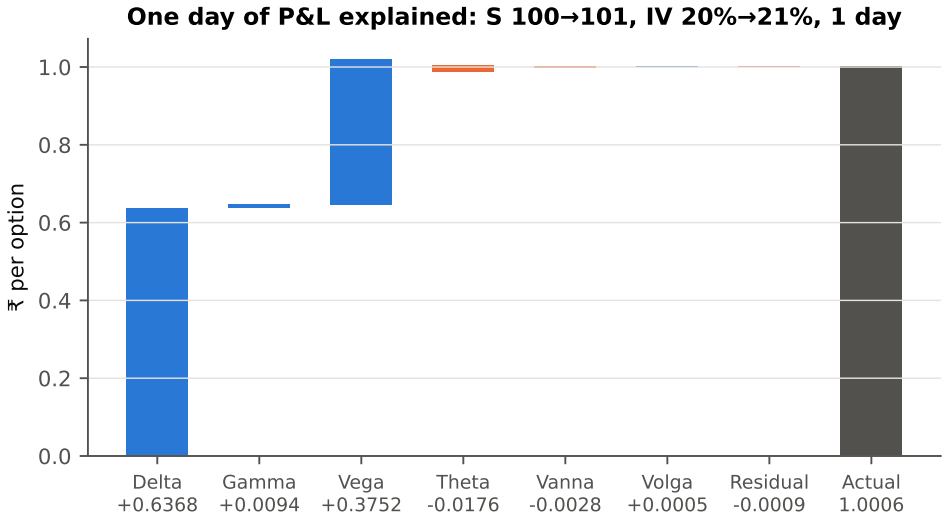


Figure 21. The day's P&L as a waterfall of Greek contributions.

Two flavours. *Risk-based* explain uses the Taylor expansion, as above. *Step-wise revaluation* moves one factor at a time and fully reprices; it has no Taylor error, but the result depends on the order of the steps, so fix and document the order.

Regulatory context. Under the Basel framework (MAR32, in force from 1 January 2023), trading desks must pass a P&L attribution test comparing "risk-theoretical" and "hypothetical" P&L using a Spearman correlation and a Kolmogorov–Smirnov test. A desk falls into the red zone if the correlation is below 0.7 or the KS statistic is above 0.12.

74 Option strategies

Individual options combine into portfolio payoffs that express a precise view on direction, volatility, or both. All premiums below are Black–Scholes values for the running example; P&L includes the premium carried forward at 5%.

74.1 Bull call spread

Buy a call at $K_1 = 100$, sell a call at $K_2 = 110$, same expiry. You are bullish, but your upside is capped at K_2. Selling the upper call cheapens the trade. Net debit $10.4506 - 6.0401 = 4.4105$ (future value 4.637). Maximum gain $10 - 4.637 = 5.363$ above 110; maximum loss 4.637 below 100; break-even 104.64. The **bear put spread** (long the higher-strike put, short the lower) is the mirror image.

74.2 Long straddle

Buy a call and a put at the same strike. You profit from a large move in *either* direction: you are trading volatility, not direction.

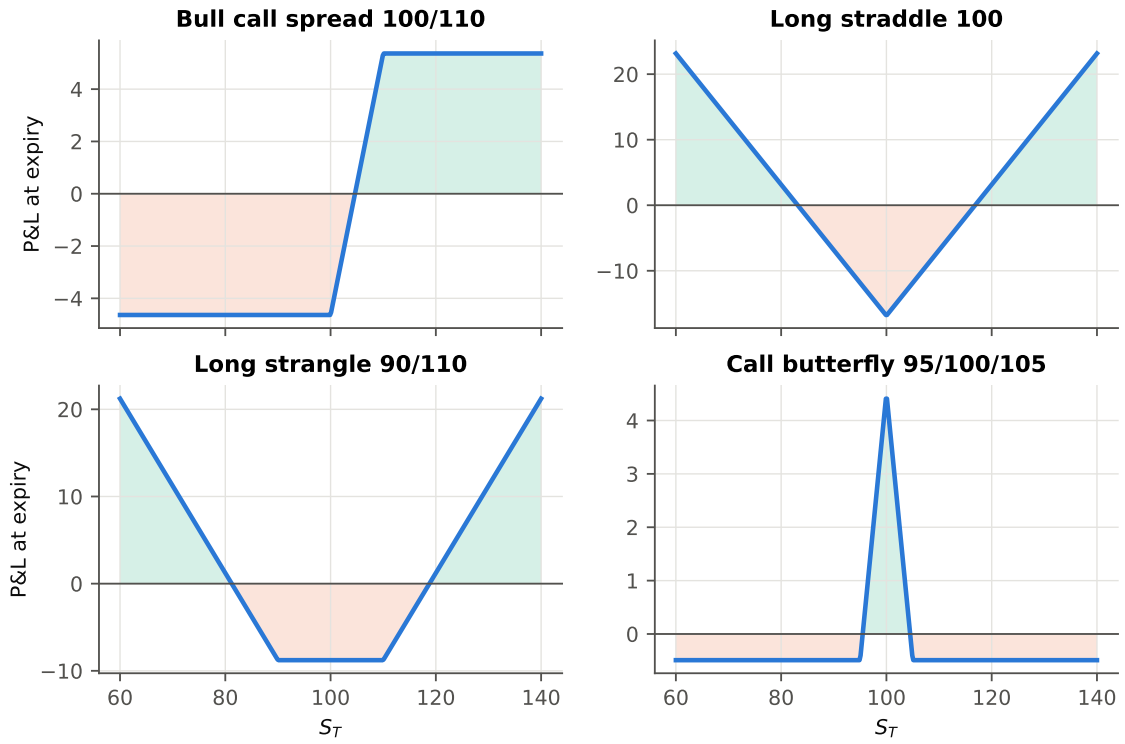


Figure 22. Expiry P&L of four classic strategies (premiums from BSM, $T = 1$).

Worked example: the notes' straddle

$K = 85$, call premium 3, put premium 2, total 5. Break-evens: $85 - 5 = \mathbf{80}$ and $85 + 5 = \mathbf{90}$. Maximum loss: ₹5, at $S_T = 85$.

S_T	70	75	80	85	90	95	100
P&L	+10	+5	0	−5	0	+5	+10

The running-example ATM straddle costs 16.02 (future value 16.85), so the stock must move more than about 17% either way in a year to profit. In Greek terms, a long straddle is long gamma and long vega, and short theta.

74.3 Long strangle

Buy an OTM put and an OTM call (e.g. 90 put and 110 call). It is cheaper than a straddle ($2.3101 + 6.0401 = 8.35$, against 16.02), but it needs a bigger move to pay: break-evens at $90 - 8.78 = 81.2$ and $110 + 8.78 = 118.8$. The notes' version with strikes 80 and 90 works the same way.

74.4 Butterfly

$$+C(K_1) - 2C(K_2) + C(K_3), \qquad K_1 < K_2 < K_3 \text{ (equally spaced)}.$$

Maximum profit is at $S_T = K_2$, and the loss is capped at the small debit. With 95/100/105: cost 0.4667, maximum profit $5 - 0.4906 = 4.51$ at 100. This is a bet that the market will *pin* near the middle strike (short vol). A deeper fact: as the wing width shrinks, the butterfly's price divided by the width squared tends to e^{-rT} times the risk-neutral density at K_2. Butterflies are how the market prices the probability of each outcome (Breeden–Litzenberger).

74.5 More shapes

- **Iron condor:** a short strangle plus long further-OTM wings (e.g. −P90 +P85 −C110 +C115). A range-bound view with defined risk: credit 2.56, maximum loss about 2.31.
- **Covered call:** long stock, short an OTM call. You collect income (6.04 for the 110 call) and give up upside above 110. Mildly bullish.
- **Protective put:** long stock, long a put. You keep the upside and insure a floor at $K-$premium (a 90 put costs 2.31).
- **Synthetics:** long call + short put = long forward (parity).

The Greeks lens. A strategy's Greeks are the sums of its legs' Greeks. Long straddles and strangles are long gamma and vega and short theta. A butterfly near its body is the reverse. A bull spread has positive delta, and its vega changes sign depending on where spot sits between the strikes.

75 Hedging

75.1 What is hedging?

Hedging means **reducing an unwanted risk**. An exporter expecting $1 million loses if the dollar falls against the rupee, so it can sell dollars forward, buy a put on USD/INR, or use a money-market hedge (§76). An options desk hedges the Greeks of its book.

75.2 Delta, gamma and vega hedging

- **Delta hedge:** trade the underlying or futures to neutralise delta. A book with $\Delta = +1{,}000$ shorts 1,000 units and has $\Delta \approx 0$. But, as before, delta $= 0$ does not mean risk $= 0$: gamma, vega, theta, rates, jumps and liquidity remain.
- **Gamma hedge:** the underlying has zero gamma, so you need another option. Choose n options so that $\Gamma_{\text{book}} + n\Gamma_{\text{hedge}} = 0$, then re-neutralise delta with the underlying.
- **Vega hedge:** the same idea, with options of the right maturity. Neutralising gamma *and* vega needs two different options: solve a 2×2 system.

Worked example: hedging gamma and vega of a short-option book

Book: short 100 of the 1-year ATM call: $\Delta = -63.68$, $\Gamma = -1.8762$, vega $= -3{,}752$ (per unit σ). Hedge instruments: a 3-month ATM call ($\Delta = 0.5695$, $\Gamma = 0.03929$, $\nu = 19.64$) and a 2-year ATM call ($\Delta = 0.6897$, $\Gamma = 0.01248$, $\nu = 49.91$). Solve

$$\begin{pmatrix} 0.03929 & 0.01248 \\ 19.64 & 49.91 \end{pmatrix} \begin{pmatrix} n_1 \\ n_2 \end{pmatrix} = \begin{pmatrix} 1.8762 \\ 3752 \end{pmatrix} \Rightarrow n_1 = 27.29,\ n_2 = 64.44.$$

After buying these, delta is -3.70, so buy 3.70 units of the underlying. The book is now delta-, gamma- and vega-neutral (to first order, at today's prices).

A subtle trap. Try to hedge with a 1-year 110-strike call instead of the 2-year one and the system has no useful solution. For options with the *same maturity*, $\nu = \Gamma\,\sigma S^2 T$ exactly (here both ratios are 2,000), so their gamma and vega are proportional and you cannot separate them. Separating gamma and vega needs options of different maturities.

Practical points. Use *bucketed* vega (by expiry and strike), because the surface does not move in parallel. Rebalance when $|\Delta|$ exceeds a band rather than on a clock: the optimal band width trades transaction costs against hedging error, scaling like $(\text{cost} \times \Gamma^2)^{1/3}$ (Whalley–Wilmott).

76 Money-market hedging for FX

Worked example: the importer (from the notes, completed)

An Indian importer must pay **$10,000 in three months**. Spot is ₹80/$, and the USD (foreign) rate is 4% a year, i.e. 1% for the quarter (simple interest).

1. Dollars to deposit today so that they grow to $10,000: $10{,}000/1.01 = \$9{,}900.99$.
2. Buy those dollars at spot: $9{,}900.99 \times 80 = 792{,}079.21$ today.
3. Deposit them at 4%. In three months they are exactly $10,000, which pays the supplier. The rupee cost is locked: no exposure to USD/INR remains.

Correction to the original notes: the INR side was missing

The notes stop at "₹792,080 today". But the importer either borrows those rupees or takes them out of its own deposits, and either way that has a cost over three months. With an illustrative INR rate of 6%:

$$792{,}079.21 \times (1 + 0.06 \times 0.25) = 803{,}960.40, \quad \text{an effective rate of } 80.396/\$.$$

That equals the covered-interest-parity forward: $F = 80 \times \frac{1.015}{1.01} = 80.396$. So the money-market hedge *replicates a forward contract*. If a bank quotes a forward rate that differs from 80.396 by more than transaction costs, one of the two routes is cheaper, and you should use that one. (With continuous compounding the dollar deposit would be $10{,}000e^{-0.01} = \$9{,}900.50$, or ₹792,039.87.)

Check yourself

- Use the spot × vol grid to find the worst scenario for a long call. Where would a short straddle's worst cell be?
- Why does delta–gamma underestimate the loss of a short call for a +20% move?
- Explain the day's P&L of an option book in terms of its Greeks. What does a persistent residual tell you?
- Draw the payoffs of a straddle, strangle, butterfly and bull spread, and state their break-evens.
- Why can't two options of the same maturity separately hedge gamma and vega?
- Show that a money-market hedge replicates a forward, and derive covered interest parity.

PART XII

Calibration, Validation and Quant Engineering

77 Calibration

77.1 What it is

A model has parameters $\theta = (\theta_1, \theta_2, \dots)$: one σ for BSM, five for Heston, four for SABR. The market gives you prices of many liquid options. **Calibration** chooses θ so that

$$\text{ModelPrice}_i(\theta) \approx \text{MarketPrice}_i \qquad \text{for all liquid instruments } i \text{ simultaneously.}$$

Your notes put it well: calibration means adjusting parameters so that theoretical prices match liquid market prices.

77.2 Why it matters

Worked example: from the notes

The market shows implied vols of 25% at strike 90, 20% at 100 and 23% at 110. Using $\sigma = 20\%$ for every strike prices the 100 option correctly and both wings badly: the model doesn't reproduce the market. A real system therefore runs:

Market options → Implied vols → Vol surface → Calibration → Pricing model → Risk

This is the step from *academic pricing* to *real-world modelling*: the model is anchored to the market before it is used to price anything that doesn't trade.

77.3 Objective functions

- **Price errors:** $\sum_i w_i (P_i^{\text{model}} - P_i^{\text{mkt}})^2$. Unweighted, this over-emphasises expensive ITM and long-dated options, whose prices are large in rupees.
- **Implied-vol errors:** $\sum_i (\sigma_i^{\text{model}} - \sigma_i^{\text{mkt}})^2$. Puts every strike on an equal footing, at the cost of an IV inversion per evaluation.
- **Vega-weighted price errors:** $w_i = 1/v_i^2$. Since $\delta P \approx v\,\delta\sigma$, this approximates the IV objective cheaply.
- **Bid-ask aware:** $w_i = 1/\text{spread}_i^2$, or zero penalty anywhere inside the bid-ask. A fit inside the spread is as good as the market can tell.

77.4 Optimisers

- **Local:** Levenberg–Marquardt or trust-region least squares (`scipy.optimize.least_squares`). Fast, given good starting values.
- **Global:** differential evolution or basin-hopping. Robust to the many local minima of models like Heston.
- **Common practice:** a global search first, then a local polish.

77.5 Overfitting and stability

More parameters fit better *today* but jump around from day to day, and a hedge built on jumping parameters churns. Remedies:

- **Regularise:** add a penalty $\lambda\|\theta - \theta_{\text{yesterday}}\|^2$.
- **Fix weakly identified parameters:** SABR's β by convention; in Heston, κ and θ trade off against each other, so consider fixing κ.
- **Test out of sample:** calibrate on some strikes, check the others.
- **Enforce no-arbitrage:** for SVI, Gatheral and Jacquier fit "slice-by-slice with a heavy penalty for calendar spread arbitrage".

Worked example: calibrating SVI to a smile

Take the five Heston-implied vols from §41 (24.05%, 21.59%, 19.21%, 17.02%, 15.27% at strikes 80–120, $F = 105.13$) as "market" data, and fit raw SVI in implied-vol terms with `least_squares`:

- a flat (BSM) vol of 19.43% gives an RMSE of **3.14 vol points**;
- SVI gives $a = 0.0150$, $b = 0.0508$, $\rho = -0.970$, $m = 0.141$, $s = 0.156$, with an RMSE of **0.0014 vol points**.

A near-perfect fit, but notice $\rho = -0.97$, almost at its bound. With five parameters and five points, the fit is *exactly determined*: it will match noise as happily as signal. In practice you would use many more strikes, add bid-ask weights, and check the no-arbitrage conditions (here the minimum total variance $a + bs\sqrt{1-\rho^2} = 0.0169 > 0$ and the wing slope $b(1 + |\rho|) = 0.10 < 2$, both fine).

Report: RMSE in vol points, the maximum error, the time series of each parameter, and the condition number of the Jacobian (a near-singular Jacobian means some parameters are not identified).

78 Model validation

A quant does not ask "does the model produce a number?" but "*does the model produce a sensible number under known conditions?*" Your project rightly makes validation its own phase. Here is the checklist, expanded from the notes' seven tests:

1. **Put–call parity.** $C - P = S_0 e^{-qT} - Ke^{-rT}$ to machine precision for every European engine (BSM, tree, PDE). For Monte Carlo, to within about 3 SE.

2. **No-arbitrage bounds.** Prices inside the bounds of §34; calls decreasing and convex in K; total variance non-decreasing in T.

3. **American $\geq$ European.** And the American call on a non-dividend stock equals the European call.

4. **Tree convergence.** Run $N = 10, 20, 50, 100, 500, 1000$. CRR should oscillate with error $\propto 1/N$; LR should converge smoothly (Table 7).

5. **Monte Carlo convergence.** Run 1k, 5k, 10k, 50k, 100k, 1M paths. The SE should fall like $1/\sqrt{N}$, and the 95% CI should contain the analytic price about 95% of the time over many seeds.

6. **PDE convergence.** Halve ΔS and Δt: the CN error should fall about $4\times$.

7. **Greek stability.** Analytic vs bump-and-revalue (central differences, sensible h) vs MC pathwise. No oscillation in CN gammas (use Rannacher). Small input changes should not produce absurd Greek jumps.

8. **Barrier behaviour.** $H \to \infty$ (knock-out) gives the vanilla; in + out = vanilla; BGK-corrected continuous price $\approx$ discrete MC.

9. **Limits.** $\sigma \to 0$ gives the discounted intrinsic value on the forward; $T \to 0$ gives the payoff; $S \to 0$ or $S \to \infty$ gives the right asymptotes.

10. **Cross-library benchmarks.** QuantLib (analytic, binomial `"lr"`, finite-difference and MC engines), py_vollib, and textbook values.

11. **Hedging simulation.** Discrete-hedge P&L standard deviation $\approx$ Derman–Kamal $\sqrt{\pi/4N}\,\nu\sigma$ (§53).

The five questions to ask of every model (from your notes, kept because they are exactly right):

1. What assumptions does it make? (BSM: constant vol, no jumps, continuous hedging.)

2. What products can it price? (BSM: European vanillas.)

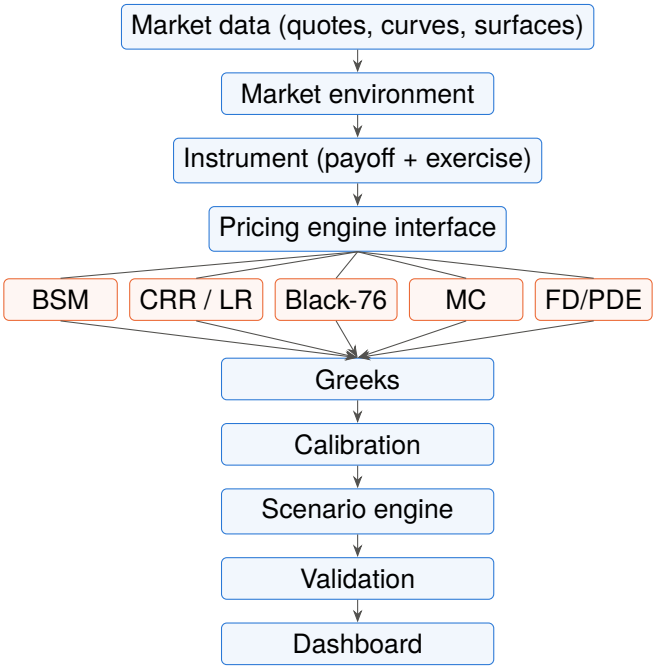
3. What inputs does it need? (S, K, T, r, q, σ.)

4. How does it fail? (Wrong vol surface, jumps, early exercise, liquidity.)

5. How do I validate it? (Analytic solutions, market prices, alternative numerical methods, convergence tests, no-arbitrage relationships.)

79 Quant software architecture

Your project is not just a mathematics exercise. It asks for a documented Python package with unit tests, a modular architecture, an interactive dashboard, a research report and a presentation. A production-style pricing system should never be one giant function.



79.1 Abstractions

Instrument (pure description, no pricing logic):

```
from dataclasses import dataclass
from typing import Callable

@dataclass(frozen=True)
class Option:
    payoff: Callable          # e.g. lambda S: np.maximum(S - K, 0)
    strike: float
    expiry: float             # in years, after day-count conversion
    exercise: str             # "european" | "american" | "bermudan"
    underlying: str           # "NIFTY", "NIFTY-FUT", ...
```

Pricing engine (interchangeable algorithms behind one interface):

```
from abc import ABC, abstractmethod

class PricingEngine(ABC):
    @abstractmethod
    def price(self, option, market) -> float: ...
    def greeks(self, option, market, h=1e-4) -> dict:
        ...  # default: bump-and-revalue with central differences; override if analytic

class BlackScholesEngine(PricingEngine): ...
class CRREngine(PricingEngine): ...
class LeisenReimerEngine(PricingEngine): ...
class Black76Engine(PricingEngine): ...
class MonteCarloEngine(PricingEngine): ...
class FiniteDifferenceEngine(PricingEngine): ...
```

Then `price(option, market)` can be computed by any engine. Pricing the *same* instrument with several engines in a loop is the core of model comparison and validation.

79.2 How QuantLib does it

QuantLib separates **market data** (observable `SimpleQuotes` wrapped in `Handles`), **term structures** (yield, dividend, volatility), **processes** (`BlackScholesMertonProcess`), **instruments** (`VanillaOption(PlainVanillaPayoff, EuropeanExercise)`) and **engines** attached with `setPricingEngine`. Because quotes are observable, changing one with `setValue` automatically re-prices everything that depends on it, which is exactly what a scenario engine needs.

```
import QuantLib as ql
today = ql.Date(3, ql.October, 2026); ql.Settings.instance().evaluationDate = today
spot = ql.SimpleQuote(100.0); dc = ql.Actual365Fixed()
r   = ql.YieldTermStructureHandle(ql.FlatForward(today, 0.05, dc))
q   = ql.YieldTermStructureHandle(ql.FlatForward(today, 0.0, dc))
vol = ql.BlackVolTermStructureHandle(ql.BlackConstantVol(today, ql.NullCalendar(), 0.20, dc)
    )
proc = ql.BlackScholesMertonProcess(ql.QuoteHandle(spot), q, r, vol)
opt = ql.VanillaOption(ql.PlainVanillaPayoff(ql.Option.Call, 100.0),
                       ql.EuropeanExercise(today + ql.Period(1, ql.Years)))
opt.setPricingEngine(ql.AnalyticEuropeanEngine(proc));         print(opt.NPV())  # ~10.45
opt.setPricingEngine(ql.BinomialVanillaEngine(proc, "lr", 201)); print(opt.NPV())
spot.setValue(105.0); print(opt.NPV())    # observer pattern re-prices automatically
```

(One year from a given date is not exactly $T = 1.0000$ under ACT/365 when the span includes 29 February, so expect tiny differences from 10.4506. It's a nice lesson in day counts.)

79.3 Design rules

- Keep instruments separate from engines (open/closed principle): adding a model should never require editing an instrument.
- Write pure, vectorised (NumPy) functions; no hidden global state.
- Put conventions (day count, theta per day, vega per point) in *one* configuration object.
- Ship unit tests with every engine, in the same commit.

A suggested package layout:

```
quantpricer/
  market/       quotes.py  curves.py  volsurface.py  daycount.py
  instruments/  vanilla.py  asian.py  lookback.py  basket.py  barrier.py
  engines/      bsm.py  black76.py  trees.py  montecarlo.py  fd.py
  risk/         greeks.py  scenarios.py  pnl_explain.py
  calibration/  iv.py  svi.py  heston.py
  validation/   checks.py
  dashboard/    app.py
tests/          test_bsm.py  test_trees.py  test_mc.py  test_fd.py  test_parity.py ...
```

80 Dashboard design

Your notes specify an option-type selector, model selector, market-data inputs, pricing output, Greeks, payoff diagram, scenario analysis, volatility smile, Monte Carlo convergence, CRR vs LR comparison and risk metrics. A page structure that covers all of it:

Page	Contents
1. Market overview	Underlying, spot, IV, rate, expiry, bid, ask, volume, open interest
2. Option pricer	Option type, model, strike, expiry, vol, rate → price, delta, gamma, vega, theta, rho (with units)
3. Payoff	Gross payoff, net P&L, break-even(s), maximum loss, maximum profit
4. Greeks	Greeks table; Greek vs spot; Greek vs vol; theta vs time; a Greek heatmap
5. Volatility	IV by strike (smile), IV by maturity (term structure), 3-D surface
6. Model comparison	BSM, CRR, LR, Monte Carlo and market price side by side, with errors; CRR vs LR convergence chart
7. Scenario analysis	Spot × vol P&L grid (as in §71); rate and time shocks; delta–gamma vs full revaluation
8. Validation	Put–call parity error; tree and MC convergence; Greek stability; no-arbitrage checks (red if violated)

Tools. Streamlit is the fastest route: sidebar sliders for S, K, T, r, σ, q, `st.cache_data` for expensive surfaces, tabs for each page. **Dash** gives more control (callbacks) for multi-user use. **Plotly** draws the charts: price and Greeks against spot for several expiries, a 3-D IV surface (`go.Surface`), a scenario heatmap, tree-vs-BSM convergence, and an MC histogram with its CI band.

Principles: show units on every Greek; display the conventions in use; never show a Monte Carlo number without its error bar; flag arbitrage violations in red; make every chart reproducible from logged inputs.

81 Research workflow and testing

- **pytest** with `@pytest.mark.parametrize` over grids of $(S, K, T, \sigma, r, q, \text{call/put})$.
- **Property-based tests** (`hypothesis`) for invariants: parity, monotonicity in σ, convexity in K, bounds.
- **Golden values:** BSM 10.4506 / 5.5735; Hull 4.7594 / 0.8086; LR $p = 0.51045$ at $n = 51$; Black-76 7.5771; American put 6.0903; up-and-out 1.1761; floating lookback 17.2168; Derman–Kamal ratios.
- **Statistical tests for MC:** assert $|\text{MC} - \text{analytic}| < 3\,\text{SE}$, not a fixed tolerance, and fix seeds.
- **Reproducibility:** pin package versions; log seeds and market-data snapshots; version calibrations; explore in notebooks but put real code in modules; run CI on every commit.
- **Mutation sanity:** deliberately break something (flip theta's sign, change the Derman–Kamal constant) and confirm the tests fail. A test suite that can't catch a planted bug can't catch a real one.

```
import pytest
from quantpricer.engines import bsm, lr_tree, mc_european

@pytest.mark.parametrize("S,K,T,r,sig,q", [(100,100,1,.05,.2,0), (100,80,.5,.03,.35,.01),
                                           (100,130,2,.07,.15,.02)])
```

```python
def test_put_call_parity(S, K, T, r, sig, q):
    c, p = bsm(S,K,T,r,sig,q,1), bsm(S,K,T,r,sig,q,-1)
    assert c - p == pytest.approx(S*np.exp(-q*T) - K*np.exp(-r*T), abs=1e-10)

def test_american_ge_european():
    assert lr_tree(100,100,1,.05,.2,cp=-1,american=True) >= bsm(100,100,1,.05,.2,cp=-1)

def test_mc_within_3se():
    est, se, _ = mc_european(100,100,1,.05,.2, N=200_000, seed=1)
    assert abs(est - bsm(100,100,1,.05,.2)) < 3*se
```

Check yourself

- What objective function would you use to calibrate Heston to a NIFTY smile, and why?
- Your SVI fit is perfect today and wild tomorrow. What went wrong, and what do you change?
- List ten validation checks for a pricing library.
- Sketch the class structure of your pricing package. Why separate instruments from engines?
- Write a test that would catch a theta sign error.

PART XIII

Putting It All Together

82 How a quant thinks about the project

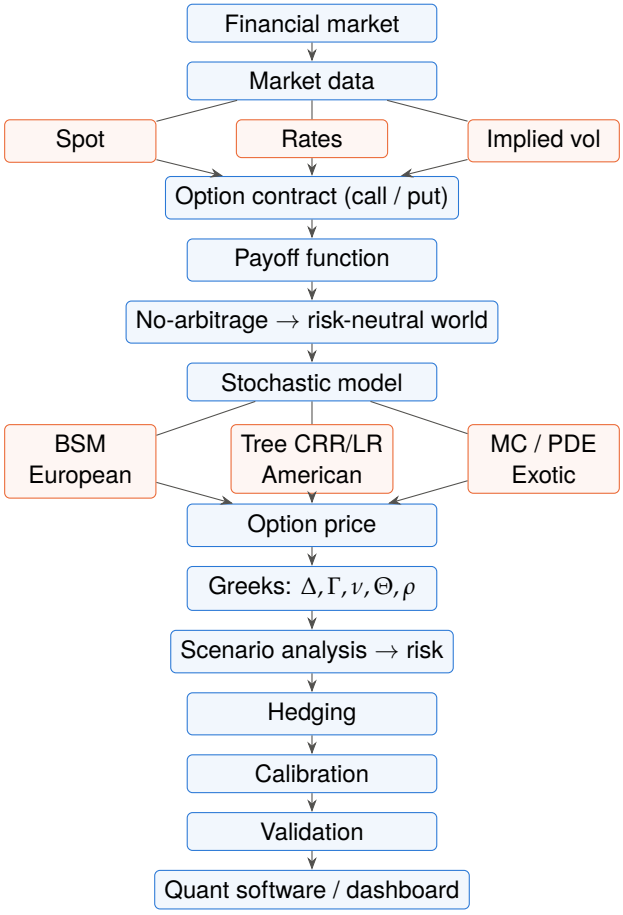
The most important shift in mindset is this. Don't think: *"I need to code Black–Scholes."* Think: *"I need to build a research and risk system that tells me whether my models are reliable under different market conditions."* The formulas are the easy part. The hard, valuable part is knowing when they apply, how they fail, and how to prove your implementation is right.

83 How all the models connect

Don't memorise the models as separate topics. Understand *why each one exists*: each solves a problem the previous one could not.

Product	The problem	Model	Why this model
European call/put	Price a payoff that depends only on S_T	BSM	Closed form; the benchmark for everything else
American call/put	Holder can exercise early	CRR / LR tree	Compare "exercise now" with "continue" at every node
Futures option	Underlying is a futures price	Black-76	F has no carry; BSM with $Se^{-qT} \to Fe^{-rT}$
Asian	Payoff depends on the average	Monte Carlo	Needs $S(t_1), \ldots, S(t_n)$: the whole path
Lookback	Payoff depends on max/min	Monte Carlo	Needs the path extremes
Basket	Several correlated assets	Monte Carlo	Needs joint paths and a correlation matrix
Barrier	Did the price cross H?	PDE / finite differences	The barrier is simply a boundary of the grid

84 The complete mental model



Once this structure is clear, Black–Scholes, Greeks, CRR, Leisen–Reimer, Monte Carlo, barrier options and PDEs stop looking like unrelated formulas. They become different mathematical and computational answers to different derivatives-pricing problems.

85 The nine conceptual connections

If you remember nothing else at first, remember these:

1. **Option → payoff.** Before pricing anything, know exactly what it pays.
2. **Payoff + no-arbitrage → pricing.** Pricing is not predicting the future.
3. **Randomness → probability → Brownian motion → GBM.** The mathematical foundation of the classical models.
4. **GBM + no-arbitrage → Black–Scholes.** A closed form for European vanillas.
5. **Option price → derivatives → Greeks.** The Greeks tell you how sensitive the price is.
6. **American exercise → tree → backward induction.** Trees make early exercise explicit.
7. **Path dependence → simulation or PDE.** Asians, lookbacks, baskets and barriers need more than a terminal payoff.
8. **Market price → inverse pricing → implied volatility.** Why root finding matters.

9. **Model → price → Greeks → risk → hedge.** Pricing is only one part of a quant system.

86 One option, end to end

Here is everything in this book applied to one contract: the running-example call ($S_0 = 100$, $K = 100$, $T = 1$, $r = 5\%$, $q = 0$, $\sigma = 20\%$).

1. **Identify the instrument.** A European call on a non-dividend stock. Model: BSM.
2. **Write the payoff.** $C_T = \max(S_T - 100, 0)$.
3. **Moneyness.** $S_0 = K$: ATM spot. The forward is 105.13, so it is about 5% OTM-forward.
4. **Intrinsic value.** $\max(100 - 100, 0) = 0$, so the whole price is time value.
5. **Theoretical price.** $d_1 = 0.35$, $d_2 = 0.15$, $C = 10.4506$. Parity check: $C - P = 4.8771$. ✓
6. **Greeks.** $\Delta = 0.6368$, $\Gamma = 0.01876$, vega 0.3752/pt, $\Theta = -0.0176$/day, $\rho = 0.5323$/1%.
7. **The market disagrees.** The market price is ₹11, not 10.45. What volatility does it imply? Newton from 20% converges in two steps to **21.46**%. The market is pricing more risk than our 20% assumption.
8. **Risk.** If spot rises to 105: delta says +3.18, delta–gamma says +3.42, full revaluation says +3.41. The spot × vol grid shows the worst cell (−10%, −10 vol) loses 8.77.
9. **Hedge.** 100 long calls have delta +63.7: short 64 shares to be delta-neutral. Gamma (+1.88) and vega (+37.5 per point) remain; hedge them with options of two other maturities if required.
10. **Validate.** Parity holds; the price lies within its bounds; the LR tree gives 10.45058 at 201 steps; Monte Carlo gives 10.446 ± 0.047 with 100,000 paths; the CN PDE converges at second order. Everything agrees.

Now you are no longer just "calculating an option price". You are running a quantitative pricing and risk system.

87 Quant versus formula-knower

A beginner says: *"Black–Scholes is* $C = SN(d_1) - Ke^{-rT}N(d_2)$*."*

A stronger answer: *"BSM derives a no-arbitrage price for European vanilla options under GBM with constant volatility, by showing the option can be replicated with the stock and a bank account. I know its inputs, assumptions and limitations, its Greeks, how to invert it for implied volatility, and how to validate it against trees, Monte Carlo, PDEs and market data."*

The strongest understanding sees the chain: model → assumptions → price → Greeks → market comparison → calibration → risk → hedging → validation. That chain is the mindset your project is trying to build.

88 What to learn mathematically, in order

1. **Basic mathematics:** algebra, exponents, logarithms (especially $\ln x$), functions, basic calculus.
2. **Probability:** random variables, distributions, expectation, variance, standard deviation, covariance, correlation, the normal and lognormal distributions, the CLT.
3. **Calculus:** derivatives, partial derivatives, the chain rule, Taylor expansion (for Greeks and numerical approximations).
4. **Stochastic processes:** Brownian motion, the Wiener process, GBM, dW, Itô's lemma.
5. **Quant models:** BSM → Greeks → implied vol → binomial → CRR → American options → LR → Black-76 → Monte Carlo → exotics → PDE.

89 A 12-week study and build plan

This merges the 10-week sequence from your notes with the build-and-validate plan from the research.

Weeks	Learn	Build and validate
1	Stocks, bonds, forwards, futures, options, payoffs, P&L, moneyness, arbitrage	Payoff functions; bounds and parity checks; day-count and compounding utilities
2	Probability, normal and lognormal, expectation, variance, covariance, correlation; partial derivatives; Taylor expansion	Historical vol (close-to-close, Parkinson, Yang–Zhang) on NIFTY OHLC data
3	Brownian motion, GBM, Itô's lemma, risk-neutral pricing, BSM derivation (Shreve Vol. I for trees, Vol. II for Itô)	A GBM path simulator; check mean/median and quadratic variation
4	BSM, Greeks, implied vol, Newton–Raphson, bisection, Brent, put–call parity	BSM price + Greeks + Brent IV; test against py_vollib/QuantLib; fix theta and vega units
5	Binomial trees, CRR, American options, Leisen–Reimer, Black-76	CRR and LR engines; the convergence experiment; American $\geq$ European; Black-76 Greeks and plots
6–7	Monte Carlo, variance reduction, MC Greeks, Asian, lookback, basket, correlation	MC engine with CIs; antithetic and control variates; Kemna–Vorst Asian; Cholesky baskets; lookback vs closed form
8	PDEs, finite differences, stability, boundary conditions, barrier options	CN solver with Rannacher; barrier via grid; in–out parity; BGK vs discrete MC
9–10	Scenario analysis, delta–gamma, P&L explain, hedging	Spot $\times$ vol grids; delta–gamma vs full revaluation; daily P&L explain; hedging simulator vs Derman–Kamal

11–12	Smile models and calibration; validation; engineering	SVI with arbitrage checks; Heston (and SABR) calibration; full validation suite; Streamlit dashboard; report and presentation

90 Questions you should be able to answer without notes

Fundamentals. What is a derivative? What is an underlying? Call vs put? Strike vs spot? European vs American? ITM/ATM/OTM, and which ATM? Intrinsic vs time value? Payoff vs P&L?

Pricing. What is no-arbitrage? Prove put–call parity. What is risk-neutral pricing, and why does μ drop out? Why discount at the risk-free rate? What is volatility? Historical vs implied? Why does the smile exist? What is a volatility surface, and what makes one arbitrage-free?

Mathematics. What is Brownian motion? What is GBM? What is dW, and why is $(dW)^2 = dt$? Why is GBM lognormal, and where does $-\frac{1}{2}\sigma^2$ come from? Derive $d \ln S$ with Itô. What do stochastic volatility and jump diffusion add?

BSM. What are d_1 and d_2? What assumptions does BSM make, and how does each fail? How is implied vol calculated, and why is it a root-finding problem? Why does BSM work for European vanillas?

Greeks. Define delta, gamma, vega, theta and rho, with units. Why is theta $\partial V/\partial t$? What is the gamma–theta relationship? Why doesn't delta-neutral mean risk-free?

Numerical methods. Why trees? How does CRR work? What is backward induction? Why do American options need early-exercise checks? What is Leisen–Reimer and why must n be odd? Why Monte Carlo, and why does it converge slowly? What is variance reduction? How do you compute an MC confidence interval and MC Greeks?

Exotics. Why do Asians and lookbacks need the path? What is a basket and why does correlation matter? What is a barrier? Knock-in vs knock-out, and in–out parity? Why are discrete and continuous barriers different? Why use PDEs for barriers?

Risk. What is scenario analysis? Delta vs delta–gamma approximation? How do you construct a payoff diagram? What is P&L attribution? How do you hedge gamma and vega together?

Quant engineering. How would you structure a pricing engine? How would you validate it, calibrate it, and test its convergence? How would you compute Greeks? What should the dashboard show? How would you compare models?

91 Curated resources

Free university material

- *Probability and stochastic calculus:* MIT OCW 18.S096 *Topics in Mathematics with Applications in Finance*; NYU Courant lecture notes (Gatheral and others).
- *Monte Carlo and Greeks:* Martin Haugh, Columbia IEOR E4703 (estimating Greeks, variance reduction) and IEOR E4602 (importance sampling).
- *Volatility smile:* Emanuel Derman's Columbia lectures on the smile; Jim Gatheral's SVI lectures.

- *Numerical methods:* Imperial College numerical-methods course materials (Jacquier); Wilmott's PDE texts.
- *Introductory finance:* MIT 15.401 Finance Theory; NPTEL courses on financial derivatives (IIT/IIM faculty).

Books, by stage

- *Beginner:* Hull, *Options, Futures and Other Derivatives*; Natenberg, *Option Volatility and Pricing* (trader intuition); Sinclair, *Volatility Trading* (realised vs implied, estimators).
- *Intermediate:* Shreve, *Stochastic Calculus for Finance I & II*; Wilmott, *Paul Wilmott Introduces Quantitative Finance*; Haug, *The Complete Guide to Option Pricing Formulas* (barriers, lookbacks, Asians).
- *Advanced:* Glasserman, *Monte Carlo Methods in Financial Engineering*; Gatheral, *The Volatility Surface*; Duffy, *Finite Difference Methods in Financial Engineering*; Ballabio, *Implementing QuantLib* and the *QuantLib Python Cookbook*.

Key papers. Black and Scholes (1973); Merton (1973, "Theory of Rational Option Pricing"; 1976, jumps); Black (1976, commodity contracts); Cox, Ross and Rubinstein (1979); Harrison and Kreps (1979); Harrison and Pliska (1981); Goldman, Sosin and Gatto (1979); Kemna and Vorst (1990); Reiner and Rubinstein (1991); Turnbull and Wakeman (1991); Conze and Viswanathan (1991); Heston (1993); Rubinstein (1994, "Implied Binomial Trees"); Leisen and Reimer (1996); Broadie, Glasserman and Kou (1997); Derman and Kamal (1999, "When You Cannot Hedge Continuously"); Yang and Zhang (2000); Hagan et al. (2002, "Managing Smile Risk"); Gatheral and Jacquier (2014, "Arbitrage-free SVI volatility surfaces"); Jäckel (2015, "Let's Be Rational").

Code. QuantLib / QuantLib-Python (the `Examples/DiscreteHedging` example is a ready-made hedging test); py_vollib (BSM, BSM with q, Black-76, Let's Be Rational); `scipy.stats.qmc` (Sobol), `scipy.optimize` (`brentq`, `least_squares`); `arch` (GARCH); Jäckel's reference implementation (jaeckel.org).

Exchange and regulator sources. NSE: NIFTY F&O product page, India VIX white paper, NSE Clearing settlement pages, lot-size circulars. FBIL: MIBOR methodology. RBI: MPC resolutions and T-bill auctions. New York Fed: SOFR. CBOE: VIX white paper. CME Group education.

Forums and practitioner commentary. Quant StackExchange (excellent for "why does my IV solver fail" and sticky-delta questions), the Wilmott forums, QuantNet, r/quant; Kris Abdelmessih (Moontower) and Euan Sinclair for trader intuition on volatility and P&L. Quality on forums and social media varies widely; cross-check against the books and papers above.

92 Caveats

- **Numbers.** Every figure printed in this book was recomputed in Python (NumPy/SciPy) while preparing it. Monte Carlo figures depend on the random seed and will differ slightly in your runs; compare them using their standard errors, not digit by digit.
- **Corrections to earlier material.** Recomputing changed two earlier figures: the P&L-explain example (actual P&L 1.0006, not 1.0013) and the Brenner–Subrahmanyam guess (19.97% applies to the ATM-*forward* option, not the ATM-spot one).
- **Dated market facts.** RBI policy rates, T-bill yields, NIFTY lot sizes and expiry days are as of autumn 2026 and change regularly. The 91-day T-bill figure comes from CEIC (sourced from

RBI) rather than RBI's own release. An RBI MPC meeting was scheduled for 5–7 October 2026.

- **A source discrepancy.** NSE's product page has described NIFTY options as "physical settled" while NSE Clearing says cash-settled. This book follows the clearing house.
- **Closed-form exotics.** The barrier and lookback formulas here are checked numerically against Monte Carlo and finite differences for the running example. For other cases (puts, rebates, dividends), implement them from Haug and validate against QuantLib rather than trusting any single transcription.
- **Illustrative assumptions.** The 6% INR rate in the money-market hedge, the Heston/SABR/Merton/SVI parameters and the basket inputs are illustrative choices, not market data.